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Chapter 24

GELFAND-P001133

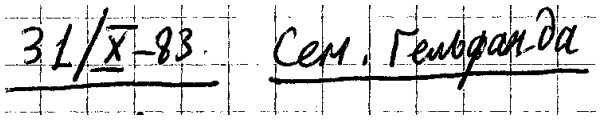


Diagram and typed description. 31 October 1983. Gelfand Seminar

Original source details: GELFAND-P001133

Zelevinsky.

Λ_n^k — homogeneous symm. polynomials in $x_1, \dots, x_n$ of degree k (over $\mathbb{Z}$).

For $n \geq k$, Λ_n^k does not depend on n.

Consider $\Lambda^k = \varprojlim_{\leftarrow n} \Lambda_n^k$ and the ring $\Lambda = \bigoplus_{k \geq 0} \Lambda^k$ of all symm. functions.

Elementary theory of symm. functions: single out 5 bases and establish the relations between them.

Partition:

$$\lambda = (\lambda_1 \geq \lambda_2 \geq \dots) \qquad \text{(all zero from some point on)}$$

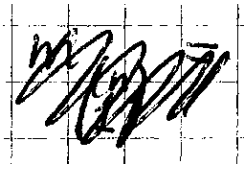
(all λ_j — nonnegative integers).

$\lambda = (4, 2, 1)$ example. λ_i — parts of the partition.

1°. Monomial functions:

$$m_\lambda(x) = x_1^{\lambda_1} x_2^{\lambda_2} \dots x_N^{\lambda_N} + \text{all possible permutations}$$

The partition can be written as $\lambda = (1^{m_1}, 2^{m_2}, \dots)$ (m_1 — the number of ones, m_2 — the number of twos, etc.).



Uncertain reading. [crossed out: $m_{\dots}$; the subscript is illegible.]

Original source details: GELFAND-P001133

$$m_{(n)} = x_1^n + x_2^n + \dots$$

$$m_{(1^n)} = x_1 x_2 \dots x_n + x_1 x_2 \dots x_{n-1} x_{n+1} + \dots$$

$m_\lambda(x)$ — a basis.

2°. $e_\lambda = e_{\lambda_1} e_{\lambda_2} \dots$

3°. $h_\lambda = h_{\lambda_1} h_{\lambda_2} \dots$

4°. $p_\lambda = p_{\lambda_1} p_{\lambda_2} \dots$

$e_n = m_{(1^n)}$ — elementary symm. functions

$h_n = \sum_{|\lambda|=n} m_\lambda$ — complete symm. function

$p_n = m_{(n)}$ — power sums

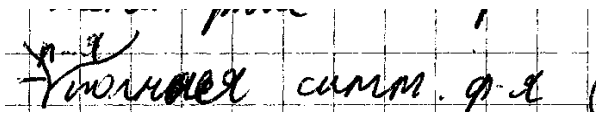
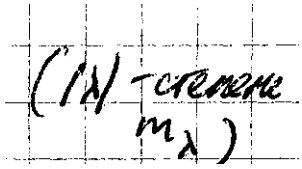


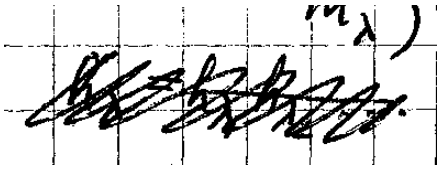
Diagram and typed description. [inserted above] nth.
The overwritten word reads «complete».

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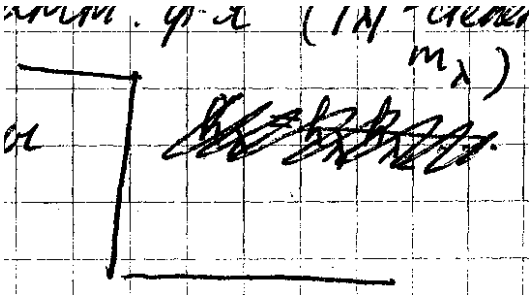
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Diagram and typed description. [At right] ($|\lambda|$ — degree of m_λ).



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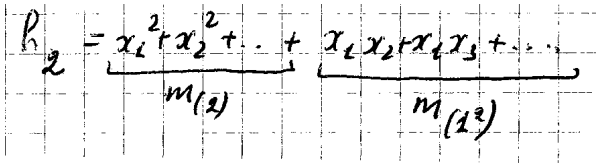
Uncertain reading. [At right, by the bracket] the crossed-out passage is illegible.



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Undeciphered passage. The writing under the strokes cannot be recovered. The right bracket is visible.

$$h_2 = \underbrace{x_1^2 + x_2^2 + \ldots +}_{m_{(2)}} \underbrace{x_1x_2 + x_1x_3 + \ldots}_{m_{(1^2)}}.$$



Original source details: GELFAND-P001133

Diagram and typed description. Brackets beneath the terms in the formula for h_2.



Original source details: GELFAND-P001133

Uncertain reading. [crossed out: h_2?]

$$h_{(1^2)} = h_1^2 = \left(\sum x_i\right)^2.$$

GELFAND-P001134

Claim.

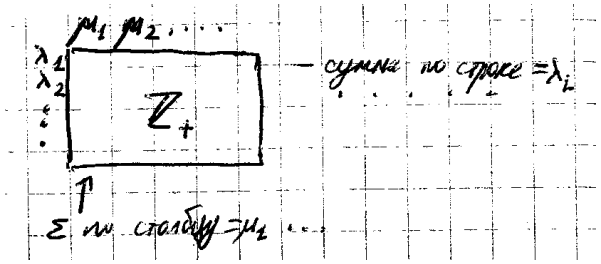
$$\Lambda = \bigoplus_\lambda \mathbb{Z} e_\lambda = \bigoplus_\lambda \mathbb{Z} h_\lambda,$$
$$\Lambda_{\mathbb{Q}} = \bigoplus_\lambda \mathbb{Q} p_\lambda.$$

[At right, beyond ||] Or:

$$\Lambda = \mathbb{Z}[e_1, e_2, \ldots] = \mathbb{Z}[h_1, h_2, \ldots],$$
$$\Lambda_{\mathbb{Q}} = \mathbb{Q}[p_1, p_2, \ldots].$$

Relation between these 4 bases.

$$h_\lambda = \sum_\mu M_{\lambda\mu} m_\mu, \qquad M_{\lambda\mu} = \text{number of matrices.}$$



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Diagram and typed description.

$\mu_1, \mu_2, \ldots$

λ_1
λ_2
$\vdots$
$\mathbb{Z}_+$

— row sum $= \lambda_i$

$\sum$ down the column $= \mu_1 \ldots$

$$e_\lambda = \sum_\mu M^0_{\lambda\mu} m_\mu, \qquad M^0_{\lambda\mu} = \text{the same matrices, but with entries 0 and 1.}$$

Exercise: express p_λ in terms of m_μ (this is done similarly).

Relate h_λ, e_λ and p_λ. This is done using generating functions and Newton's formulas.

$$E(t) = \sum_{n \geq 0} e_n t^n, \qquad H(t) = \sum_{n \geq 0} h_n t^n, \qquad P(t) = \sum_{n \geq 1} p_n t^n.$$

$$E(t) = \prod_i (1 + x_i t), \qquad H(t) = \prod_i (1 - x_i t)^{-1}, \qquad P(t) = \frac{tH'(t)}{H(t)} = \sum_i \frac{t x_i}{1 - t x_i}.$$

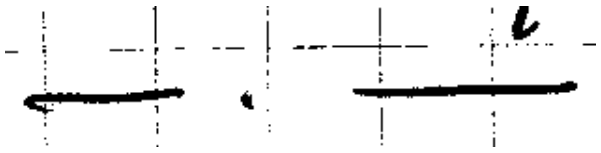


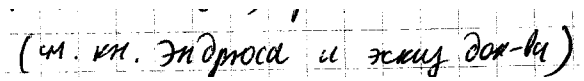
Diagram and typed description. — · —

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I.M.G. In Ramanujan

$p(n)$ — the number of partitions of n into a sum $\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_n \geq 0$ ($|\lambda| = n$).

There is an explicit (!!!) formula — the result from the asymptotics, but with more terms. (see Andrews's book and [unclear abbreviation] proofs).



Uncertain reading. The abbreviation between "and" and "proofs" is unclear.

Original source details: GELFAND-P001134

$$\sum_{n\geq 0} p(n)t^n = \prod_{i\geq 1}(1-t^i)^{-1} = \frac{1}{(1-t)(1-t^2)(1-t^3)\dots}.$$

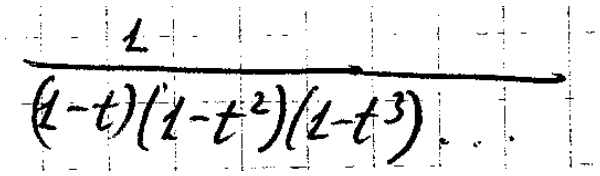


Diagram and typed description. Ellipsis under the fraction bar after $(1-t^3)$.

Original source details: GELFAND-P001134

Newton's formulas:

$$H(t)E(-t) = 1.$$

Hence

$$\sum_{r=0}^{n}(-1)^r e_r h_{n-r} = 0 \qquad \text{for } n \geq 1.$$

$$P(t)H(t) = tH'(t)$$

$$nh_n = \sum_{r=1}^{n} p_r h_{n-r}, \qquad n \geq 1.$$

$$a_\alpha = \det \begin{vmatrix} x_1^{\alpha_1} & x_1^{\alpha_2} & \dots & x_1^{\alpha_n} \\ x_2^{\alpha_1} & x_2^{\alpha_2} & \dots & x_2^{\alpha_n} \\ \dots & \dots & \dots & \\ x_n^{\alpha_1} & x_n^{\alpha_2} & \dots & x_n^{\alpha_n} \end{vmatrix}$$

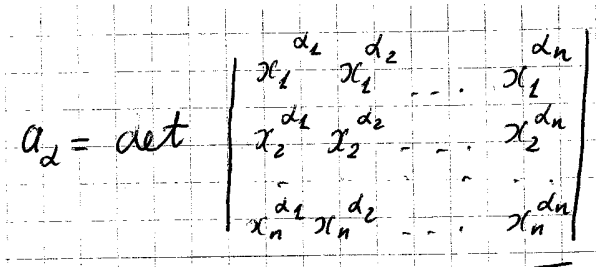


Diagram and typed description. The manuscript has "det" before the vertical bars and a line below the determinant.

Original source details: GELFAND-P001134

$$a_\delta = \prod_{i<j}(x_i - x_j)$$

[At right:] Schur function:

$$s_\lambda = a_{\lambda+\delta}/a_\delta.$$



Uncertain reading. [Before δ an illegible sign is crossed out.]

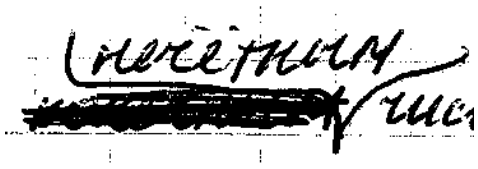
Original source details: GELFAND-P001134

$$\delta = (n-1, n-2, \dots).$$

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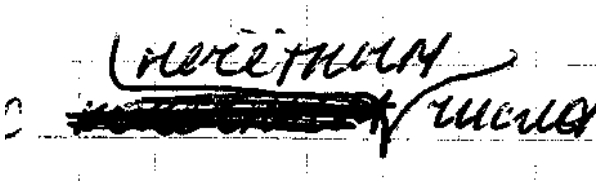
[source number: 4]

On Witten's work (nonexistence of gauge theories with [crossed out; inserted above: an odd] number of fermions).



Uncertain reading. Above the crossed-out word is «an odd».

Original source details: GELFAND-P001135



Undeciphered passage. The word beneath the crossing-out is illegible.

Original source details: GELFAND-P001135

$$\mathcal{L}(\Psi, A)$$

$$\int d\Psi\, d\bar{\Psi}\, dA\ \exp(-\mathcal{L}),$$



Uncertain reading. [After the comma crossed-out writing.]

Original source details: GELFAND-P001135



Undeciphered passage. The crossed-out signs after the comma cannot be recovered.

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$$\mathcal{L} = \int (\bar{\Psi}\, iD\Psi + \text{[crossed out]}\ \operatorname{Tr} F_{\mu\nu}^2)\,.$$



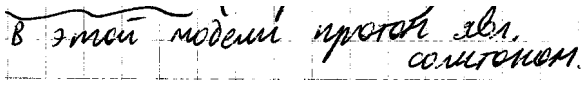
Uncertain reading. [Between + and Tr crossed-out writing.]

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There arises $(\det D)^{+1/2}$, where

$$D = \sum \gamma_i \left(A_i + \frac{\partial}{\partial x_i} \right).$$

Perelomov. On Witten's work on the σ-model (of the proton).

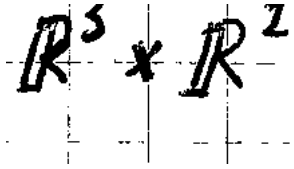


Original source details: GELFAND-P001135

Diagram and typed description. [At right below the heading] In this model the proton is a soliton.

$$S = S_0 + S_1 + S_2$$

$M_4 = \mathbb{R}^3 \times \mathbb{R}^1$ — space/time.



Original source details: GELFAND-P001135

Diagram and typed description. $\mathbb{R}^3 \times \mathbb{R}^1$.

$G = SU(2)$ or $SU(3)$

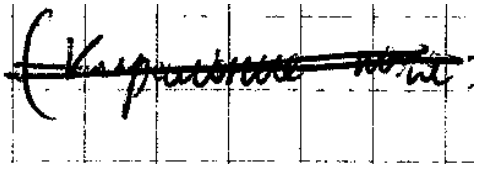
$$\varphi \colon M^4 \longrightarrow G, \qquad g(x), \qquad \varphi_\mu = \partial_\mu g \cdot g^{-1}.$$



Original source details: GELFAND-P001135

Uncertain reading. [After φ before the colon crossed-out writing.]

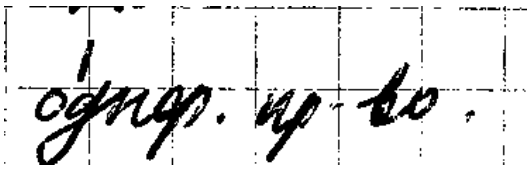
$$S_0 = \int d^4x \, \langle \varphi_\mu, \varphi_\mu \rangle.$$



Original source details: GELFAND-P001135

Diagram and typed description. [crossed out] ~~(Chiral field.~~

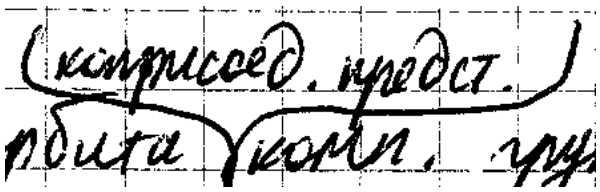
More general case: $\varphi \colon M^4 \longrightarrow X$.



Original source details: GELFAND-P001135

Diagram and typed description. [Under X] homogeneous space.

2-dimensional theory, X — orbit of a compact group [inserted above: (coadjoint repr.)] $\Longrightarrow$ all instantons have been found (Perelomov).



Original source details: GELFAND-P001135

Diagram and typed description. [inserted above:] (coadjoint repr.)

$X = S^2$ — Belavin, Polyakov (1975).

For chiral fields there is an analogue of d'Alembert's representation (local solutions found — Krichever). Soliton solutions have been written down.

$$S_2 = \int d^4x \, \langle [\varphi_\mu, \varphi_\nu], [\varphi_\mu, \varphi_\nu] \rangle.$$

$$M^4 \longrightarrow S^3 \times S^1 \qquad \text{(compactification)}$$



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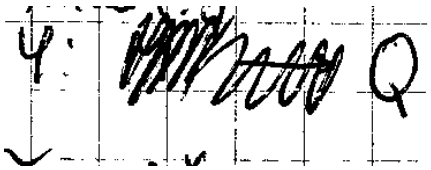
Uncertain reading. The crossed-out mark resembles φ?

$$M^4 = \partial Q, \qquad Q = S^3 \times D^2.$$

$$M^4 \longrightarrow SU(3)$$

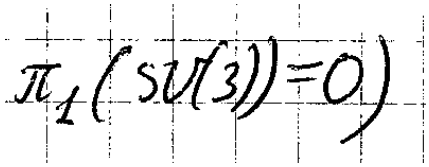
Extend:

$$\varphi \colon Q \longrightarrow SU(3) \qquad \text{(for } \pi_1(SU(3)) = 0\text{)}.$$



Original source details: GELFAND-P001135

Uncertain reading. [Between $\varphi\colon$ and Q crossed-out writing.]



Original source details: GELFAND-P001135

Diagram and typed description. $\pi_1(SU(3)) = 0$.

$$\widetilde{\omega} = \varphi^* \omega$$

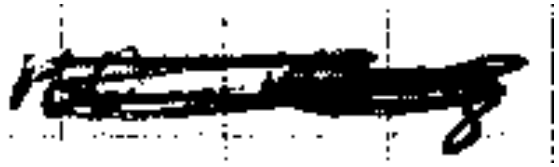
ω — generator of homology in the corresponding dimension.

$(1 + t^3)(1 + t^5)$ — Poincaré polynomial.

$$S_1 = N \int_Q \varphi^* \omega.$$

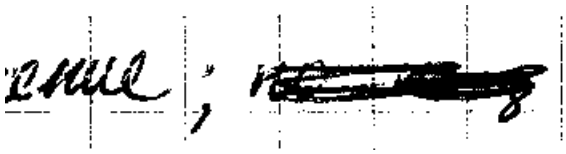
GELFAND-P001136

Wess and Zumino wrote down a complicated expression; [crossed out]



Uncertain reading. A passage is crossed out after the semicolon. A lower loop is visible at the end.

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Undeciphered passage. The letters under the strokes are illegible.

Original source details: GELFAND-P001136

in particular, integrality was previously unknown.

S_1 — a special case of S. P. Novikov's multivalued functionals.



Diagram and typed description. — · —

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GELFAND-P001137

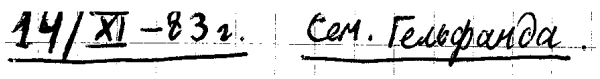
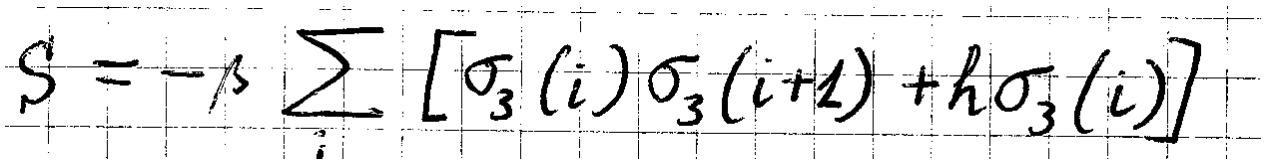


Diagram and typed description. 14 November 1983. Gelfand Seminar.

Original source details: GELFAND-P001137

I.M.G. Ising model Phys. Rev. D17, 1978

$$S = -\beta \sum_i [\sigma_3(i)\sigma_3(i+1) + h\sigma_3(i)]$$



Original source details: GELFAND-P001137

Diagram and typed description. First expression for S.

(with periodic conditions [crossed out] — on $\mathbb{Z}/N\mathbb{Z}$).

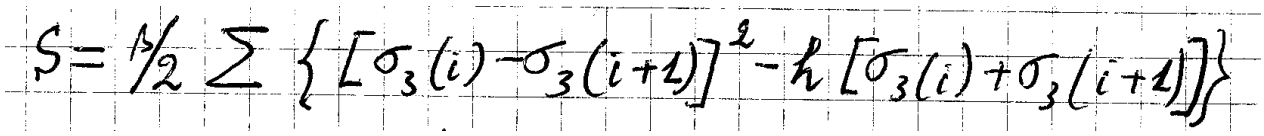


Uncertain reading. Crossed-out mark, possibly a bracket.

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$$\sigma_3(i) = \pm 1$$

$$S = \beta/2 \sum \{[\sigma_3(i) - \sigma_3(i+1)]^2 - h[\sigma_3(i) + \sigma_3(i+1)]\}$$



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Diagram and typed description. Squared-difference expression for S.

$$\mathcal{L}(i, i+1) = \frac{\beta}{2} \{[\sigma_3(i) - \sigma_3(i+1)]^2 - h[\sigma_3(i) + \sigma_3(i+1)]\}$$

$$S = \sum_i \mathcal{L}(i, i+1)$$

$$Z = \sum_\sigma e^{-S[\sigma]} = \sum_\sigma \prod_i e^{-\mathcal{L}(i,i+1)}.$$



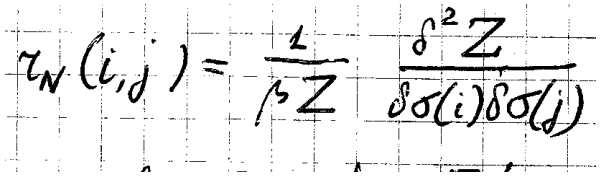
Original source details: GELFAND-P001137

Diagram and typed description. [In the left margin] Chulaevsky.

Correlation fn.:

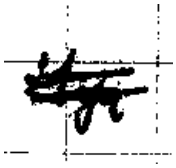
$$\tau_N(i,j) \stackrel{\text{def}}{=} E[\sigma(i)\sigma(j)]$$

$$\tau_N(i,j) = \frac{1}{\beta Z} \frac{\delta^2 Z}{\delta\sigma(i)\delta\sigma(j)}.$$



Original source details: GELFAND-P001137

Diagram and typed description. $\frac{1}{\beta Z} \frac{\delta^2 Z}{\delta\sigma(i)\delta\sigma(j)}$.



Original source details: GELFAND-P001137

Uncertain reading. [In the left margin] The crossed-out writing is illegible.

Probability space $\Omega_N = \{\sigma : \mathbb{Z}/N\mathbb{Z} \longrightarrow \pm 1\}$.

$$P_N\{\sigma\} = \frac{1}{Z_N} e^{-S[\sigma]}.$$

GELFAND-P001138

Zelevinsky.

$p(n) =$ the number of partitions of n into a sum of positive integers.

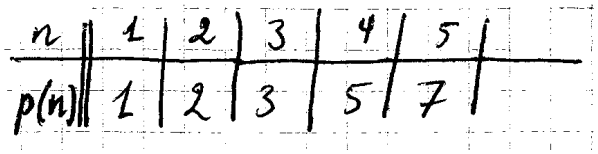


Diagram and typed description.

n	1	2	3	4	5	
$p(n)$	1	2	3	5	7	

Original source details: GELFAND-P001138

$$\sum p(n)t^n = \frac{1}{(1-t)(1-t^2)\dots}$$

$$\begin{aligned} p(n) &= (\text{number of conjugacy classes of elements in } S_n) \\ &= (\text{number of irreducible representations of } S_n) \end{aligned}$$

$$p(5n+4) \equiv 0 \pmod 5$$

$$p(7n+5) \equiv 0 \pmod 7$$

Dyson's problem (unsolved), connected with the first congruence.

$$\text{Invariant} \stackrel{\text{def}}{=} (\text{largest part of the partition}) - (\text{number of parts})$$

$$8 = 6+1+1 \rightsquigarrow 3\ (= 6-3)$$



Diagram and typed description. Retraced digit 3.

Original source details: GELFAND-P001138

$$8 = \underbrace{1+\dots+1} \rightsquigarrow -7$$

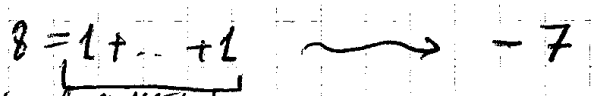


Diagram and typed description. Straight bracket below $1+\dots+1$.

Original source details: GELFAND-P001138

The remainder [inserted above (of the invariant)] on division by 5 gives a split into equal parts.

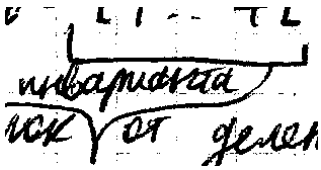
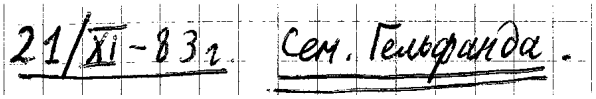


Diagram and typed description. [inserted above] (of the invariant).

Original source details: GELFAND-P001138

Proved, but by a noncombinatorial argument.



Original source details: GELFAND-P001139

Diagram and typed description. 21/XI-83 yr. Gelfand Sem..

A. M. Vershik. (on work with Kerov).



Original source details: GELFAND-P001139

Uncertain reading. [An illegible entry is struck out on the left.]

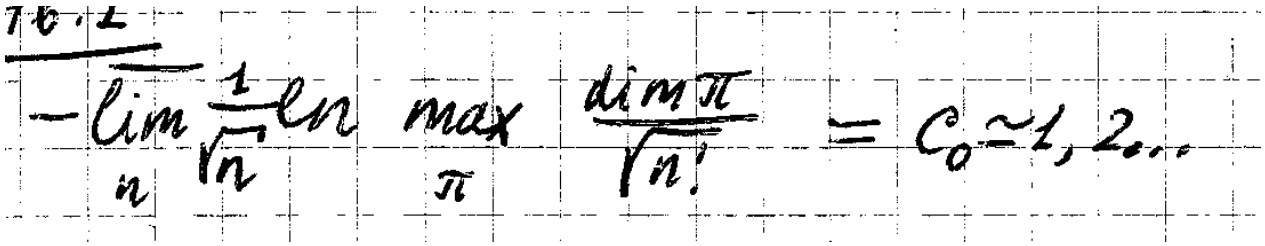
S_n is the symm. group.

Consider its representations (irred.) of max. dimension.

Question: asymptotics

$$\max_{\pi\in\widehat{S}_n} \dim \pi \qquad \text{as } n \longrightarrow \infty. \tag{1}$$

Prop. 1



Original source details: GELFAND-P001139

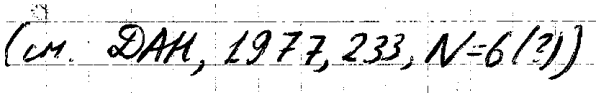
Diagram and typed description.

$$-\overline{\lim_n} \frac{1}{\sqrt{n}} \ln \max_\pi \frac{\dim \pi}{\sqrt{n!}} = c_0 \simeq 1.2\ldots \tag{2}$$

or

$$\max \dim \pi \simeq \sqrt{n!}\, e^{-1.2\sqrt{n}(1+o(1))} \tag{2'}$$

Sol. obtained by analysing Plancherel measure

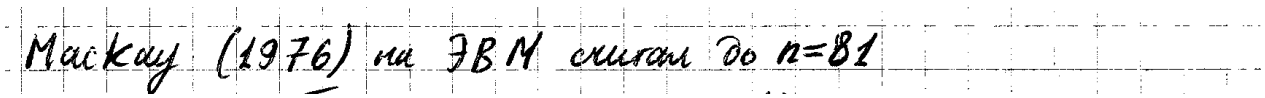


Original source details: GELFAND-P001139

Diagram and typed description. (see DAN, 1977, 233, $N = 6(?)$).

There, the asympt. of a typical Young diagram.

Idea: there is a limit of the Young diagrams (and it can be found!).

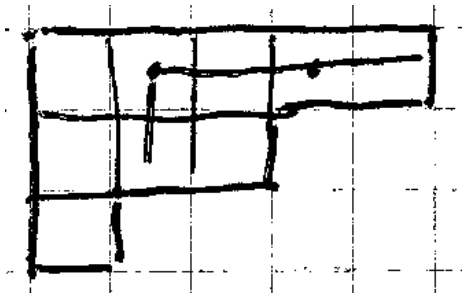


Original source details: GELFAND-P001139

Diagram and typed description. MacKay (1976) computed up to $n = 81$.

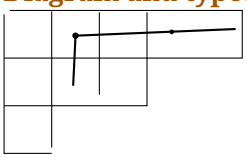
His conjecture was that there is a representation with $\frac{\dim \pi}{\sqrt{n!}} > \frac{1}{n}$.

It is false.



Original source details: GELFAND-P001139

Diagram and typed description.



Two heavier marks on the horizontal stroke. It is unclear whether they indicate points.

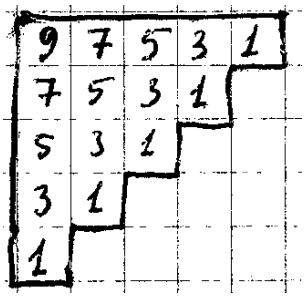
Dimension of the representation from the Young diagram.

Construct a hook at $\forall$ points.

$$\dim = \frac{n!}{\prod(\text{hook lengths})}.$$

$$\sum \left(\frac{n!}{\prod \dots}\right)^2 = n! \qquad \text{(consequence of a theorem of representation theory)}$$

Hook lengths:

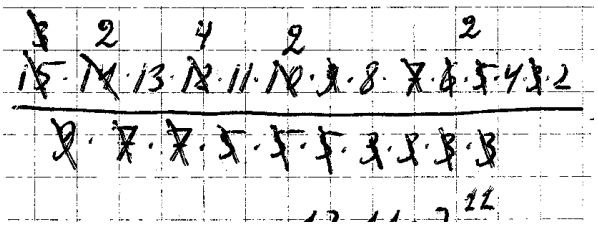


Original source details: GELFAND-P001139

Diagram and typed description.

9	7	5	3	1
7	5	3	1	
5	3	1		
3	1			
1				

$$\begin{aligned}\dim &= \frac{15!}{9 \cdot 7^2 \cdot 5^3 \cdot 3^4} \\ &= \frac{\overset{3}{\cancel{15}} \cdot \overset{2}{\cancel{14}} \cdot 13 \cdot \overset{4}{\cancel{12}} \cdot 11 \cdot \overset{2}{\cancel{10}} \cdot \cancel{9} \cdot 8 \cdot \cancel{7} \cdot \overset{2}{\cancel{6}} \cdot \cancel{5} \cdot 4 \cdot \cancel{3} \cdot 2}{\cancel{9} \cdot \cancel{7} \cdot \cancel{7} \cdot \cancel{5} \cdot \cancel{5} \cdot \cancel{5} \cdot \cancel{3} \cdot \cancel{3} \cdot \cancel{3} \cdot \cancel{3}} \\ &= \underline{13 \cdot 11 \cdot 2^{11}}.\end{aligned}$$

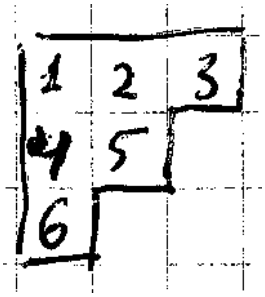


Original source details: GELFAND-P001139

Uncertain reading. The crossed-out digit above 15 reads as 3. The last superscript 2 sits to the right of 6.

or dim = the number of Young tableaux of the given shape; the tableau increases along rows and columns.

The numbers $1, 2, \ldots, k$, where k is the number of cells in the tableau.



Original source details: GELFAND-P001139

Diagram and typed description.

1	2	3
4	5	
6		

16 ways to fill it.

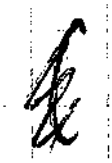
GELFAND-P001140

Fact: diagrams of representations of [~~max~~?] dimension



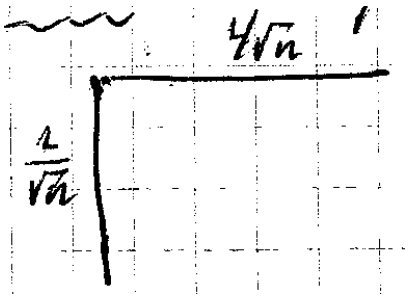
Original source details: GELFAND-P001140

Uncertain reading. The crossed-out entry tentatively reads “max”.



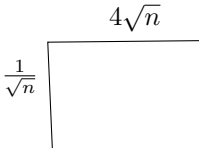
Original source details: GELFAND-P001140

Uncertain reading. [Left margin] The crossed-out entry resembles $\sqrt{n}$.

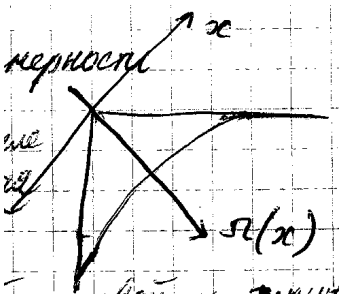


Original source details: GELFAND-P001140

Diagram and typed description.

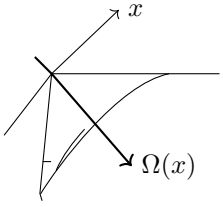


normalize: in the limit, a curve.



Original source details: GELFAND-P001140

Diagram and typed description.



A thin parallel stroke appears near the bottom of the arc. Its purpose is unclear.

The max representation tends to this curve too, as do typical ones.



Original source details: GELFAND-P001140

Diagram and typed description. The beginning of “tends” is retraced. The strokes underneath do not yield a separate reading.

$$\Omega(x) = \begin{cases} \frac{2}{\pi}\left(x \arcsin x + \sqrt{1-x^2}\right), & |x| \le 1, \\ |x|, & |x| > 1. \end{cases}$$

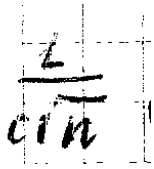
Histogram: according to Plancherel measure: $\frac{(\dim \Lambda)^2}{n!}$ — probability of a representation.

Almost all representations in this measure tend to this curve. Typical representations have a large fluctuation.

Prop. 2. Max. representations are so close to it $\Rightarrow$ computer calculation.

There are $p(n)$ conjugacy classes of elements. Their asymptotics

$$p(n) \simeq \frac{1}{c\sqrt{n}}\, e^{\frac{2\pi}{\sqrt{6}}\sqrt{n}}. \tag{3}$$



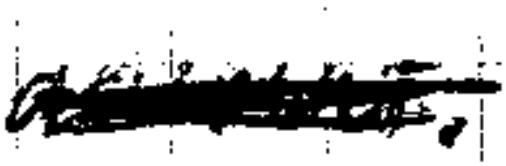
Original source details: GELFAND-P001140

Diagram and typed description. The manuscript gives the prefactor $\frac{1}{c\sqrt{n}}$.

This follows from

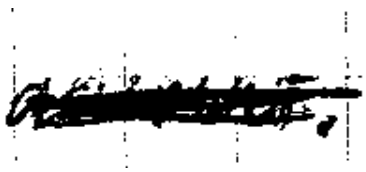
$$\sum_n p(n)x^n = \frac{1}{\prod_k (1-x^k)}.$$

([~~asympt.~~?], we estimate the integral)



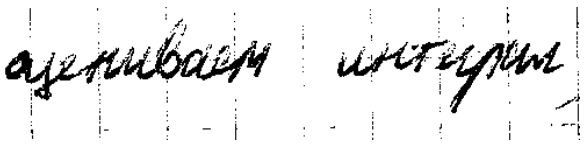
Original source details: GELFAND-P001140

Uncertain reading. Crossed-out writing, tentatively “asympt.”



Original source details: GELFAND-P001140

Undeciphered passage. Strokes obscure the writing after the initial “a”.

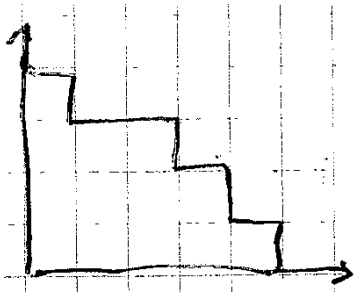


Original source details: GELFAND-P001140

Diagram and typed description. we estimate the integral

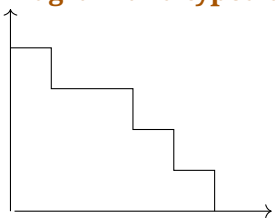
Rademacher (book on analyt. number theory) — asympt. of $p(n)$ (full) — $\simeq 40$ pp..

$p(n)$ — the number of curves with a given area.



Original source details: GELFAND-P001140

Diagram and typed description.



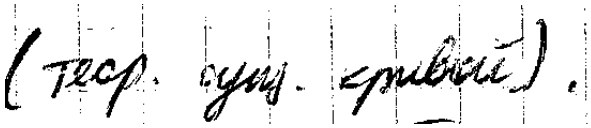
Consider tableaux with n cells and let each have probability $\frac{1}{p(n)}$ (all equally likely).

Typical diagram? (again shrink by a factor of $\sqrt{n}$)

Prop. 3. The resulting curve (in the same axes).

$$e^{-\frac{\pi}{\sqrt{6}}x} + e^{-\frac{\pi}{\sqrt{6}}y} = 1. \tag{4}$$

Idea: law of large numbers (theory of a random curve).



Original source details: GELFAND-P001140

Diagram and typed description. theory of a random curve

The sections of curve (4) are independent (in a weak sense).

GELFAND-P001141

[source number: 21]

Haar asympt.:

$$n = n_1 + \cdots + n_k.$$

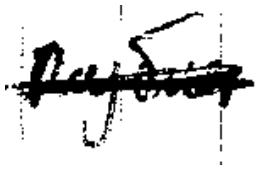
Probability proportional to the number of elements in the class. (a different statistic).

There is no limit here!

Bell statistics.

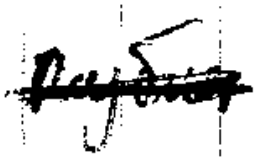
n individual objects split into subgroups

[~~Partit~~...] Number of partitions.



Original source details: GELFAND-P001141

Uncertain reading. The crossed-out entry begins “Partit...” [«Разби...»].



Original source details: GELFAND-P001141

Undeciphered passage. The stroke obscures the final letters.

Characters.

S_∞ — finitely supported permutations of the natural numbers.

$U(\infty)$

E. Thoma (FRG) — described the representations of S_∞. 1964; Vershik et al. 1981, FA, 15:4.



Diagram and typed description. FA, 15:4.

Original source details: GELFAND-P001141

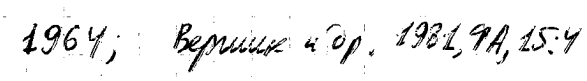


Diagram and typed description. 1964; Vershik et al. 1981, FA, 15:4.

Original source details: GELFAND-P001141

$U(\infty)$ Voiculescu 1976 and Vershik and Co. 1982 (DAN).

Earlier, in the 50s, in other terms (M. G. Krein, Gantmacher, Schoenberg, Karlin — totally positive functions)

Analytical problem — description of totally positive kernels

(Edrei [written «Эдрейн»] — by methods of the theory of entire functions)



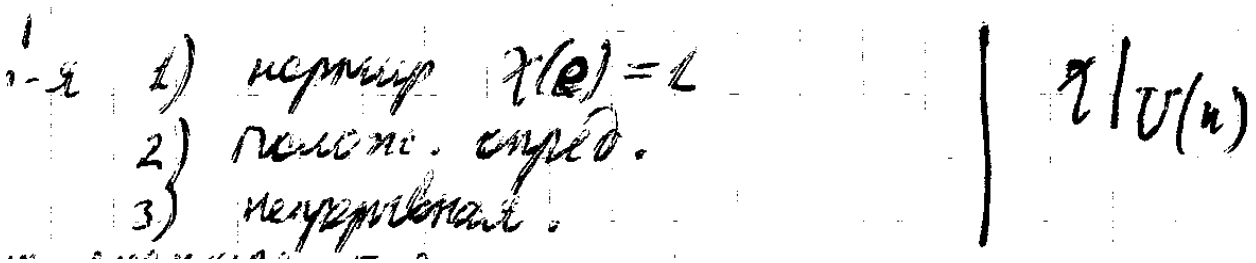
Diagram and typed description. Edrei [written «Эдрейн»]

Original source details: GELFAND-P001141

$$\begin{matrix} U(\infty) \\ \downarrow \\ g \end{matrix} \qquad \chi(g) \text{ — indecomposable character.}$$

Function:

1. normalized $\chi(e) = 1$
2. positive definite
3. continuous.



Original source details: GELFAND-P001141

Diagram and typed description. [At right, beyond the vertical stroke:] $\left| \chi|_{U(n)} \right.$.

$\Rightarrow \chi$ — a function of the spectrum, i.e.

$$\chi(g) = \prod_{\lambda_i \in \operatorname{Spec} g} \varphi(\lambda_i), \qquad \varphi(1) = 1.$$

Condition on φ: total positivity in the sense of Krein et al..

$$\varphi(\lambda) = \sum_{n \in \mathbb{Z}} c_n \lambda^n.$$

Answer: parameters $\gamma^{\pm}, \alpha_n^{\pm}, \beta_n^{\pm}$.

$$\varphi(z) = e^{\gamma^+(z-1)+\gamma^-(\bar{z}-1)} \prod_k \frac{(1-\beta_k^+ + \beta_k^+ z)}{(1+\alpha_k^+) - \alpha_k^+ z} \cdot \prod_k \frac{(1-\beta_k^- + \beta_k^- \bar{z})}{(1+\alpha_k^-) - \alpha_k^- \bar{z}}.$$

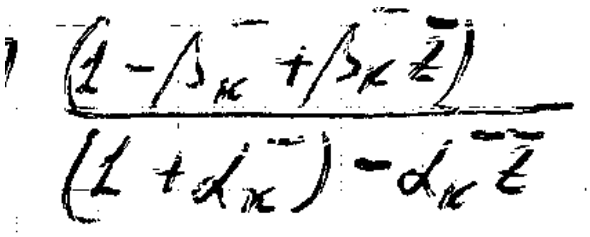


Diagram and typed description. The index in both terms of the denominator of the negative product reads k.

Original source details: GELFAND-P001141

$$\alpha_1^{\pm} \geq \cdots \geq \alpha_n^{\pm} \geq \cdots \geq 0, \qquad \gamma^{\pm} \geq 0,$$
$$\beta_1^{\pm} \geq \cdots \geq \beta_n^{\pm} \geq \cdots \geq 0,$$
$$\sum \alpha_k^+ + \alpha_k^- + \beta_k^+ + \beta_k^- < \infty,$$
$$\beta_1^+ + \beta_1^- \leq 1.$$

GELFAND-P001142

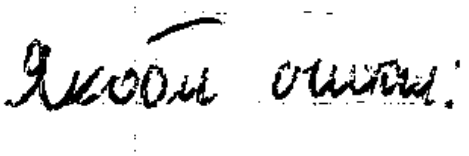
S_∞ — the difference is small, but the functions are analytic (without negative powers).



Uncertain reading. There is a retraced stroke beneath “without”. No distinct entry can be read beneath it.

Original source details: GELFAND-P001142

α_j, β_j — limiting parameters of Young diagrams.
Jacobi [interpretation?]:



Uncertain reading. The word after “Jacobi” tentatively reads “interpretation” [«смысл»]. The final strokes also permit «симв.» (“symbol”).

Original source details: GELFAND-P001142

$$\frac{\begin{vmatrix} x_1^{\lambda_1} & x_2^{\lambda_1} & \cdots & x_n^{\lambda_n} \\ x_1^{\lambda_2+1} & x_2^{\lambda_2+1} & \cdots & x_n^{\lambda_2+1} \\ \cdot & \cdot & \cdot & \cdot \end{vmatrix}}{\begin{vmatrix} 1 & \cdots & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \\ x_1^2 & x_2^2 & \cdots & x_n^2 \\ \cdot & \cdot & & \\ x_1^{n-1} & \cdots & \cdots & x_n^{n-1} \end{vmatrix}}, \qquad \lambda_1 \le \lambda_2 \le \cdots \le \lambda_n. \tag{5}$$

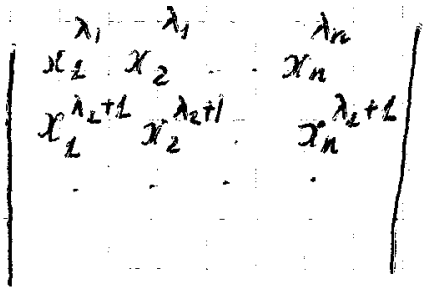


Diagram and typed description. The upper-right entry of the numerator is written $x_n^{\lambda_n}$.

Original source details: GELFAND-P001142

Suppose there exists $\lim_{n\to\infty} \frac{\lambda_k^{(n)}}{n} = \alpha_k \Rightarrow$ there exists a limit of det (5) and it is the generating function of the characters.

Prop. Let $|x_i| = 1$ and from some point on they are $= 1$.

Miwa, Jimbo: for each diagram, a differential operator.

K-functor:

Proc. Sem. LOMI, [crossed out] vol. 123



Uncertain reading. A mark before “vol. 123” is crossed out. It may be a number sign.

Original source details: GELFAND-P001142

— · —

$K^0(S_\infty)$ is constructed from projective modules over $\mathbb{C}(S_\infty)$
(finitely generated modules are direct summands).

$K^0(S_n)$ — additive group of characters.

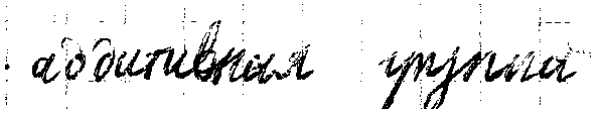
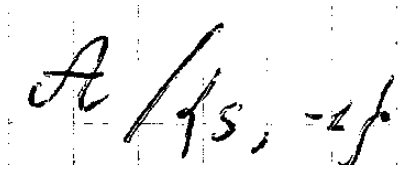


Diagram and typed description. additive group.

Original source details: GELFAND-P001142

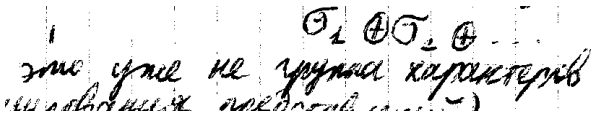
Thm.:

$$\varinjlim_{n\to\infty} K^0(S_n) = K^0(S_\infty) = A/\{s_1 - 1\}.$$



Uncertain reading. The subscript on s tentatively reads 1. The shape also allows i.

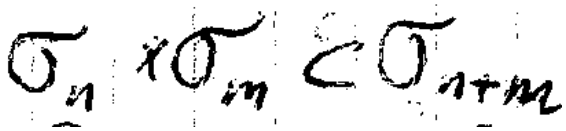
Original source details: GELFAND-P001142



Original source details: GELFAND-P001142

Diagram and typed description. [Beneath A:] $\sigma_1 \oplus \sigma_2 \oplus \cdots$.
[Beneath $K^0(S_\infty)$, marked by a vertical stroke:] this is no longer a group of characters.

(embedding by means of induced representations).



Original source details: GELFAND-P001142

Diagram and typed description. [At right:]

$$\sigma_n \times \sigma_m \subset \sigma_{n+m}.$$

A description of the positive elements? It is given in terms of differential operators.
Positivity — positive combinations of characters.

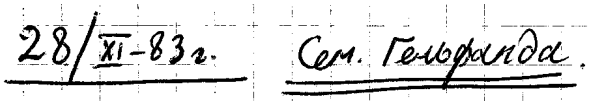


Original source details: GELFAND-P001142

Uncertain reading. The beginning of the abbreviation “positive” is retraced. No distinct entry can be read under the strokes.

The cone of positive elements has a nontrivial structure.

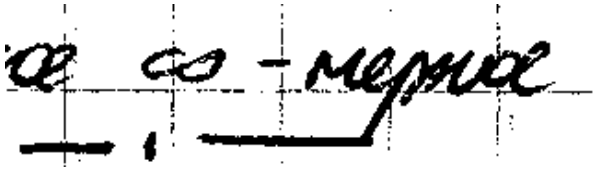
GELFAND-P001143



Original source details: GELFAND-P001143

Diagram and typed description. 28/XI-83 yr. Gelfand Sem..

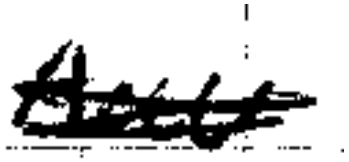
I. M. G. It is unclear what an ∞-dimensional space is.
Repeats Vershik



Original source details: GELFAND-P001143

Diagram and typed description. Strokes under “∞-dimensional”. The right stroke turns up toward the word.

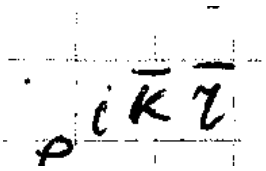
A. Migdal. [~~If~~?] The case of a deviation from the CLT (Gaussian limit theorem) is of interest



Original source details: GELFAND-P001143

Uncertain reading. The crossed-out word may read “If” [«Если»].

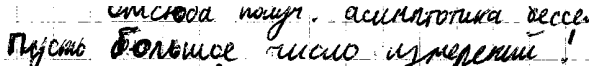
$$\int d\vec{\tau}\,\delta(1-\vec{\tau}^2)\,e^{i\bar{k}\bar{\tau}} \sim J_n(k) \tag{1}$$



Original source details: GELFAND-P001143

Diagram and typed description. The exponent has separate bars over k and τ: $i\bar{k}\bar{\tau}$.

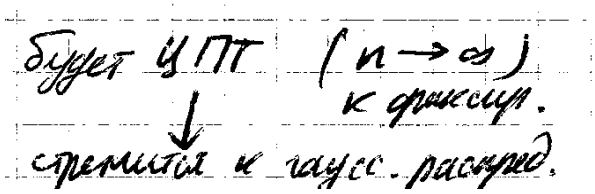
From this we get the asymptotics of the Bessel functions



Original source details: GELFAND-P001143

Diagram and typed description. Let there be a greater number of dimensions!

If k is held fixed, there will be



Original source details: GELFAND-P001143

Diagram and typed description.

CLT $\quad (n \to \infty)$

$\downarrow \quad k$ fixed

tends to a Gaussian distribution

Proof: Replace δ by a Fourier $\int$; integrate over τ.

$$\int d\alpha \int d\vec{\tau}\, e^{\alpha(1-\tau^2)+ik\tau}$$



Original source details: GELFAND-P001143

Uncertain reading. An illegible formula fragment after the first $\int$ and before $d\alpha$ is crossed out.

$\alpha \in i\mathbb{R}$ $\quad \| \quad$ evaluate the integral over $\tau \Rightarrow$ Gaussian distribution.

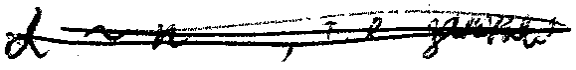
GELFAND-P001144

We obtain

$$\int \frac{d\alpha}{2\pi i}\,\alpha^{-n/2} e^{\alpha - \frac{k^2}{4\alpha}}.$$

Take $k \sim n$!! replace α by $n\alpha$ in the integral.

[~~$\alpha \sim n$~~, illeg.]



Original source details: GELFAND-P001144

Uncertain reading. The line is crossed out. The opening reads $\alpha \sim n$; the following signs and word remain unclear.

$$\int \frac{d\alpha}{2\pi i}\,\alpha^{-\frac{n}{2}} e^{n\alpha-\frac{k^2}{4\alpha n}}.$$

α fluctuates weakly $\Rightarrow$ replace it by a const for small k

n is large, $k \sim n \Rightarrow$ obtain some kind of arcsin.

An analogous problem arises for the
Fourier integral over the unitary group

$$\int du\ \exp\bigl(\beta\,\mathrm{tr}(\bar{n}u + u^{+}n)\bigr), \qquad u^{+}u = I \tag{2}$$

($n, \bar{n}$ — constant matrices)

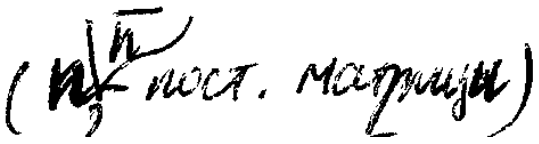


Diagram and typed description. [$\bar{n}$ is inserted above the line with an arrow to the position after n.]

Original source details: GELFAND-P001144

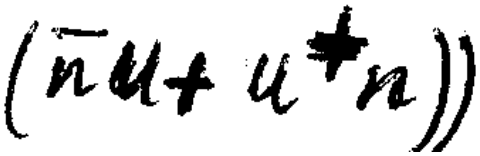


Diagram and typed description. The letter for the matrix reads n.

Original source details: GELFAND-P001144

Nontrivial asymptotics of large groups (there are no asymptotics).

Again.

$$\int d\vec{\tau}\,\delta(1-\vec{\tau}^{\,2})e^{i\vec{k}\vec{\tau}} \qquad 2\pi i\,\delta(x) = \int_{-i\infty}^{i\infty} e^{\alpha x}\,d\alpha$$

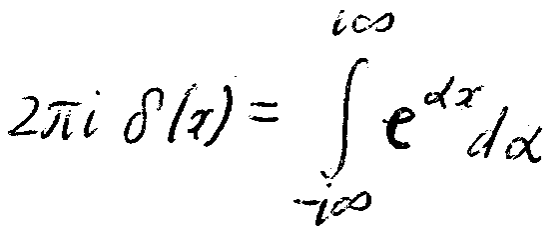


Diagram and typed description. $2\pi i\,\delta(x) = \int_{-i\infty}^{i\infty} e^{\alpha x}\,d\alpha$.

Original source details: GELFAND-P001144

||



Uncertain reading. The crossed-out entry before $\frac{1}{2\pi i}$ may read $\int d\alpha$.

Original source details: GELFAND-P001144

$$\frac{1}{2\pi i}\, n \int d\alpha \int d\vec{\tau}\, e^{n\alpha(1-\vec{\tau}^{\,2})+i\vec{k}\vec{\tau}} =$$
$$= \int d\alpha \, \exp\left(-\frac{n}{2}\ln\alpha + n\alpha - \frac{k^2}{4\pi\alpha}\right).$$

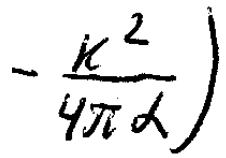


Diagram and typed description. The last denominator is written $4\pi\alpha$.

Original source details: GELFAND-P001144

$k \sim n \Rightarrow$ apply the saddle-point method

Entropy: n-fold integral with an exponential containing n, equals $e^{n(\ldots)}$

$\Rightarrow$ ln of the number of states $=$ ln of the integral $\sim n$.

Try this for (2) — replace it with $\quad u \in U(N)$

$$\int \underbrace{du\, du^+\, \delta\overbrace{\left(I - u^+u\right)}^{\text{matrix}}}_{\substack{\text{inv. measure} \Rightarrow \\ \text{measure of Haar}}} \exp\left(N\beta\,\mathrm{tr}(\bar{n}u + u^+n)\right).$$



Uncertain reading. An illegible mark is crossed out after the first $\int$ and before du.

Original source details: GELFAND-P001144



Uncertain reading. A crossed-out entry between du^+ and δ may read $d\alpha$.

Original source details: GELFAND-P001144

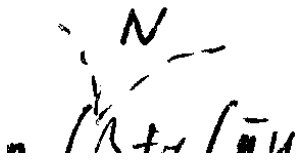


Diagram and typed description. N is inserted above β. Two dashed arrows point to the position before β.

Original source details: GELFAND-P001144

$$\int du\, du^+\, d\alpha\, \exp\left(N\,\mathrm{tr}(\alpha - u^+u\alpha) + N\beta\,\mathrm{tr}(\bar{n}u + u^+n)\right).$$

GELFAND-P001145

[source number: 28]

$\int d\alpha$ cannot be evaluated by the saddle-point method.

Entropy associated with α, is of the same order:

N^2 terms and N^2 integrals $\Rightarrow$ each integral ~ 1 and there are no asymptotics

(for the 1-dimensional integral)

(Rule of stat. phys.:

Multiple integral $\sim e^k$, where k — the number of dimensions)

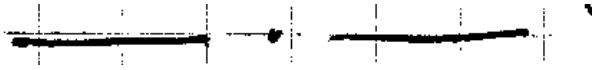


Diagram and typed description.

Original source details: GELFAND-P001145

Ambula.

Consider the problem

$$\int dA \, \exp\bigl(+N\beta \, \mathrm{Sp}(V(A))\bigr) \, \mathrm{Sp}(e^{At}) \tag{3}$$

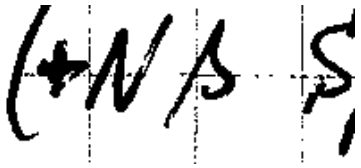


Diagram and typed description. Sign in (3): $+N\beta$.

Original source details: GELFAND-P001145

A — [inserted above: skew-]Hermitian matrices (alg. of the group $SU(N)$) $\quad A^+ = -A$.

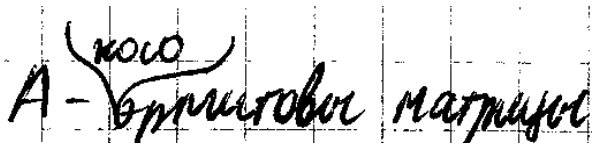
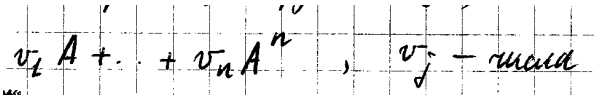


Diagram and typed description. “skew-” above the line, with an insertion mark before “Hermitian”.

Original source details: GELFAND-P001145

$$\underset{\text{— polynomial}}{V(A)} = v_1 A + \ldots + v_n A^n, \qquad v_j \text{ — numbers}$$



Uncertain reading. The coefficient symbol is read as v; ν is also possible.

Original source details: GELFAND-P001145

$$\Psi(t) = \frac{(3)}{\int dA \, \exp\bigl(-N\beta \, \mathrm{Sp}(V(A))\bigr)} \tag{4}$$

(generating function for moments)

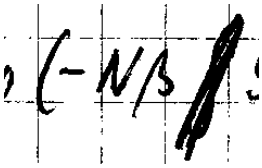


Diagram and typed description. Sign in the denominator of (4): $-N\beta$.

Original source details: GELFAND-P001145

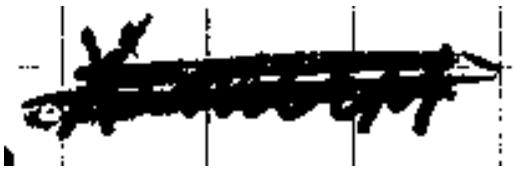


Original source details: GELFAND-P001145

Uncertain reading. An illegible mark is crossed out between $N\beta$ and Sp.

A more complicated object: the Wilson average

$$\operatorname{Sp} T \exp\left(\int_C A_\alpha \, dx_\alpha\right). \qquad (T \text{ — matrix})$$



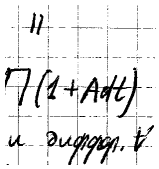
Original source details: GELFAND-P001145

Uncertain reading. An illegible word is crossed out. A branching initial stroke is visible.

Consider.

$$0 = \int dA \, \frac{\partial}{\partial A_{ij}} \left[e^{At} \exp\bigl(N\beta \operatorname{Sp} V(A)\bigr) \right] \tag{5}$$

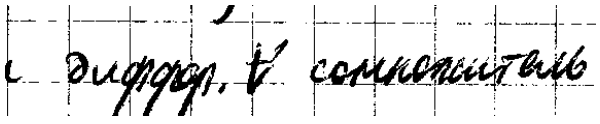
$$\sum_{i,j} \frac{\partial}{\partial A_{ij}} \underset{\parallel}{(e^{At})_{ij}} = \int_0^t d\tau \, \operatorname{Sp}(e^{A\tau}) \operatorname{Sp}(e^{A(t-\tau)})$$
$$\prod(1+A\,dt)$$



Original source details: GELFAND-P001145

Diagram and typed description. Beneath $(e^{At})_{ij} \parallel \prod(1 + A\,dt)$.

and differentiate the t' factor
(what comes before collects into exp;
what comes after also collects into exp).



Original source details: GELFAND-P001145

Uncertain reading. There is a slanted stroke above t, tentatively read as t'.

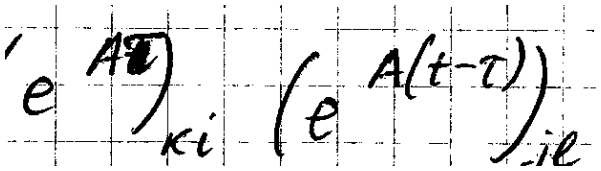
More generally:



Original source details: GELFAND-P001145

Uncertain reading. A summation sign before the derivative in (6) is crossed out, possibly $\sum_{i,j}$.

$$\frac{\partial}{\partial A_{ij}}(e^{At})_{k\ell} = \int_0^t d\tau\,(e^{A\tau})_{ki}\,(e^{A(t-\tau)})_{j\ell} \tag{6}$$



Uncertain reading. The last mark in the first exponent on the right, $A\tau$, is overwritten. The earlier mark is illegible.

Original source details: GELFAND-P001145

$$\frac{\partial}{\partial A_{ij}} \operatorname{Sp} V(A) = (V'(A))_{ji} \tag{7}$$

(V — polynomial)

Using (6), (7), differentiate in (5).

GELFAND-P001146

One calculates

$$\frac{\int dA\ \exp(N\beta \operatorname{Sp} V(A))\operatorname{Sp}(e^{At})/N}{\int dA\ \exp(N\beta \operatorname{Sp} V(A))} \overset{\text{def}}{=} \left\langle \frac{\operatorname{Sp} e^{At}}{N} \right\rangle \tag{8}$$



Diagram and typed description. Above the equals sign is written $\overset{\text{def}}{=}$.

Original source details: GELFAND-P001146

From (5)–(7) we find

$$0 = \left\langle \int_0^t d\tau\ \operatorname{Sp} e^{A\tau} \operatorname{Sp} e^{A(t-\tau)} + N\beta \operatorname{Sp}(V'(A)e^{At}) \right\rangle \tag{9}$$

$N\beta$ [crossed out ~~Sp~~?] $V'\left(\frac{\partial}{\partial t}\right)\operatorname{Sp}(e^{At})$



Uncertain reading. A crossed-out sign after $N\beta$ tentatively reads Sp.

Original source details: GELFAND-P001146

$$\operatorname{Sp} \mapsto \frac{\operatorname{Sp}}{N} \qquad \text{(mean of the eigenvalues)}$$

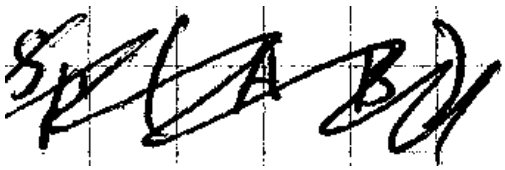


Diagram and typed description. [crossed out] ~~Sp(AB)~~.

Original source details: GELFAND-P001146

$$\langle \mathrm{Sp}\, A \cdot \mathrm{Sp}\, B \rangle \sim \langle \mathrm{Sp}(A \cdot B) \rangle$$

(statist. fact.)

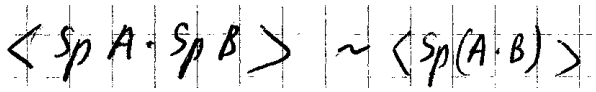


Diagram and typed description. The right-hand side is written $\langle \mathrm{Sp}(A \cdot B) \rangle$.

Original source details: GELFAND-P001146

We obtain

$$0 = \beta V' \left(\frac{\partial}{\partial t} \right) \Psi(t) + \int_0^t d\tau\, \Psi(\tau) \Psi(t - \tau) \tag{10}$$

Solved by the Fourier (Laplace) transform



Uncertain reading. The abbreviation before “Fourier” tentatively reads «пробр.» (“transform”).

Original source details: GELFAND-P001146

$$\beta V'(\omega) \Psi(\omega) + \Psi^2(\omega) = K(\omega) \tag{11}$$

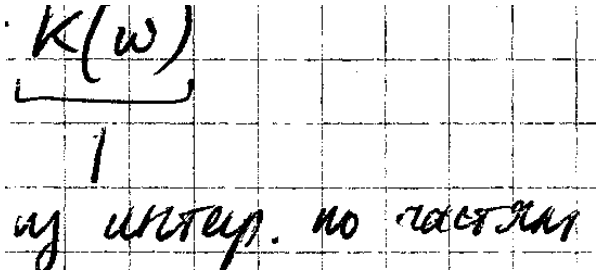


Diagram and typed description. [Under the bracket from $K(\omega)$:] from integration by parts.

Original source details: GELFAND-P001146

Hence

$$\Psi = -\frac{\beta}{2} V' + \sqrt{\frac{\beta^2}{4} V'^2 - K(\omega)} \tag{12}$$

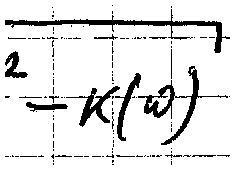


Diagram and typed description. Under the radical is written $-K(\omega)$.

Original source details: GELFAND-P001146

$K(\omega)$ is chosen so that the radical can almost be evaluated.

For the $\int$ over the group, analogously. The equation looks like this:

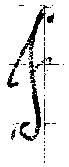
$$\int du\,[F(u) - F(u(1+\delta))] = 0 \qquad \text{(right-invariance condition for Haar measure)} \tag{13}$$

$$\int du \left(\frac{\partial F}{\partial u_{ij}} u_{jk}\right) = 0$$

and so on.

This can be done for arbitrary N and so on. $\Rightarrow$ hence an algorithm to find the series in powers of $1/N$.

Field theory — analogous equations, but functional.



Uncertain reading. [At the bottom] An isolated sign resembling $\int$.

Original source details: GELFAND-P001146

GELFAND-P001147

[source number: 29]

$$\frac{\int e^{N\beta \operatorname{Sp} V(A)} \frac{1}{N} \operatorname{Sp} e^{At}\, dA}{\int e^{N\beta \operatorname{Sp} V(A)}\, dA} = \Psi_N(t) = \Psi(t) + \frac{1}{N^2}\Psi_1(t) + \cdots \tag{14}$$

(asympt. expansion in $1/N$)

Consider also

$$\left\langle \frac{\operatorname{Sp} e^{At}}{N} \right\rangle = \frac{\int dA\, \exp\left(N\beta \operatorname{Sp} V(A) + \int J(t)\, dt\right) \operatorname{Sp} e^{At}/N}{\int dA\, \exp\left(N\beta \operatorname{Sp} V(A) + \int dt\, J(t) \operatorname{Sp}(e^{At})\right)} \tag{15}$$

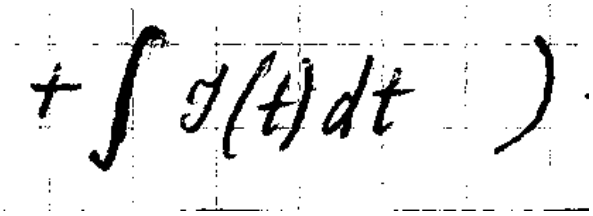
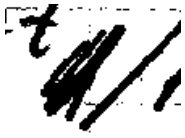


Diagram and typed description. The numerator exponent contains $\int J(t)\, dt$.

Original source details: GELFAND-P001147



Uncertain reading. An illegible mark is crossed out after $\operatorname{Sp} e^{At}$ and before $/N$.

Original source details: GELFAND-P001147

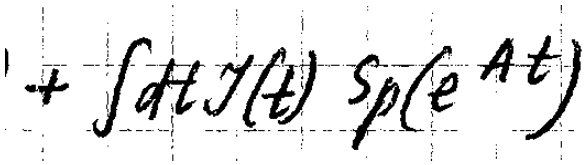


Diagram and typed description. The denominator exponent contains $\int dt\, J(t) \operatorname{Sp}(e^{At})$.

Original source details: GELFAND-P001147

The resulting equation: (something like the Fourier equation)

$$\beta V'\left(\frac{\partial}{\partial t}\right)\Psi(t,J) = \int_0^t d\tau\,\Psi(\tau)\Psi(t-\tau) + \\ + \frac{1}{N}\left\{\int_0^t d\tau\,\frac{\delta\Psi(t-\tau)}{\delta J(\tau)} + \int_0^\infty d\tau\,J(\tau)\Psi(t+\tau)\right\} \tag{16}$$

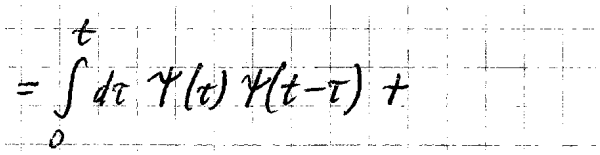


Diagram and typed description. In (16) the integral immediately follows the equals sign. The + at the end of the first line is repeated at the beginning of the next.

Original source details: GELFAND-P001147

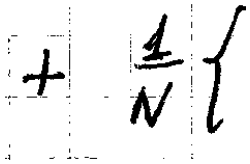


Diagram and typed description. Before the brace is written $\frac{1}{N}$.

Original source details: GELFAND-P001147

Field theory.

$$V(A) = F_{\mu\nu}^2, \qquad V'(A) \equiv \nabla_\mu F_{\mu\nu}$$

$$V'\left(\frac{\partial}{\partial t}\right) \mapsto \int_0^1 ds_1 \int_{s_1-0}^{s_1+0} ds_2\,\frac{\delta^2}{\delta x_\mu(s_1)\delta x_\nu(s_2)}.$$

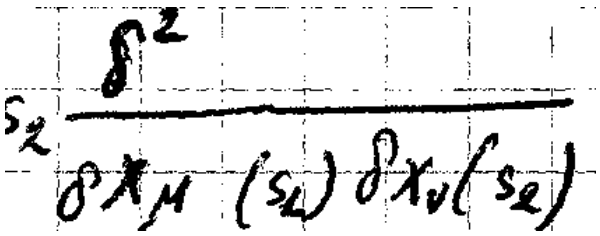


Diagram and typed description. The denominator reads $x_\mu(s_1)$, $x_\nu(s_2)$.

Original source details: GELFAND-P001147

Wiegmann P.

Exact sol. of the principal chiral field in 2-dimensional space

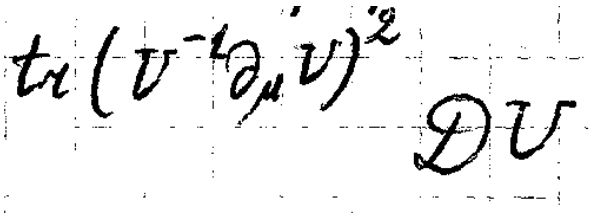
(jointly with Polyakov, following the programme he proposed 10 years ago)

$$\int e^{-\frac{1}{T}\int d^2x\ \mathrm{tr}(U^{-1}\partial_\mu U)^2}\,\mathcal{D}U \tag{1}$$



Diagram and typed description. The coefficient of the integral reads minus $\frac{1}{T}$.

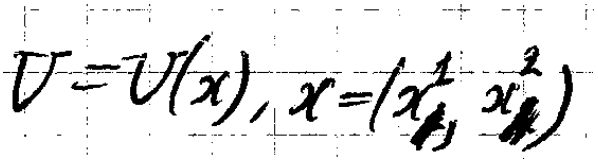
Original source details: GELFAND-P001147



Uncertain reading. The same letter in the action, the measure and the field definition is read as U. Its angular shape also allows V.

Original source details: GELFAND-P001147

$U = U(x), \quad x = (x^1, x^2)$



Uncertain reading. Illegible subscripts are crossed out below x^1, x^2.

Original source details: GELFAND-P001147

(from $n = 2$ to $n = 4$ there are ideas by A. B. Zamolodchikov, so far without success)

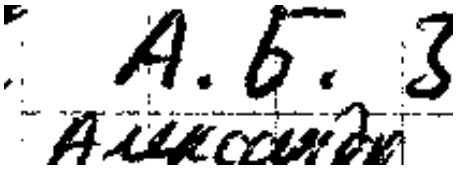
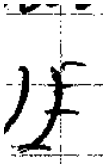


Diagram and typed description. [Under «A. B.»:] Alexander.

Original source details: GELFAND-P001147

What does it describe? π and K-mesons (if in $\mathbb{R}^4$)



Uncertain reading. After the parenthesis is a mark resembling a crossed-out $\int$.

Original source details: GELFAND-P001147

$(1+1)$ QFT or equivalent 2-dimensional stat. problems

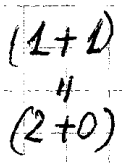


Diagram and typed description. [Under $(1+1)$:] $\|$ and below: $(2+0)$.

Original source details: GELFAND-P001147

Many 2-dimensional quantum models have been solved recently. This is one of them.
Glossary of the field theory terms needed

1. Renormalization group
2. Asymptotic freedom
3. Quantum anomaly of the axial current.
4. Bethe ansatz (Bethe) $\cong$ quantum inverse scattering method
5. $\mathrm{Det}(i\partial_\mu\gamma^\mu + A_\mu^{(x)}\gamma^\mu) = W\{A\}, \qquad A_\mu = \sum_a A_\mu^a X^a$

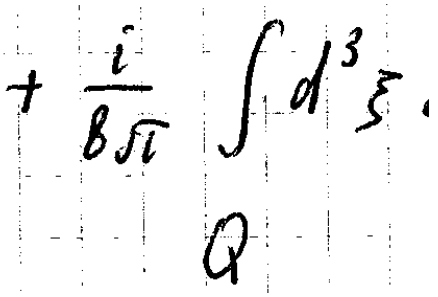
GELFAND-P001148

$$A_+ = a^{-1}\partial_+ a, \qquad A_- = b^{-1}\partial_- b.$$

$$g = ab^{-1}$$

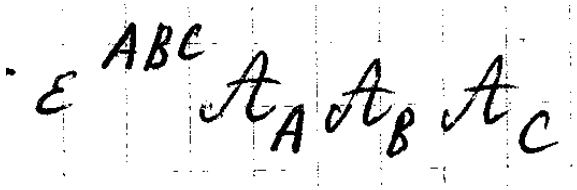
$$\mathcal{A}_\mu = g^{-1}\partial_\mu g$$

$$W\{A\} = \frac{1}{2}\int \operatorname{tr}\mathcal{A}_\mu^2\, d^2x + \frac{i}{8\pi}\int_Q d^3\xi\, \varepsilon^{ABC}\mathcal{A}_A\mathcal{A}_B\mathcal{A}_C$$



Original source details: GELFAND-P001148

Diagram and typed description. Coefficient of the cubic term: $\frac{i}{8\pi}$.

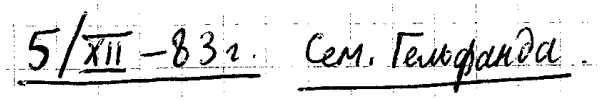


Original source details: GELFAND-P001148

Diagram and typed description. $\varepsilon^{ABC}\mathcal{A}_A\mathcal{A}_B\mathcal{A}_C$.

$$\partial Q = S^2,$$

GELFAND-P001149



Original source details: GELFAND-P001149

Diagram and typed description. 5/XII-83 yr. Gelfand Sem..

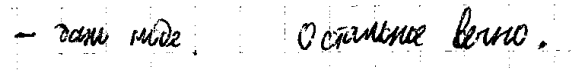


Original source details: GELFAND-P001149

Diagram and typed description. P. Wiegmann.

What does one need to know?

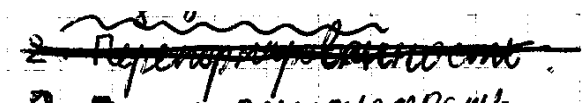
1. Bethe ansatz. — basic idea. The rest [вно.?].



Original source details: GELFAND-P001149

Uncertain reading. After «осн.» the manuscript reads «иде.» (“idea”). The abbreviation after “The rest” is uncertain and may read «вно.».

~~2. Renormalizability.~~



Original source details: GELFAND-P001149

Diagram and typed description. ~~2. Renormalizability.~~

2. Renormalizability.

3. Critical phenomena.

4. Asymptotic freedom.

5. Gauge theories: axial anomaly,

[crossed out] Instantons and topology, θ-vacuum.



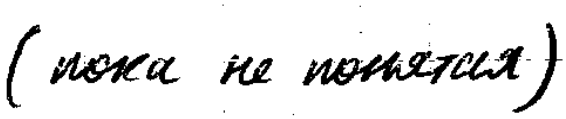
Original source details: GELFAND-P001149

Uncertain reading. An illegible mark to the left of “Instantons” is crossed out, possibly an item number.

6. String theory, which does not yet exist, the Wilson loop, minimal surfaces, statistics of random surfaces.

7. Virasoro algebra.

8. Solid-state physics: there is Anderson localization (as yet not understood).



Original source details: GELFAND-P001149

Diagram and typed description. (as yet not understood).

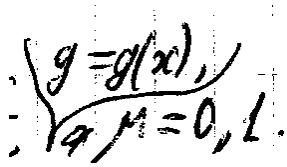


Original source details: GELFAND-P001149

Diagram and typed description. [Under «localization»:] (phenomenon).

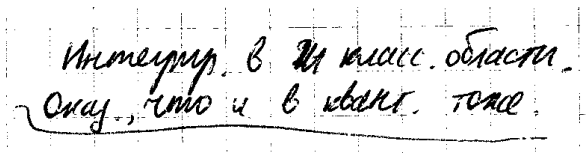
The principal chiral field in two dimensions.

$$S = \frac{1}{2\lambda}\int d^2x\ \mathrm{tr}(g^{-1}\partial_\mu g)^2, \qquad g \in G, \quad \begin{matrix} g = g(x), \\ \mu = 0, 1. \end{matrix} \tag{1}$$



Original source details: GELFAND-P001149

Diagram and typed description. [An insertion to the right of $g \in G$] $g = g(x); \quad \mu = 0, 1.$



Original source details: GELFAND-P001149

Uncertain reading. [At right] Integr. in the [crossed out] classical domain. It turns out this also holds in the quantum domain.

Quantum integrability: the possibility of calculating the integral

$$Z = \int e^{-S}\,\mathcal{D}g. \tag{2}$$

or diagonalizing the corresponding Hamiltonian

There will be standard calculations for this $\longrightarrow$ Bethe ansatz.

One needs to know an eigenstate (possibly other than the ground state). Such a state is unknown for the bosonic field. One must take the next step:

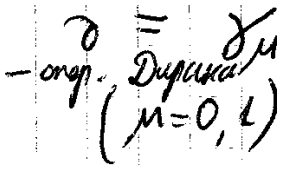
(3) Prop. The theory is equivalent to a fermionic theory with Lagrangian

$$\mathcal{L} = \bar{\psi} i \hat{\partial} \psi + \lambda j^a_\mu j^\mu_a, \qquad S = \int \mathcal{L}\, d^2x \tag{4}$$

$$\psi = \begin{pmatrix} \psi_R \\ \psi_L \end{pmatrix} \quad \text{(Dirac spinor)},$$

transforms according to $O(2)$ (Lorentz group in the 2-dimensional case)

$$\hat{\partial} = \gamma_\mu \partial^\mu, \qquad \{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu}$$



Original source details: GELFAND-P001149

Diagram and typed description. [Under $\hat{\partial}$:] — Dirac operator $(\mu = 0, 1)$.

γ_0, γ_1

GELFAND-P001150

[source number: 36]

Isotopic indices

$$\alpha = 1, \dots, N \quad \text{— colour.}$$
$$(g \in SU(N)) \qquad f \quad \text{— scent = aroma (flavour)}$$
$$f = 1, \dots, N_f$$

$$j^a_\mu = \sum_f \bar{\psi}^\alpha_f \gamma_\mu X^a_{\alpha\beta} \psi^\beta_f, \qquad X^a \text{ — generators of the group } SU(N) \tag{5}$$

The interaction is constructed from the currents.

(6) Prop. As $N_f \longrightarrow \infty$ these theories are equivalent.

(theory (4) goes over into theory (1))

The colour groups must coincide.

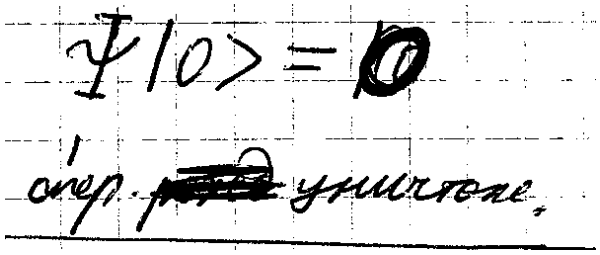
There is left and right invariance.

$$g \longrightarrow gG \quad \text{and so on; group } G \times G$$

For theory (4): $\underbrace{G \times F}$

Right invariance is absent and is restored at ∞ flavours.

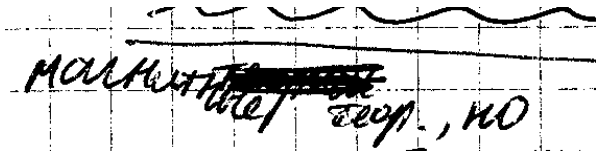
For (4) there is an eigenvector — the vacuum $|0\rangle$: $\Psi|0\rangle = 0$



Uncertain reading. [At the stroke under Ψ:] operator [~~creation~~?] annihil..

Original source details: GELFAND-P001150

(1) describes 2-dimensional magnetic [the ending is inserted below] [inserted below] theor., but such problems resemble 4-dimensional gauge theories (there are arguments, but only verbal ones so far).



Uncertain reading. The end of the original word is crossed out. The ending «ные» and then «теор.» (“theor.”) are written below.

Original source details: GELFAND-P001150

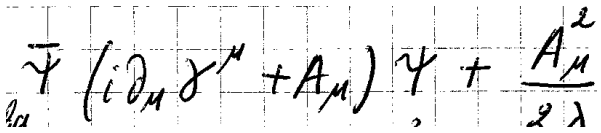
Proof of equivalence.



Uncertain reading. An illegible sign is crossed out at left under the heading.

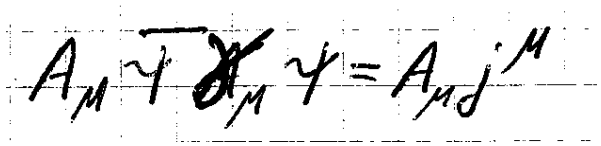
Original source details: GELFAND-P001150

$$(4) \Longrightarrow \bar{\psi}(i\partial_\mu \gamma^\mu + A_\mu)\psi + \frac{A_\mu^2}{2\lambda} \tag{7}$$



Original source details: GELFAND-P001150

Diagram and typed description. In (7) no γ^μ is written beside A_μ.

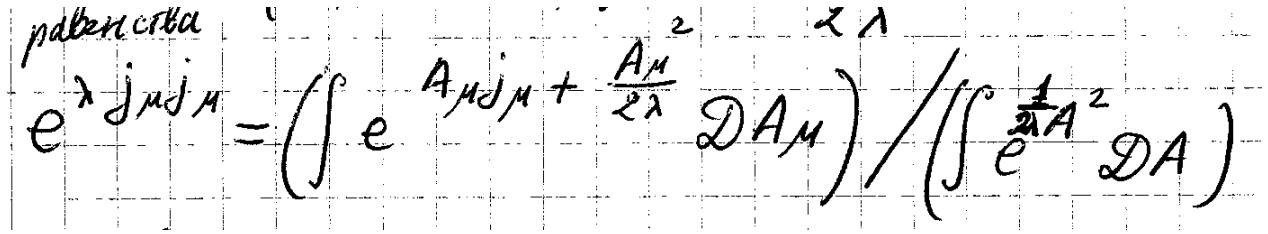


Original source details: GELFAND-P001150

Diagram and typed description. [At right] $A_\mu \bar{\psi}\gamma_\mu\psi = A_\mu j^\mu$. The γ is retraced.

from the equality

$$e^{\lambda j_\mu j_\mu} = \left(\int e^{A_\mu j_\mu + \frac{A_\mu^2}{2\lambda}} \mathcal{D}A_\mu\right) \Big/ \left(\int e^{\frac{1}{2\lambda}A^2} \mathcal{D}A\right) \tag{8}$$

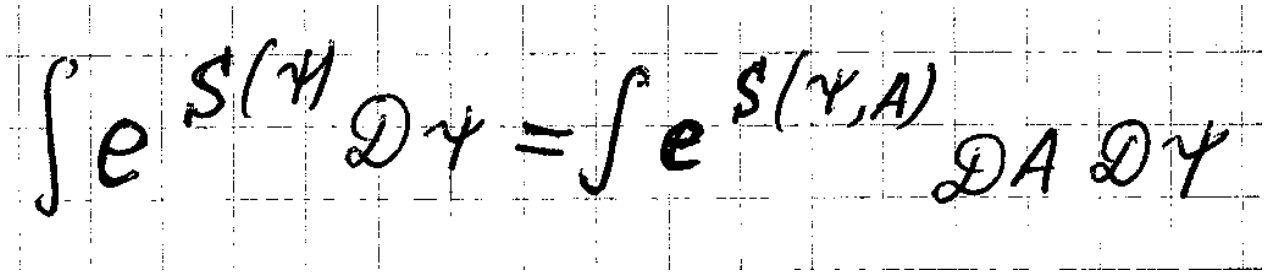


Original source details: GELFAND-P001150

Diagram and typed description. Both quadratic terms in (8) are written with positive signs.

(Gaussian integral with an intermediate field)

$$\int e^{S(\psi)}\,\mathcal{D}\psi = \int e^{S(\psi,A)}\,\mathcal{D}A\,\mathcal{D}\psi \tag{9}$$



Original source details: GELFAND-P001150

Diagram and typed description. The exponents in (9) are written $S(\psi)$, $S(\psi, A)$.

The index f is omitted for simplicity.

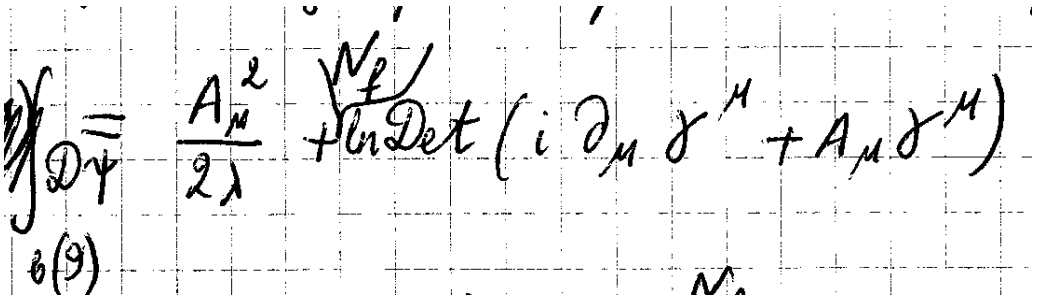
(Sometimes called the Stratonovich–Hubbard formula)



Original source details: GELFAND-P001150

Uncertain reading. An illegible sign is crossed out before the integral.

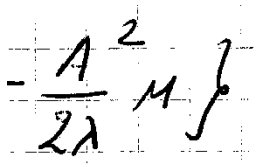
$$\int_{\mathcal{D}\psi} = \frac{A_\mu^2}{2\lambda} + \overset{N_f}{\ln}\,\mathrm{Det}(i\partial_\mu\gamma^\mu + A_\mu\gamma^\mu)$$



Original source details: GELFAND-P001150

Uncertain reading. Beneath the integral is written “in (9)”. N_f is inserted above ln Det. The left-hand side is in shorthand.

$$S(\psi, A) = \int d^2x \left\{ \sum_{f=1}^{N_f} \bar{\psi}_f (i\partial_\mu\gamma^\mu + \gamma^\mu A_\mu)\psi_f + \frac{A^2}{2\lambda}\,\mu \right\} \tag{10}$$



Uncertain reading. The μ is placed after the fraction and before the brace. It may be a displaced index on A^2.

Original source details: GELFAND-P001150

GELFAND-P001151

This gives

$$S\{A\} = \int \left[\frac{A_\mu^2}{2\lambda} + N_f \ln \mathrm{Det}(i\partial_\mu\gamma^\mu + A_\mu\gamma^\mu) \right] d^2x \tag{11}$$

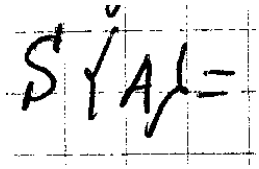


Diagram and typed description. $S\{A\}$.

Original source details: GELFAND-P001151

One needs to calculate



Uncertain reading. After “calculate” is a looped mark, possibly a crossed-out $\int$.

Original source details: GELFAND-P001151

$$\int e^{-S(A)}\,\mathcal{D}A_\mu \tag{12}$$



Diagram and typed description. [At right:] A_μ is already a bosonic field.

Original source details: GELFAND-P001151

and then $N_f \longrightarrow \infty$.

$\Longrightarrow$ in the limit, the trajectories that contribute are those where min ln Det

This gives a constraint on the fields A_μ: namely, min on the fields

$$A_\mu = g^{-1}\partial_\mu g \tag{13}$$

(pure gauge field); these are exactly the ones on which $F_{\mu\nu} = 0$

$$\lim_{N_f\to\infty} \int e^{-S(A)}\,\mathcal{D}A_\mu = \int e^{-S(A_\mu)}\delta(F_{\mu\nu})\,\mathcal{D}A_\mu \tag{14}$$

$$F_{\mu\nu} = \partial_\nu A_\mu - \partial_\mu A_\nu + [A_\mu, A_\nu]$$

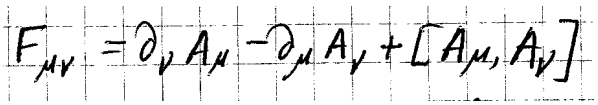
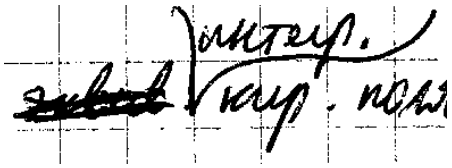


Diagram and typed description. The manuscript reads $\partial_\nu A_\mu - \partial_\mu A_\nu + [A_\mu, A_\nu]$.

Original source details: GELFAND-P001151

The problem with a finite number of flavours is integrable too. It is the one that is integrated, followed by passage to the limit.

Problem of classical integr. $\sim$ [~~equiv~~...] [inserted above: integr.] of the chiral field with the addition of a Wess–Zumino term.



Uncertain reading. The crossed-out word appears to begin “equiv...”. “Integr.” is inserted above the line.

Original source details: GELFAND-P001151

Gauge transform.:

$$A_\mu \longrightarrow U^{-1} A_\mu U + U^{-1} \partial_\mu U \tag{15}$$

$(13) \sim 0$ under this, or $F_{\mu\nu} = 0$ on (13).

det — gauge invariant; all eigenvalues of the Dirac operator are gauge inv.

$$(i\partial_\mu \gamma^\mu + A_\mu \gamma^\mu)\Psi = \lambda \Psi$$

The gauge transform. must be performed on A_μ and simultaneously on Ψ

(16) min — on fields gauge equiv. to 0.

Take the Abelian case for simplicity:

$$A_\mu = \partial_\mu \varphi + \epsilon_{\mu\nu} \partial_\nu \Phi \tag{17}$$

Transform. $A_\mu \longmapsto \partial_\mu \varphi + A_\mu$

In this case:

$$\ln \mathrm{Det}(i\partial_\mu \gamma^\mu + A_\mu \gamma^\mu) = \int d^2x \, (\partial_\mu \Phi)^2 \qquad \text{(Schwinger, 1961)} \tag{18}$$

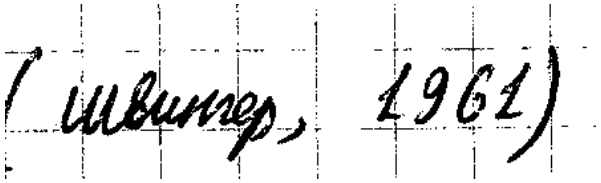


Diagram and typed description. (Schwinger, 1961).

Original source details: GELFAND-P001151

Hence the variation of det > 0, if there is Φ.

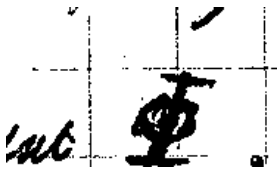


Diagram and typed description. The Φ is retraced. No separate reading can be established beneath the strokes.

Original source details: GELFAND-P001151

Moral: $A_\mu = \partial_\mu \varphi$ and so on.

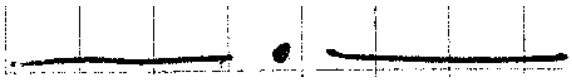


Diagram and typed description.

Original source details: GELFAND-P001151

GELFAND-P001152

[source number: 37]

Bethe ansatz method.

Simplify by dropping the flavours (leave 1), with colours $SU(2)$.

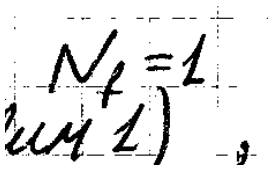
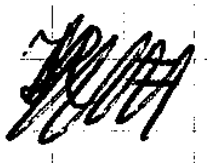


Diagram and typed description. [Above "leave 1"] $N_f = 1$.

Original source details: GELFAND-P001152

It will be a nonrelativistic problem. (for simplicity)

On diagonalizing the following fermionic Hamiltonian.

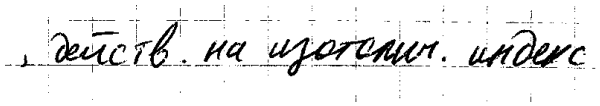


Original source details: GELFAND-P001152

Uncertain reading. A sign before (19) is crossed out, possibly $\mathcal{H}$.

$$\mathcal{L} = \bar{\psi}\, i\hat{\partial}\psi + \lambda\bar{\psi}\gamma_\mu\bar{\sigma}\psi\bar{\psi}\gamma_\mu\bar{\sigma}\psi \tag{19}$$

$\bar{\sigma}$ — the Pauli matrices (group $SU(2)$), acting on the isotopic index.

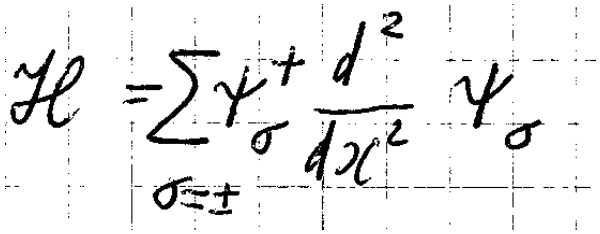


Original source details: GELFAND-P001152

Diagram and typed description. acting on the isotopic index.

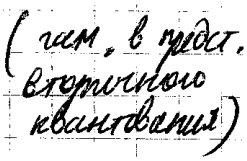
Hamiltonian density: (nonrelativ.):

$$\mathcal{H} = \sum_{\sigma=\pm} \psi_\sigma^+ \frac{d^2}{dx^2}\psi_\sigma + \lambda(\psi_\sigma^+\psi_\sigma)(\psi_{\sigma'}^+\psi_{\sigma'}) \tag{20}$$



Original source details: GELFAND-P001152

Diagram and typed description. The kinetic term in (20) is written with $\frac{d^2}{dx^2}$.



Original source details: GELFAND-P001152

Diagram and typed description. [At right of (20):] (Ham. in the second-quantized representation)

σ — spin.

The spectrum of $\mathcal{H}$ is already bounded below (unlike (19)) — this is in effect a regularization of (19) and the problems are equivalent up to this regularization.

$\psi_\sigma = \psi_\sigma(x)$.

Hamiltonian:

$$H = \int \mathcal{H}\, dx \qquad \psi_\sigma = \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix}. \tag{21}$$

ψ — a fermionic field, i.e.

$$\{\psi_\sigma^+(x), \psi_{\sigma'}(y)\} = \delta(x-y)\delta_{\sigma\sigma'} \tag{22}$$

Problem: find all eigenvalues and eigenfunctions of the Hamiltonian H.

$$H\Phi = \lambda\Phi. \tag{23}$$

$$N = \int \psi^+\psi\, dx \tag{24}$$

— number of particles. (commutes with H)

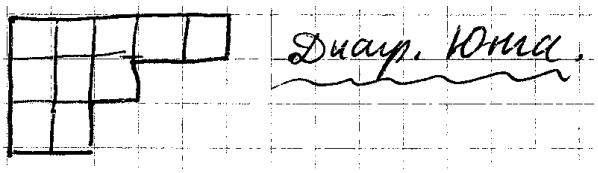
System of wave functions

$$\Phi(x_1, \dots, x_N, \sigma_1, \dots, \sigma_N)$$

$$S^z = \sum_\sigma \int \sigma\psi_\sigma^+\psi_\sigma\, dx \tag{25}$$

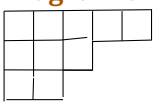
Consider a representation of the permutation group (of the pairs (x_j, σ_j)))

Fermions are antisymm..



Original source details: GELFAND-P001152

Diagram and typed description.

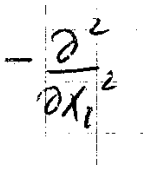


Young diagram.

$\Rightarrow$ on the n-particle subspace

$$\left[\sum_{i=1}^n -\frac{\partial^2}{\partial x_i^2} + \lambda \sum_{i<j} \delta(x_i - x_j)\right] \Phi = E\Phi \tag{26}$$

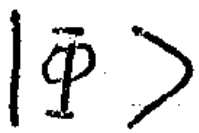
(in the first-quantized representation)



Original source details: GELFAND-P001152

Diagram and typed description. The kinetic term in (26) is written with $-\frac{\partial^2}{\partial x_i^2}$.

$$|\Phi\rangle = \int \Phi(x_1, \ldots, x_N) \prod_{i=1}^{N} \psi^+(x_i)|0\rangle \tag{27}$$



Uncertain reading. There is a stroke over the Φ in the ket, possibly part of the letterform.

Original source details: GELFAND-P001153

$|0\rangle$ — the vacuum, $\psi|0\rangle = 0$.

Φ — the same wave fn. as in (26).

Φ satisfies (26) $\Rightarrow |\Phi\rangle$ — a state vector.



Uncertain reading. Before “satisfies”, strokes overlap the Φ, one resembling a closing bracket.

Original source details: GELFAND-P001153

δ because the potential is of this form. In the general case:

$$\lambda\psi_\sigma^+(x)\psi_\sigma(x)V(x-y)\psi^+(y)\psi_\sigma(y)$$

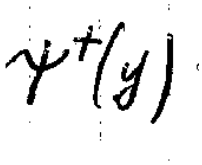
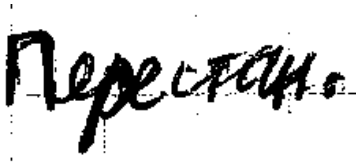


Diagram and typed description. The third field factor $\psi^+(y)$ has no subscript.

Original source details: GELFAND-P001153



Uncertain reading. Commutes

Original source details: GELFAND-P001153

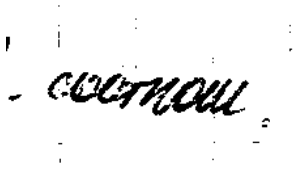
with the particle number ⇒ always a Schrödinger eq..

Integrability; Bethe ansatz — form of the wave fn..

1931 — Bethe wrote down the form of the wave fn. for our problem.

$$\Phi(x_1,\dots,x_N)=\sum_{\substack{\sigma_1,\dots,\sigma_N\\ P}} A_{\sigma_{q_1}\dots\sigma_{q_N}}(Q\cdot P)(-1)^P e^{i\sum_{j=1}^N x_j k_{P_j}} \tag{28}$$

k_{P_j} — arbitrary (characterize the state vector), $\sum k_j^2 = E$ — the only relation?



Uncertain reading. relation

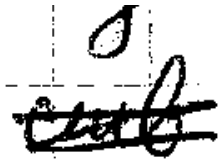
Original source details: GELFAND-P001153

Q, P — permutations of the numbers $\{1,\dots,N\}$, $(-1)^P$ — parity of the permutation, reflecting its fermionic nature.

(28) — in the region $X_Q = \{x_{q_1} < x_{q_2} < \dots < x_{q_N}\}$.

On passing [inserted above the line, circled and linked to this place: from one region X_Q to another] a collision occurs, the particles exchange places. A subtle point:

order the particles, there it is just a wave; we pass to another region, [crossed out: join?]



Uncertain reading. join

Original source details: GELFAND-P001153

joining the solns., will the momenta there be the same??

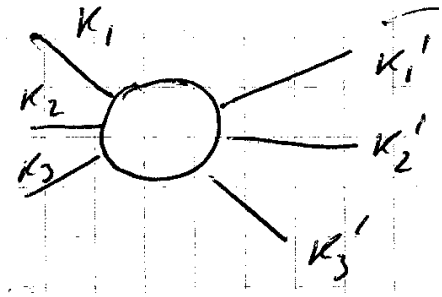
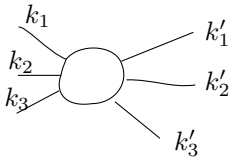


Diagram and typed description.



Original source details: GELFAND-P001153

[At the right of the diagram:] The laws of conservation of momentum and energy determine everything only for two.

Integrability: there are several more conservation laws.

Matching conditions — multiply by a matrix!

There is an analogous Bethe ansatz for any integrable system.

Nonintegrable case: $O(N)$ even with a δ-like potential.

GELFAND-P001154

12/XII-83 Gelfand Sem.

V. Ginzburg.

On a paper by Witten containing an attempt to unify gravitation and the electromagn. field.

I. $M = \mathbb{R}^4 \times S^1$, and S^1 — a small circle.



Uncertain reading. A crossed-out mark before the second S^1, possibly "and".

Original source details: GELFAND-P001154

$$R_{ij} - \frac{1}{2} g_{ij} R = 0 \qquad \text{(Einstein's eq.)} \tag{1}$$

(write it on M)

g_{ij} — a 5×5 matrix (block matrix).

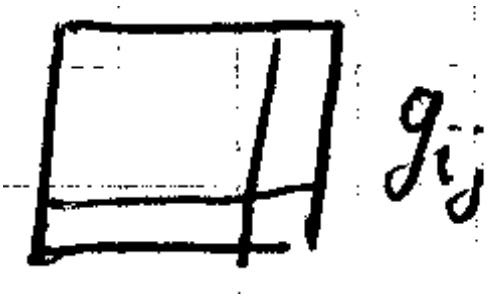


Diagram and typed description. 

Original source details: GELFAND-P001154

Expand g on S^1 in a Fourier series in $\varphi \in S^1$.

$$g = \sum g_k(x) e^{\frac{ik\varphi}{r}} \tag{2}$$

(r — radius of the small circle)

Grav. field is small $\rightsquigarrow$ Laplace's eq..

Write Laplace's eq.:

$$0 = \left(\Box_4 + \frac{\partial^2}{\partial \varphi^2} \right) g \tag{3}$$

from this and from (2):

$$\Box_4 g_k = \frac{k^2}{r^2} g_k \tag{4}$$

$\Rightarrow$ the kth term — a particle with mass $\frac{k^2}{r^2}$.

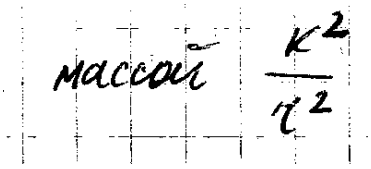


Diagram and typed description. The manuscript reads "mass" followed by $\frac{k^2}{r^2}$.

Original source details: GELFAND-P001154

$\underset{\sim}{k \neq 0} \Rightarrow$ with $r \sim$ radius of the electron one gets $m \sim 10^{37}$ [crossed out: illegible[?]]



Uncertain reading. illegible

Original source details: GELFAND-P001154

eV, which does not occur

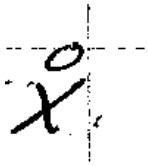


Diagram and typed description. After “does not occur” in the right margin, a slanted oval above two crossing strokes. The lower mark resembles x or χ.

Original source details: GELFAND-P001154

Therefore $g = g(x)$ (but this does not save it either).

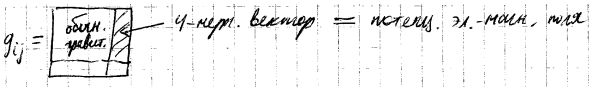


Diagram and typed description.

(5) $g_{ij} =$ ordinary grav. — 4-dimensional vector = potential of the electromagn. field

Original source details: GELFAND-P001154

(Kaluza–Klein theory)

The charge also comes out large (too large).

II part[?]



Uncertain reading. part

Original source details: GELFAND-P001154

Positivity of the energy of the gravitational field.

P. Wiegmann (continued).

Quantum integrable systems by example, or
On a one-dimensional Fermi gas with δ-like attraction.

$$\mathcal{H} = -\sum_{\sigma} \psi_{\sigma}^{+}(x) \frac{d^2}{dx^2} \psi_{\sigma} + c \left(\sum_{\sigma} \psi_{\sigma}^{+} \psi_{\sigma} \right) \left(\sum_{\sigma'} \psi_{\sigma'}^{+} \psi_{\sigma'} \right) \tag{1}$$

$\sigma = \pm 1$.

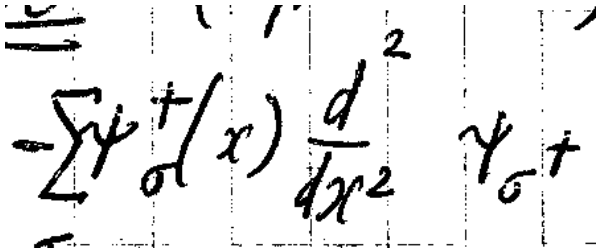
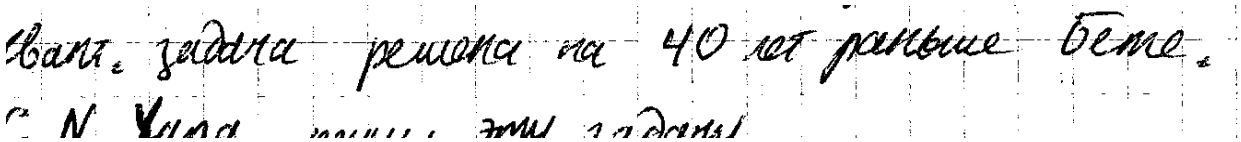


Diagram and typed description. Here the kinetic term has a minus before the sum.

Original source details: GELFAND-P001154

Classical analogue — NLS (nonlinear Schrödinger eq.), completely integr. (Zakh.–Shabat).
The quantum problem was solved 40 years before Bethe.



Original source details: GELFAND-P001154

Diagram and typed description. The quantum problem was solved 40 years before Bethe.
C. N. Yang solved this problem.

$$\psi_\sigma = \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix} = \begin{pmatrix} \psi_\uparrow \\ \psi_\downarrow \end{pmatrix}$$

Complex field — there is also ψ^+.

GELFAND-P001155

[source number: 42]

$$\begin{aligned} \{\psi^+(x), \psi(y)\} &= \delta(x-y) \qquad \text{(fermions)} \\ \{\psi^+_\sigma(x), \psi_{\sigma'}(y)\} &= \delta(x-y)\delta_{\sigma\sigma'} \end{aligned} \tag{2}$$



Original source details: GELFAND-P001155

Uncertain reading. In the spin version of (2), the argument of the first operator, possibly (x), is unclear under overlapping strokes.

ψ^+ — creation op., ψ — annihilation op.

contents.

1. Bethe substitution and conservation laws
2. Integrability conditions (triangle eqs.)
3. Diagonalization of the transfer matrix (cumbersome computations)
   — the simplest method — Faddeev et al.

We have to solve the Schrödinger eq.:

$$\mathcal{H}|\phi\rangle = E|\phi\rangle \tag{3}$$

$$N = \int dx \sum_\sigma \psi^+_\sigma(x)\psi_\sigma(x) \qquad \text{(particle-number operator)} \tag{4}$$

$$[N, \mathcal{H}] = 0, \quad [\mathcal{H}, S] = 0, \quad \text{where} \tag{5}$$



Original source details: GELFAND-P001155

Uncertain reading. The $\mathcal{H}$ in the second commutator is overwritten.

$$S \text{ — spin op.:} \qquad S = \int dx \sum_{\sigma} \sigma \psi_{\sigma}^{+}(x) \psi_{\sigma}(x) \tag{6}$$

[crossed out: illegible mark[?].]

Uncertain reading. illegible mark



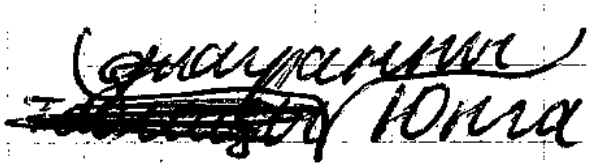
Original source details: GELFAND-P001155

(5_2) — spin conservation ($SU(2)$ symmetry)

We shall look for $|\phi\rangle$ with fixed particle number and spin.

We shall classify the repr. of the permutation group with the help of [crossed out: illegible word[?];

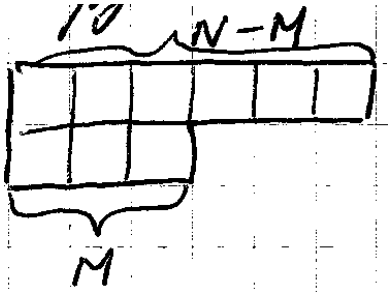
Uncertain reading. illegible word



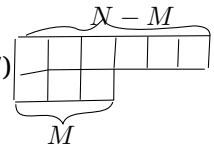
Original source details: GELFAND-P001155

inserted above the crossed-out word: “(diagram)”] of Young

Diagram and typed description.



(7)

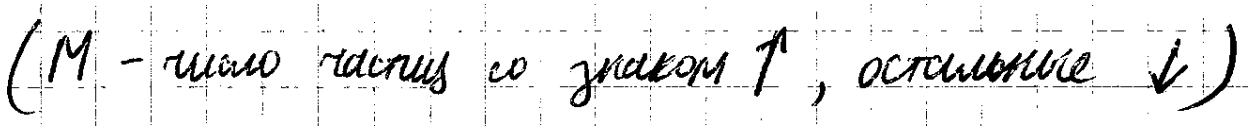


Original source details: GELFAND-P001155

Here the total spin

$$S = N - M - M = N - 2M \tag{8}$$

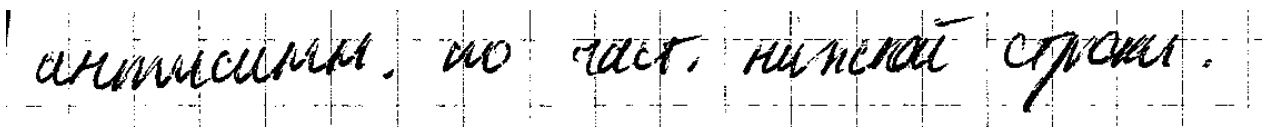
(M — number of particles with sign ↑, the others ↓)



Original source details: GELFAND-P001155

Diagram and typed description. The arrows in the line after (8) point up and down, respectively.

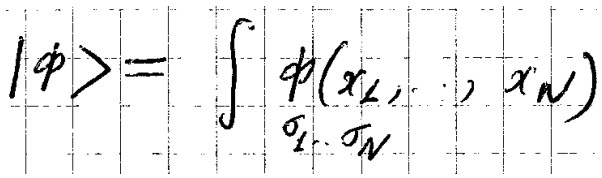
(9) ‖ Wave fn. symm. in the particles corresponding to the upper row and antisymm. in the particles of the lower row.



Original source details: GELFAND-P001155

Diagram and typed description. The manuscript reads “lower row”.

$$|\phi\rangle = \int \phi_{\sigma_1 \ldots \sigma_N}(x_1, \ldots, x_N) \psi^+_{\sigma_1}(x_1) \ldots \psi^+_{\sigma_N}(x_N) \, dx_1 \ldots dx_N \, |0\rangle \tag{10}$$



Original source details: GELFAND-P001155

Diagram and typed description. Equation (10) begins with an integral sign and no explicit spin-index sum.

$$|0\rangle \text{ — the vacuum;} \qquad \psi|0\rangle = 0 \tag{11}$$

(12) $\phi_{\sigma_1 \ldots \sigma_N}(x_1, \ldots, x_N)$ called the wave function.

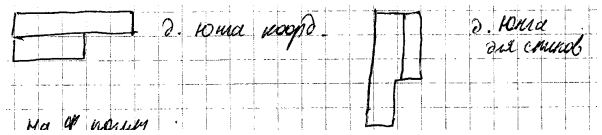
[In the right margin:] $\|$ antisymm. on swapping (x_i, σ_i) with (x_j, σ_j)

It must belong to repr. (7) of the symmetric group.

Expression (1) — second-quantized repr..

First quantization — an eq. for $\phi_{\sigma_1 \ldots \sigma_N}(x_1, \ldots, x_N)$

GELFAND-P001156



Original source details: GELFAND-P001156

Diagram and typed description.

(13) 

Young d., coordinates 

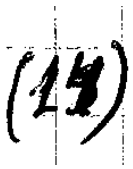
Young d. for spins

A short slanting stroke at the left of the first diagram resembles an arrowhead.

(width 2 because 2 spin values ± 1)

For ϕ we get:

$$\left\{ -\sum_{i=1}^{N} \frac{\partial^2}{\partial x_i^2} + c \sum_{i>j} \delta(x_i - x_j) \right\} \Phi = E\Phi \tag{14}$$



Original source details: GELFAND-P001156

Uncertain reading. The last digit of (14) is corrected and its previous form is illegible.

Explanation:

$\Phi_{\sigma_1 \dots \sigma_N}(x_1, \dots, x_N)$ — antisymm. under permutation of the pairs $(x_i, \sigma_i), (x_j, \sigma_j)$.

What happens if only the coordinates are permuted for a given set of σ_i?

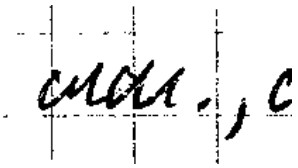
$$\phi = \sum f(x_1, \dots, x_N)\chi(\sigma_1, \dots, \sigma_N)$$

(sum of tensors; this can always be done; then we introduce only pairs f, χ with certain symmetries)

$$\Phi(x, \sigma) = \sum_{\tau_1, \tau_2} f_{\tau_1}(x) g_{\tau_2}(\sigma)$$

τ_1, τ_2 — repr. of the symmetric group.

Φ antisymm. $\Rightarrow$ only the terms[?],



Uncertain reading. terms

Original source details: GELFAND-P001156

corresp. to representations that complement one another are present.

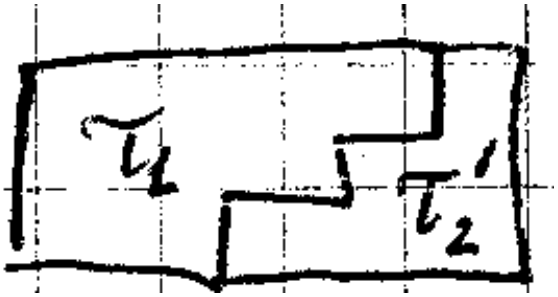
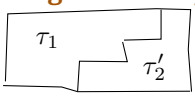


Diagram and typed description.



Original source details: GELFAND-P001156

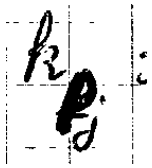
(see § in the chapter «Spin» book by Landau and Lifshitz, vol. 3)

Consider the region

$$X_Q = \{x_{q_1} < \dots < x_{q_N}\} \qquad (Q \text{ — permutation of } (1, 2, \dots, N)). \tag{15}$$

The answer in X_Q — Bethe substitution:

$$\Phi = \sum A_{\sigma_{q_1} \dots \sigma_{q_N}} (Q \cdot P)(-1)^P \exp i \sum_{j=1}^{N} k_{P_j} x_j \tag{16}$$



Uncertain reading. The subscript on k is overwritten, possibly P_j or p_j.

Original source details: GELFAND-P001156

$(k_1, \ldots, k_N)$ — momenta specifying the wave fn..

$$E = \sum_j k_j^2.$$

Nontriviality of formula (16):

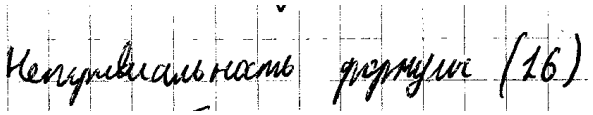


Diagram and typed description. Nontriviality of formula (16).

Original source details: GELFAND-P001156

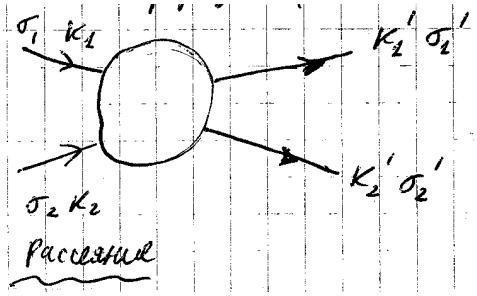
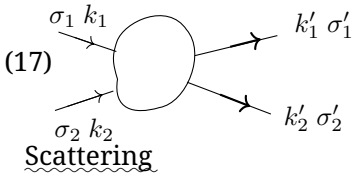


Diagram and typed description.



Original source details: GELFAND-P001156

GELFAND-P001157

[source number: 43]

$$\begin{aligned} k_1 + k_2 &= k_1' + k_2' \qquad \text{(cons. of momentum)} \\ k_1^2 + k_2^2 &= k_1'^2 + k_2'^2 \qquad \text{(cons. of energy)} \end{aligned} \tag{18}$$

(19) This is enough (in the 1-dimensional case) for fixing (k_1, k_2) (up to permutation). Only a change of spin can occur.

S-matrix (will be

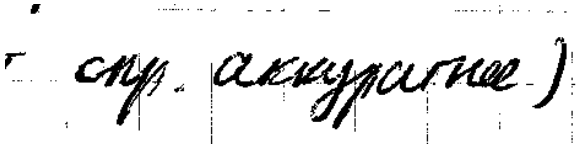
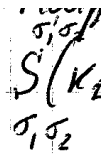


Diagram and typed description. defined

Original source details: GELFAND-P001157

more carefully later)

$$S_{\sigma_1\sigma_2}^{\sigma_1'\sigma_2'}(k_1, k_2) = bI + cP \qquad (P \text{ — rearrangement op.}) \tag{20}$$

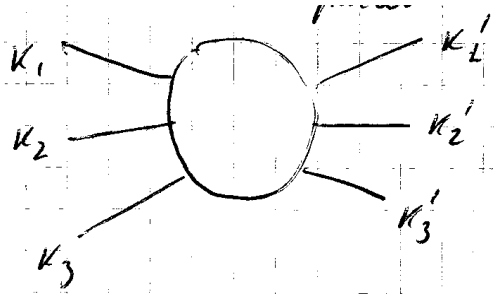


Uncertain reading. The first upper index in (20) is tightly written next to σ, possibly σ_1'.

Original source details: GELFAND-P001157

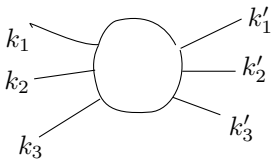
(σ_1, σ_2) — initial spins, (σ'_1, σ'_2) — final

Consider 3-particle scattering:



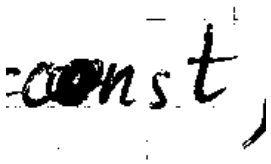
Original source details: GELFAND-P001157

Diagram and typed description.



A short angled stroke at the outer end of the upper left line has unclear significance.

$$\sum k_j = \text{const}, \qquad \sum k_j^2 = \text{const} \qquad \text{— insufficient} \tag{21}$$



Original source details: GELFAND-P001157

Uncertain reading. A letter in the first const in (21) is overwritten.

But because of integrability, here:

$$\sum k_j^n = \text{const}, \qquad n = 1, 2, 3, \ldots \tag{22}$$

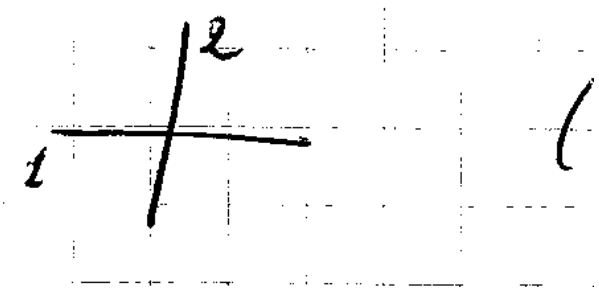
(absence of multiple production; the only event — a reshuffling of the particle spins)

$$\begin{matrix} \sigma_1 & k_1 \\ \vdots & \vdots \\ \sigma_n & k_n \end{matrix} \longmapsto \begin{matrix} \sigma_{q_1} & k_1 \\ \vdots & \vdots \\ \sigma_{q_n} & k_n \end{matrix} \tag{23}$$

Def. of the scattering matrix:

$$A(Q) = \widehat{S} A(Q') \tag{24}$$

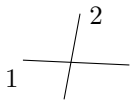
Suppose 2 change places:



Original source details: GELFAND-P001157

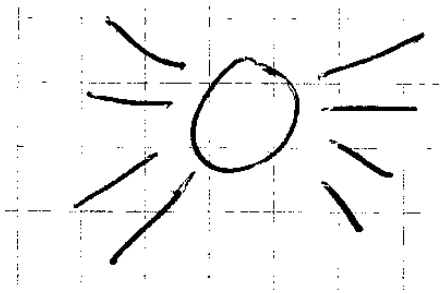
Diagram and typed description.

$$S^{\sigma'_1 \sigma'_2}_{\sigma_1 \sigma_2}(k_1, k_2) = \tag{25}$$



(notation)

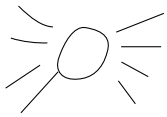
n-particle:



Original source details: GELFAND-P001157

Diagram and typed description.

$$S^{\sigma'_1 \dots \sigma'_n}_{\sigma_1 \dots \sigma_n}(k_1, \dots, k_n) = \quad (26)$$

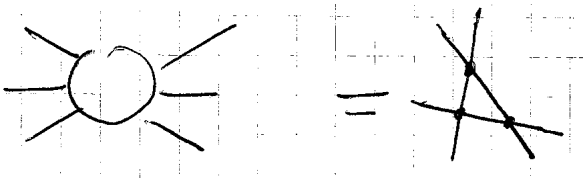


(27) Claim: (26) = a product of 2-particle scattering matrices.



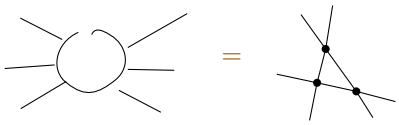
Original source details: GELFAND-P001157

Uncertain reading. Letters in “2-particle” have been corrected, obscuring their previous form.



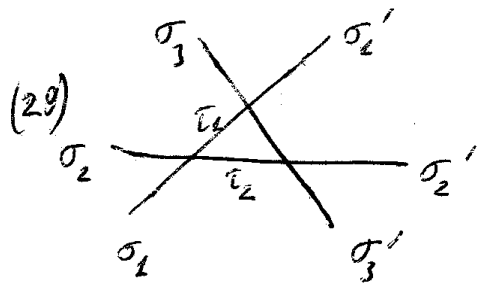
Original source details: GELFAND-P001157

Diagram and typed description.



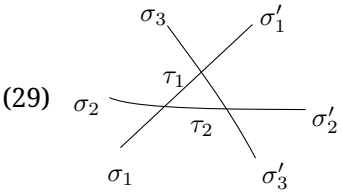
(28)

GELFAND-P001158



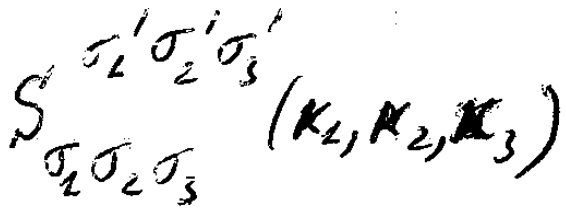
Original source details: GELFAND-P001158

Diagram and typed description.



(29)

$$S^{\sigma'_1 \sigma'_2 \sigma'_3}_{\sigma_1 \sigma_2 \sigma_3}(k_1, k_2, k_3) = \sum_{\tau} S^{\sigma'_2 \sigma'_3}_{\tau_2 \tau_3}(k_2, k_3) \\ S^{\sigma'_1 \tau_3}_{\tau_2 \sigma_3}(k_1, k_3) S^{\tau_1 \tau_2}_{\sigma_1 \sigma_2}(k_1, k_2) \quad (30)$$



Original source details: GELFAND-P001158

Uncertain reading. The momentum letters in the arguments on the left of (30) are heavily retraced or corrected.

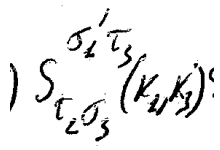


Diagram and typed description. The lower indices on the middle factor read $\tau_2\sigma_3$.

Original source details: GELFAND-P001158

Not every object [inserted below, circled and linked to “object”: (of type S)] is represented this way (by 2-particle scattering events).

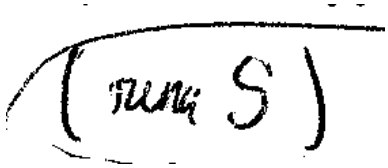


Diagram and typed description. (of type S)

Original source details: GELFAND-P001158

But this is so in integr. systems!!

Everything is as in a sys. of colliding hard balls, etc.

Condition for existence of repr. (30) — the $\triangle$ equation (nec. and suff.)

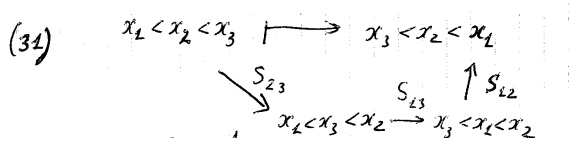


Diagram and typed description.

(31)

$$\begin{array}{ccc} x_1<x_2<x_3 & \longmapsto & x_3<x_2<x_1 \\ \downarrow S_{23} & & \uparrow S_{12} \\ x_1<x_3<x_2 & \xrightarrow{S_{13}} & x_3<x_1<x_2 \end{array}$$

Original source details: GELFAND-P001158

(via pairwise steps)

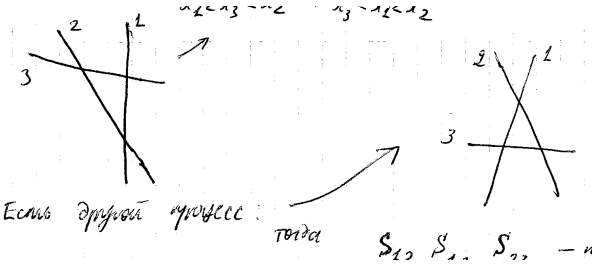
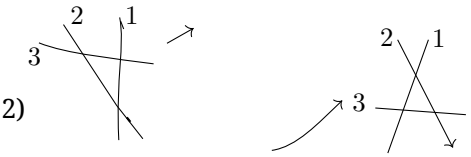


Diagram and typed description.

(32)



There is another process:
then

Original source details: GELFAND-P001158

In the left drawing, the short stroke at line 1 and the thickening on line 2 may be traces of arrowheads.

S_{12} S_{13} S_{23} — order of collisions

(33) Condition: the result of scattering must be the same in all ways: symbolically this is written as follows

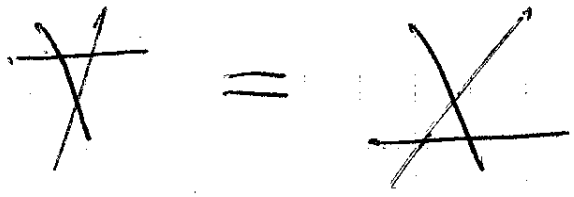
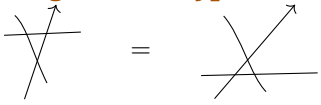


Diagram and typed description.



Original source details: GELFAND-P001158

Without isotopic indices: $S = e^{i\varphi}$ and everything holds automatically.



Uncertain reading. After i in the exponent is a looped sign, probably φ.

Original source details: GELFAND-P001158

With indices — a nontriv. relation:

$$\begin{aligned} &S_{23}(k_2, k_3) S_{13}(k_1, k_3) S_{12}(k_1, k_2) \\ &= S_{12}(k_1, k_2) S_{13}(k_1, k_3) S_{23}(k_2, k_3) \end{aligned} \tag{34}$$

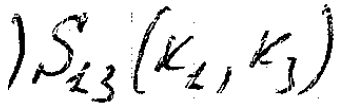


Diagram and typed description. On the right, the middle factor reads $S_{13}(k_1, k_3)$.

Original source details: GELFAND-P001158

This is a nec. and suff. condition for integrability of quantum systems. All quantum integr. sys. are described this way.

It is interesting that they exist!! (illegible abbreviation beginning with не?)



Uncertain reading. illegible abbreviation beginning with не

Original source details: GELFAND-P001158

They are still not fully classified.

GELFAND-P001159

[source number: 44]

Yang found the first soln..

2×2 Baxter found all solns.

(35) Claim. $\exists$ a parametrization $k(\theta)$, such that $S(k, k') = S(\theta - \theta')$

$$\left(\begin{array}{ll} \text{corresp.} & k = m \operatorname{sh} \theta \\ & \varepsilon = m \operatorname{ch} \theta \end{array} \right)$$

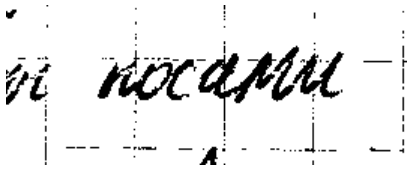
[At right:] $\| \ \theta - \theta'$ corresp. to the Galilei group and the Lorentz group

(36) Belavin, Drinfeld studied the quasi-class. $\triangle$ eqs.: $S = 1 + \hbar r$

(solved the corresp. quadr. eq.)

Tetrahedron: in multidimensional problems one must study strings, surfaces, etc.

Tetrahedra are sent? toward one another.



Uncertain reading. are sent

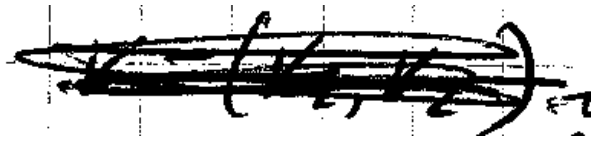
Original source details: GELFAND-P001159

No nontriv. solutions have been found so far (and it is suspected there are no nontriv. solutions; if there were, they would give integr. systems in high dimension)

The solns. found by Zamolodchikov describe a free particle.

Their existence would lead to multiple scattering (nontriv.)

Solution: [crossed out: $S(k_1, k_2)$ (?)]

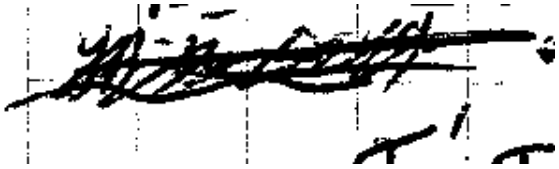


Uncertain reading. $S(k_1, k_2)$ (?)

Original source details: GELFAND-P001159

of the eqs. (34) for matrices of size 2×2 with the cond. of $SU(2)$ invariance for the upper and lower index pairs

[Under "Solution", crossed out: eqs. (34) (?)]



Uncertain reading. eqs. (34) (?)

Original source details: GELFAND-P001159

$$S^{\sigma_1'\sigma_2'}_{\sigma_1\sigma_2}(k) = I \cdot k + \text{[crossed-out coefficient]}\, \hat{P}, \qquad \hat{P} \text{ — permutation op.} \tag{37}$$



Uncertain reading. Between the plus and $\hat{P}$ is a heavily crossed-out, illegible coefficient.

Original source details: GELFAND-P001159

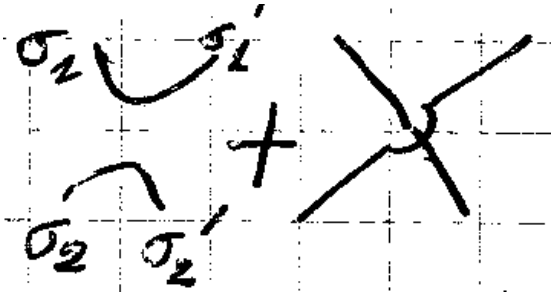
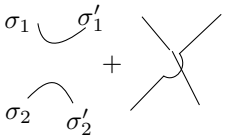


Diagram and typed description.

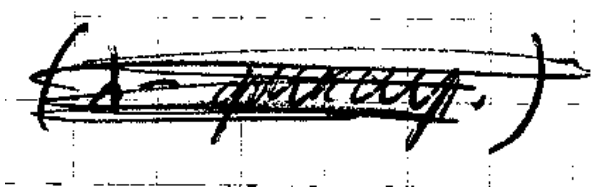


Original source details: GELFAND-P001159

A small bend is visible in the rising line at the crossing.

(here $k = k_1 - k_2$)

[crossed out: (λ — parameter.) (?)]



Uncertain reading. (λ — parameter.) (?)

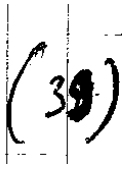
Original source details: GELFAND-P001159

$$I = \delta_{\sigma_1\sigma_1'}\delta_{\sigma_2\sigma_2'}, \qquad \hat{P} = \delta_{\sigma_1\sigma_2'}\delta_{\sigma_1'\sigma_2} \tag{38}$$

This is the $SU(2)$-invariant solution of the $\triangle$ eqs..

Compare with the soln. of the 2-particle problem.

$$\left[-\frac{\partial^2}{\partial x_1^2} - \frac{\partial^2}{\partial x_2^2} + V_{\sigma_1\sigma_2}^{\sigma_1'\sigma_2'}\delta(x_1 - x_2)\right]\Phi = E\Phi \tag{39}$$



Original source details: GELFAND-P001159

Uncertain reading. The last digit of (39) is overwritten.

V — a matrix taking $\sigma_1\sigma_2$ to $\sigma_1'\sigma_2'$

$$S = \frac{k + i\hat{V}\cdot\hat{P}}{k + i\hat{V}} \tag{40}$$

(this is from the soln. of eq. (39))

(Cayley transform of the potential $\hat{V}$).

Question: for which $\hat{V}$ does integrability hold?

(41) Solutions for $SU(2)$: $\hat{V} = I$ or $\hat{V} = \hat{P}$

(but they coincide after relabelling)

For other groups, other solutions and other potentials.

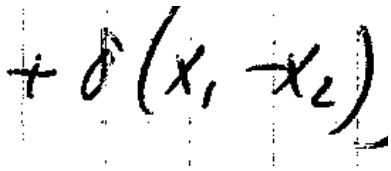
Therefore $\hat{V} = I$ and the system

GELFAND-P001160

$$\left[-\frac{\partial^2}{\partial x_1^2} - \frac{\partial^2}{\partial x_2^2} + \delta(x_1 - x_2)\right] \tag{42}$$

is integrable. Its 2-particle S-matrix has the form

$$S(k) = \frac{k + ic\hat{P}}{k + ic} \tag{43}$$



Original source details: GELFAND-P001160

Diagram and typed description. In (42) there is no visible coefficient before $\delta(x_1 - x_2)$, while (43) has c.

(44) For $\hat{V} \neq I, \hat{P}$ the system is nonintegrable!!

[Question: what then is the complete system of eigenfns.?].

(43) — S-matrix of bare particles.

We are interested in real particles [crossed out: illegible mark?].



Uncertain reading. illegible mark

Original source details: GELFAND-P001160

Spectrum — ?. Consider periodic boundary conditions (period $2L$)

$$\Phi(x_1, \ldots, x_i, \ldots, x_N)|_{x_i=-L} = \phi(x_1, \ldots, x_i, \ldots, x_N)|_{x_i=L}\,, \quad i = 1, \ldots, N \tag{45}$$

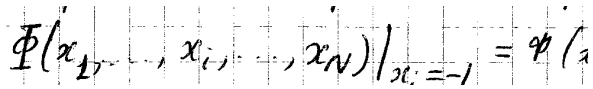


Diagram and typed description. In (45) the left has uppercase Φ and the right has lowercase ϕ.

Original source details: GELFAND-P001160

The conditions must be formulated in terms of the S-matrix { return to the initial phase on a segment of length $2L$):

$$\prod_i \hat{S}(k - k_i) \tag{46}$$

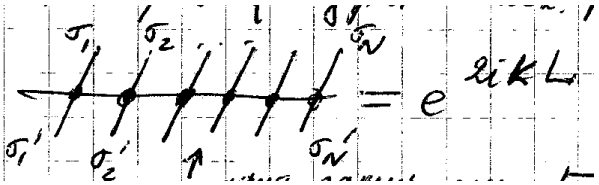
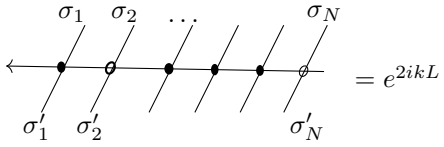


Diagram and typed description.



Original source details: GELFAND-P001160

Small openings remain inside the second and last crossings.

[Below the diagram, arrow to the diagram ↑:] graphical notation for the op. $\prod_{j \neq i} \hat{S}_{ji}(k_j - k_i)$

$$e^{ik_jL}A = \prod_{j \neq i} \hat{S}_{ji}(k_j - k_i)A \tag{47}$$

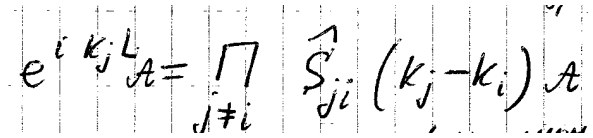


Diagram and typed description. Equation (47) writes e^{ik_jL} and has $j \neq i$ below the product sign.

Original source details: GELFAND-P001160

(actually a $2^N \times 2^N$ matrix, the initial amplitudes go into the final ones)
[An arrow from the parenthetical note points to the next line ↙]
A — coeff. (vector) in the Bethe wave fn., standing before the exponential.
Spectrum: quantization of momenta in a box.
Now solve the eigenvalue problem (47)

$$\prod_{j \neq i} \hat{S}_{ji}(k_j - k_i) \stackrel{\text{df}}{=} \text{transfer matrix} \tag{48}$$

It transports [crossed out: Φ (?)]

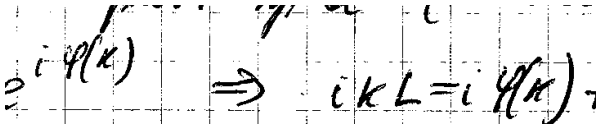


Uncertain reading. Φ (?)

Original source details: GELFAND-P001160

from one wall of the box to the other.
We have to diagonalize simultaneously N operators.
(49) A — a vector in a 2^N-dimensional space (eigenvector); $A = A_{\sigma_1 \ldots \sigma_N}$
Ordinary case:

$$e^{ikL} = e^{i\varphi(k)} \quad \Rightarrow \quad ikL = i\varphi(k) + i2\pi N \tag{50}$$



Uncertain reading. Both phase signs resemble φ, with a long right-hand loop.

Original source details: GELFAND-P001160



Uncertain reading. The first digit of (50) is heavily retraced or corrected.

Original source details: GELFAND-P001160

Here the operator has to be diagonalized.
[crossed out:]

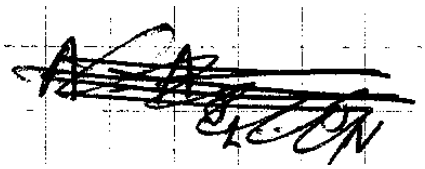
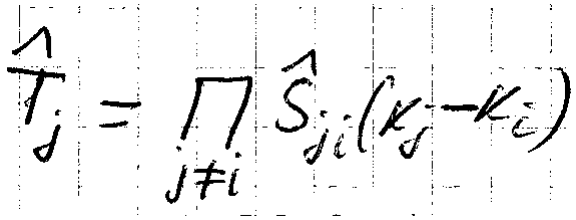


Diagram and typed description. $A = A_{\sigma_1 \ldots \sigma_N}$.

Original source details: GELFAND-P001160

$$\begin{cases} e^{ik_jL} = T_j(k_j, k_1, \dots, k_N), \\ \quad j = 1, \dots, N. \end{cases} \qquad \hat{T}_j = \prod_{j \neq i} \hat{S}_{ji}(k_j - k_i) \tag{51}$$



Original source details: GELFAND-P001160

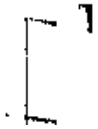
Diagram and typed description. On the right of (51), j appears both on $\hat{T}$ and below the product sign.



Original source details: GELFAND-P001160

Uncertain reading. A mark above T_j in the first line is crossed out, possibly a hat.

[Above the formula, a diagonal mark pointing to T_j:] eigenvalue of the op. $\hat{T}_j$

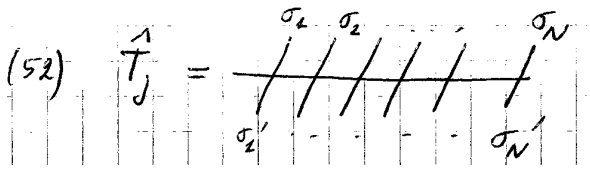


Original source details: GELFAND-P001160

Uncertain reading. In the upper right, a short angled mark by the edge of the sheet.

GELFAND-P001161

[source number: 45]



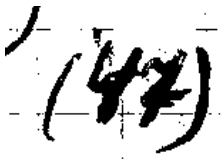
Original source details: GELFAND-P001161

Diagram and typed description.

$$\hat{T}_j = \quad \sigma_1 \quad \sigma_2 \quad \dots \quad \sigma_N \qquad \sigma'_1 \quad \dots \quad \sigma'_N \tag{52}$$

$$[\hat{T}_j, \hat{T}_{j'}] = 0 \tag{53}$$

(consequence of the eqs. for $\triangle$s) $\Rightarrow$ consistency of (47)



Original source details: GELFAND-P001161

Uncertain reading. The reference number (47) is overwritten.

Diagonalization of $\hat{T}$ (the Leningrad group devised the most suitable method)

Yang was the first to solve it in 1967. Faddeev et al. started working on it in 1978. Their method is simpler than Yang's, but still cumbersome.

$\hat{T}$ corresp. to the Lax op. L ([crossed out: illegible?.]

Uncertain reading. illegible



Original source details: GELFAND-P001161

this is a quantum generalization of the Lax op. L)

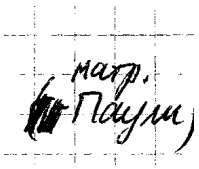
$$\begin{aligned} S^{\sigma_1'\sigma_2'}_{\sigma_1\sigma_2}(k) &= b(k)I + c(k)\hat{P} \\ &= \begin{pmatrix} b+\frac{1}{2}c+\frac{1}{2}c\sigma^z & c\sigma^- \\ c\sigma^+ & b+\frac{1}{2}c-\frac{1}{2}c\sigma^z \end{pmatrix}, \end{aligned} \tag{54}$$

where

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \sigma^+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

([crossed out: m.?;

Uncertain reading. m.?

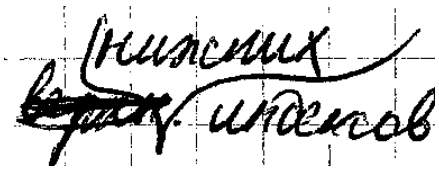


Original source details: GELFAND-P001161

inserted above the line: matrices] Pauli)

The table is in the space of [crossed out: upper?;

Uncertain reading. upper?

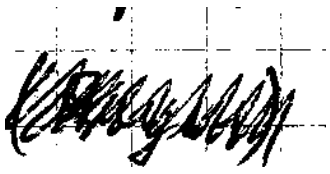


Original source details: GELFAND-P001161

inserted above the line: lower] indices

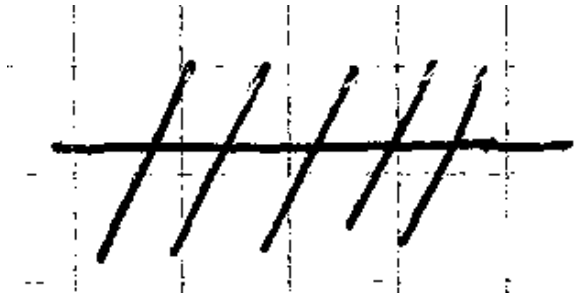
The matrices $\sigma^{\pm}, \sigma_z$ [crossed out: (Pauli) (?)]

Uncertain reading. (Pauli) (?)



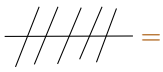
Original source details: GELFAND-P001161

act in the space of the upper indices.



Original source details: GELFAND-P001161

Diagram and typed description.

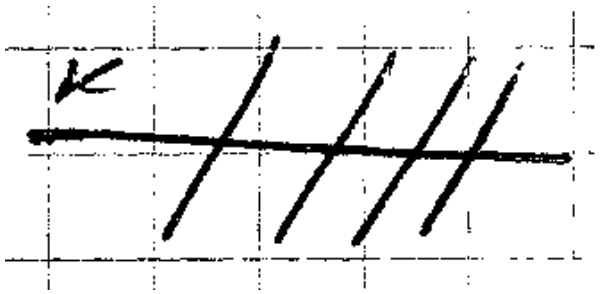


$$= \prod_i S_{ji}(k_j - k_i) = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \tag{55}$$

(notation accounting for the open ends)

$k_j = k$ is singled out. (analogue of the spectral param. in the Lax repr.)

$$\begin{pmatrix} A(k) & B(k) \\ C(k) & D(k) \end{pmatrix} = \prod_i S_{*i}(k - k_i) \tag{56}$$



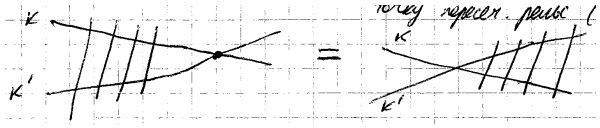
Original source details: GELFAND-P001161

Diagram and typed description.



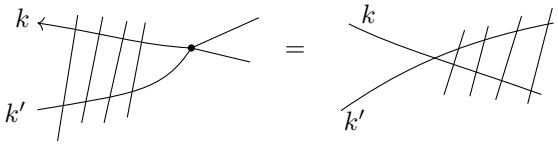
$T(k) \otimes T(k')$

All the sleepers can be moved through the point where the rails cross (eq. for $\triangle$s)



Original source details: GELFAND-P001161

Diagram and typed description.



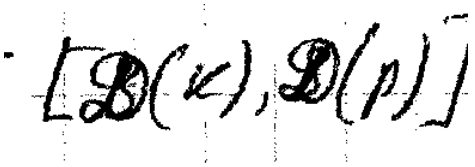
$$T(k) \otimes T(k') \cdot S(k - k') = S(k - k')T(k') \otimes T(k) \tag{57}$$

We get commut. relations between the entries A, B, C, D. They have the form:

GELFAND-P001162

Commut. relations in the entries A, B, C, D:

$$[A(k), A(p)] = [B(k), D(p)] = 0 \tag{58}$$



Original source details: GELFAND-P001162

Uncertain reading. The letters in the second commutator have different forms, with $B(k), D(p)$ a possible reading.

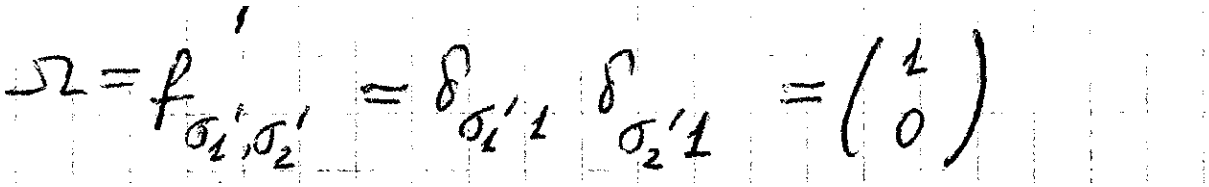
$$b(k-p)A(k)B(p) = B(p)A(k) - c(k-p)B(k)A(p) \tag{59}$$

and one more by analogy.

Here $b(k), c(k)$ are the same as in (54).

Consider the vector

$$\Omega = f_{\sigma_1',\sigma_2'} = \delta_{\sigma_1'1}\delta_{\sigma_2'1} = \begin{pmatrix}1\\0\end{pmatrix} \tag{60}$$



Original source details: GELFAND-P001162

Diagram and typed description. On the right of (60) is a single two-entry column $\begin{pmatrix}1\\0\end{pmatrix}$.

S-matrix [inserted above the line: (54).] on the vector Ω becomes triangular



Original source details: GELFAND-P001162

Diagram and typed description. inserted above the line: (54).

Hence the product of S-matrices has the eigenvector

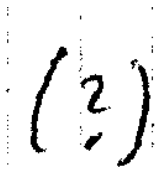
$$\Omega = \begin{pmatrix}1\\0\end{pmatrix} \otimes \begin{pmatrix}1\\0\end{pmatrix} \otimes \dots \tag{61}$$

Since

$$b = \frac{k}{k+ic}, \qquad c = \frac{ic}{k+ic} \tag{62}$$

(from unitarity and so on),

we find the eigenvalue: $b(k-k_1)b(k-k_2)\dots$ (?) (63)

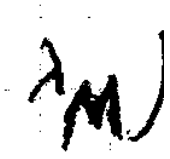


Original source details: GELFAND-P001162

Diagram and typed description. At the right of the product the manuscript has (?)

Other eigenvectors: we seek them in the form:

$$\Omega(\lambda_1, \dots, \lambda_M) = B(\lambda_1)\dots B(\lambda_N)\Omega \tag{64}$$



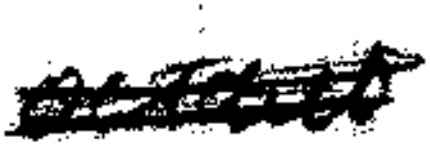
Original source details: GELFAND-P001162

Diagram and typed description. The last index on the left is overwritten and reads as M.

[Below the last $\Omega\uparrow$:] vacuum.
Now we choose $\lambda_1, \dots, \lambda_N$.

$$\Omega \text{ — eigenv. for } A \text{ with eigenval. } 1 \tag{65}$$

We apply A and use the commut. relations $\Rightarrow$ for the 1st term, always eigen.
For [crossed out: illegible word?.]



Uncertain reading. illegible word

Original source details: GELFAND-P001162

the 2nd [inserted above the line: in (59).] it is no longer so ("unwanted terms")
Eigenvalues:

$$T = \prod_{\alpha} \frac{1}{b(k_j - \lambda_\alpha)} \tag{66}$$

Result:

$$e^{2ik_jL} = \prod_{\alpha=1}^{M} \frac{k_j - \lambda_\alpha + ic}{k_j - \lambda_\alpha - ic}, \qquad j = 1, \dots, N. \tag{67}$$

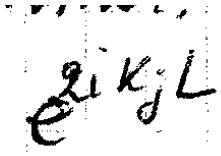


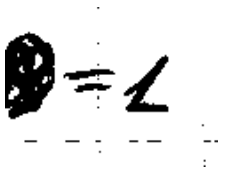
Diagram and typed description. The exponent on the left in (67) is $2ik_jL$.

Original source details: GELFAND-P001162

GELFAND-P001163

[source number: 46]

$$\prod_{j=1}^{N} \left(\frac{\lambda_\alpha - k_j + ic}{\lambda_\alpha - k_j - ic} \right) = \prod_{\beta=1}^{M} \left(\frac{\lambda_\alpha - \lambda_\beta + i2c}{\lambda_\alpha - \lambda_\beta - i2c} \right), \qquad \alpha = 1, \dots, M, \tag{68}$$



Uncertain reading. The letter in the lower limit of the product on the right is retraced and reads as β.

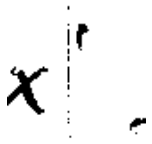
Original source details: GELFAND-P001163

We get a decomposition of the op. into a Clebsch–Gordan series



Diagram and typed description. $\square \times \square \times \dots \times \square = \sum$ 

Original source details: GELFAND-P001163



Original source details: GELFAND-P001163

Uncertain reading. After the second $\times$ there is a short raised mark of unclear significance.

These are algebr. eqs. for $k_1, \dots, k_N$, $E = \sum k_j^2$.

λ_j — auxiliary numbers to be determined.

We have to find λ and k from (67)–(68). (and we only need k)

We want to study the limit as $\begin{matrix} N \to \infty, \\ M \to \infty. \end{matrix}$

In the limit they can be solved!!

Yang proved the solvability of these eqs., the number of solutions — 2^N and so on.

In the limit — integral eqs., boundary-value problems and so on.

GELFAND-P001164

[source number: 56]



Original source details: GELFAND-P001164

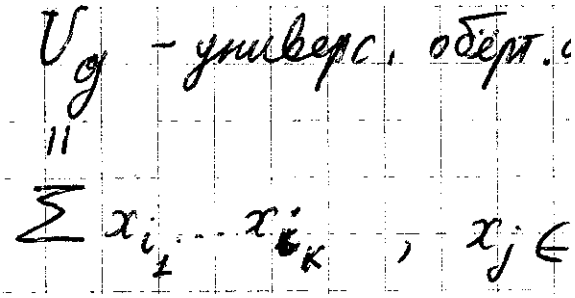
Uncertain reading. At upper left is a mark resembling $\downarrow$. The middle of the mark overlaps a thick vertical band.

20/II-84 Cherednik. Nonlinear representation theory

$\mathfrak{g}$ — Lie alg. over $\mathbb{C}$.

$U_\mathfrak{g}$ — univ. enveloping alg.

$$\| \sum x_{i_1} \cdots x_{i_k}, \quad x_j \in \mathfrak{g}.$$



Original source details: GELFAND-P001164

Uncertain reading. Two vertical strokes join $U_\mathfrak{g}$ to the sum. The last index is retraced and reads as i_k.

$$U_n = \left\{ \sum_{x_j \in \mathfrak{g}} x_1 \cdots x_n \right\}.$$

$$\mathrm{Gr} = \bigoplus U_n / U_{n-1} = S^\infty(\mathfrak{g})$$

(commut. alg., generated by $\mathfrak{g}$ as a vect. space)

$$[U_n, U_m] \subset U_{n+m-1} \quad \Rightarrow \quad \mathrm{Gr} \text{ is commutative.}$$

[ ] generates the Poisson bracket on $S^\infty(\mathfrak{g})$. (“quasiclassical limit”).

New def.

{ U — graded [inserted below: and filtered] with struct. in the param. $\overline{h}$ }

1. Zamolodchikov alg.
2. Sklyanin alg.
3. On some [inserted: finite] reprs.

$$\mathfrak{g} = \mathfrak{gl}_n, \qquad V = \bigoplus_{\substack{\alpha\in\mathbb{Z}_n^2\\ \lambda\in\mathbb{C}}} A_\alpha(\lambda).$$

$A_\alpha(\lambda)$ — a symbol (generator). [Inserted above "generator": basis of a vect. space.]

$\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$.

(very infinite-dimensional)

$T(V)$ — the tensor alg. of the space V.

In $\mathfrak{gl}_n$ choose a basis $\{I_\alpha, \alpha \in \mathbb{Z}_n^2\}$ (any for now).

$$L(\lambda) = \sum_{\alpha\in\mathbb{Z}_n^2} A_\alpha(\lambda) I_\alpha \in \mathfrak{gl}_n(T(V)).$$

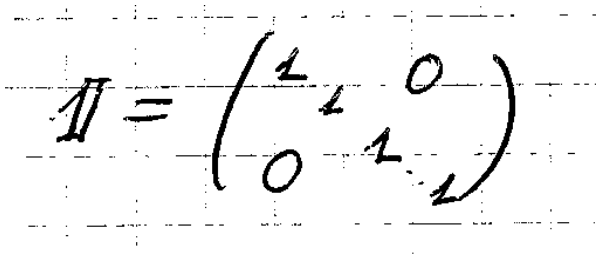
(matrix [struck-out mark] with entries from $T(V)$)



Original source details: GELFAND-P001164

Uncertain reading. After "matrix" there is a heavily struck-out mark. One possible reading is n.

$$L^1 = L \otimes \mathbf{1}, \qquad L^2 = \mathbf{1} \otimes L.$$



Original source details: GELFAND-P001164

Diagram and typed description.

$$\mathbf{1} = \begin{pmatrix} 1 & & & & 0 \\ & 1 & & & \\ & & 1 & & \\ & & & \ddots & \\ 0 & & & & 1 \end{pmatrix}$$

Introduce a matrix [above the line: $n^2 \times n^2$]:

$$R(\lambda, \mu) \in \mathfrak{gl}_n \otimes \mathfrak{gl}_n.$$

Relations on $A_\alpha(\lambda)$:

$$R(\lambda,\mu)L^1(\lambda)L^2(\mu) = L^2(\mu)L^1(\lambda)R(\lambda,\mu). \tag{1}$$



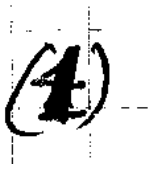
Original source details: GELFAND-P001164

Uncertain reading. The numeral in the label is overwritten. The reading (1) follows the order of the formulas on the leaf.

$$L^1(\lambda)L^2(\mu) \neq L^2(\mu)L^1(\lambda)$$

(in general) because of the noncommutativity of $T(V)$.

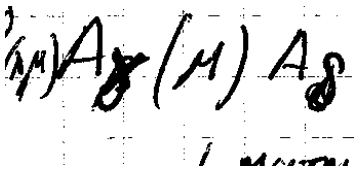
It follows from (1) that



Original source details: GELFAND-P001164

Uncertain reading. The numeral in the reference is overwritten in the same way as the label on the preceding formula.

$$A_\alpha(\lambda)A_\beta(\mu) = \sum r^{\gamma\delta}_{\alpha\beta}(\lambda,\mu)A_\gamma(\mu)A_\delta(\lambda). \tag{2}$$



Original source details: GELFAND-P001164

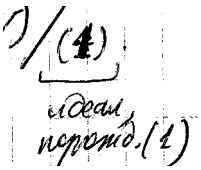
Uncertain reading. The subscript of the first A on the right is overwritten. The reading γ agrees with the coefficient's index pair $\gamma\delta$.

($r^{\gamma\delta}_{\alpha\beta}$ is not written out) (the matrix $R(\lambda,\mu)$ is assumed invertible)

GELFAND-P001165

(3) Thm. Let $A = T(V)/\underbrace{(1)}_{\substack{\text{ideal}\\ \text{generated by } (1)}}$. Then the conditions are equivalent

a) All the monomials $A_\alpha(\lambda_1)A_\beta(\lambda_2)\cdots A_\delta(\lambda_k)$ are linearly indep. in A (for fixed $\lambda_1,\lambda_2,\dots,\lambda_k$ and for arbitrary fixed k).



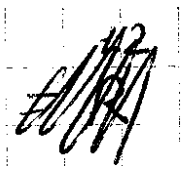
Original source details: GELFAND-P001165

Uncertain reading. The numeral in the denominator is overwritten. The note underneath reads "ideal generated by (1)."

b) R is factorizable, i.e. the Yang–Baxter or $\triangle$ relations hold.

$$\overset{12}{R}(\lambda,\mu)\overset{13}{R}(\lambda,\nu)\overset{23}{R}(\mu,\nu) = \overset{23}{R}(\mu,\nu)\overset{13}{R}(\lambda,\nu)\overset{12}{R}(\lambda,\mu). \tag{4}$$

$$R = \sum \rho_{\alpha\beta} I_\alpha \otimes I_\beta \quad \Rightarrow \quad \overset{12}{R} = R \otimes \mathbf{1}$$



Original source details: GELFAND-P001165

Uncertain reading. Between I_β and the arrow, $\overset{12}{R}$ is struck out, possibly preceded by $=$.

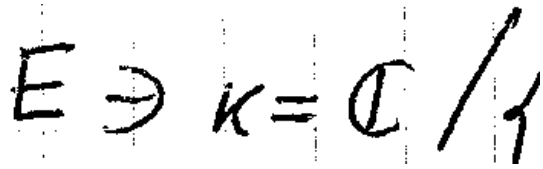
$$\overset{13}{R} = \sum \rho_{\alpha\beta} I_\alpha \otimes \mathbf{1} \otimes I_\beta \qquad \text{and simil. for } \overset{23}{R}.$$

Idea: move across using rel. (1) and so on.

$$(4) \;\Rightarrow\; \underset{\lambda,\mu}{\overset{12}{R}}\; \underset{\mu,\lambda}{\overset{21}{R}} \sim \mathbf{1} \otimes \mathbf{1}. \tag{5}$$

Example connected with the ell. curve E.

$$E \ni \kappa = \mathbb{C}/\underbrace{\begin{matrix}\{\mathbb{Z}+\tau\mathbb{Z}\} \\ \operatorname{Im}\tau > 0\end{matrix}}_{\Lambda}, \qquad \begin{matrix} E_n = \frac{1}{n}\mathbb{Z} + \frac{1}{n}\tau\mathbb{Z} \mod \Lambda \\ \| \\ \mathbb{Z}_n^2. \end{matrix}$$



Uncertain reading. The letter after $\ni$ reads as κ or k.

Original source details: GELFAND-P001165

$w_\alpha(\lambda)$ — a function [inserted above: doubly per.] which

1) has poles of ord. ≤ 1 at the points $-\eta + E_n$

$$2) \quad w_\alpha(\lambda + \underset{\uparrow}{\underset{\mathbb{Z}_n^2 = E_n}{\beta}}) = \omega^{\alpha_1\beta_2 - \beta_1\alpha_2} w_\alpha(\lambda), \qquad \text{where } \omega^n = 1 \quad \text{(primitive root)}$$

$$3) \quad w_\alpha(0) = 1.$$

$w_\alpha(\lambda)$ is unique.

Let

$$I_\alpha I_\beta = \omega^{-\alpha_2\beta_1} I_{\alpha+\beta} \qquad \text{(we choose this basis)}. \tag{6}$$

($n = 2 \;\Rightarrow$ Pauli matrices;

$n > 2 \;\Rightarrow$ diag. + Coxeter element

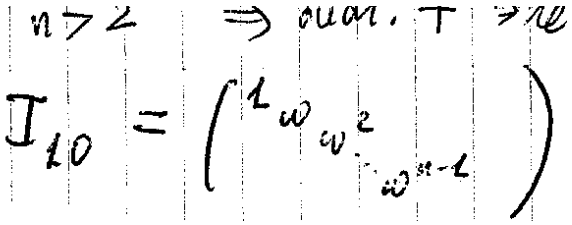


Diagram and typed description.

$$I_{10} = \begin{pmatrix} 1 & & & & \\ & \omega & & & \\ & & \omega^2 & & \\ & & & \ddots & \\ & & & & \omega^{n-1} \end{pmatrix}$$

Original source details: GELFAND-P001165

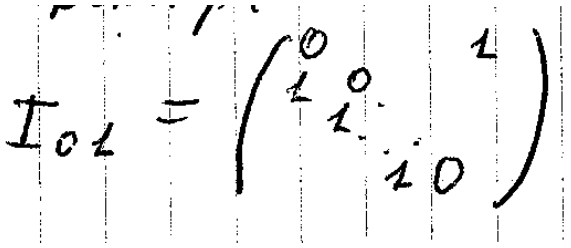


Diagram and typed description.

$$I_{01} = \begin{pmatrix} 0 & & & & 1 \\ 1 & 0 & & & \\ & 1 & \ddots & & \\ & & \ddots & \ddots & \\ & & & 1 & 0 \end{pmatrix}$$

Original source details: GELFAND-P001165

and the rest by property (6)

GELFAND-P001166

[source number: 57]

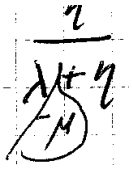
Prop.

$$R(\lambda,\mu)=\sum_{\alpha\in\mathbb{Z}_n^2} w_\alpha(\lambda-\mu)I_\alpha\otimes I_\alpha^{-1}. \tag{7}$$

Satisfies the Yang–Baxter rels. (Belavin's thm.).

Another example

$$\begin{aligned} R&=\mathbf{1}\otimes\mathbf{1}\frac{\lambda-\mu}{\eta+\lambda-\mu}+\frac{\eta}{\lambda-\mu+\eta}P,\\ P&=\frac{1}{n}\sum_{\alpha\in\mathbb{Z}_n^2} I_\alpha\otimes I_\alpha^{-1}. \end{aligned} \tag{8}$$



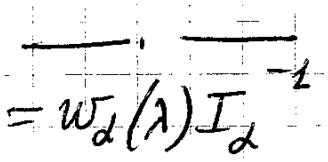
Uncertain reading. In the second denominator $-\mu$ is inserted from below.

Original source details: GELFAND-P001166

(9) Prop.

$$A_\alpha(\lambda)=\overline{w_\alpha}(\lambda)\overline{I_\alpha^{-1}} \quad\Rightarrow\quad \text{gives a finite-dimensional representation}$$

$$\mathcal{A}_R\longrightarrow \mathfrak{gl}_n.$$



Uncertain reading. Two separate horizontal strokes and a short mark between them appear above the right-hand side. Their purpose is unclear.

Original source details: GELFAND-P001166

Def. Sklyanin alg.

$$\begin{aligned} L(\lambda)&=\sum_{\alpha\in\mathbb{Z}_n^2} S_\alpha w_\alpha(\lambda)I_\alpha,\\ A_\alpha(\lambda)&=S_\alpha w_\alpha(\lambda). \end{aligned} \tag{10}$$

Reasonable rels.

$$\sum_{\alpha,\beta}\underset{\mathbb{C}}{\ell_{\alpha\beta}}S_\alpha S_\beta=0.$$

$$W=\bigoplus_{\alpha\in\mathbb{Z}_n^2}\mathbb{C}S_\alpha.$$

Important: there is a series of fin.-dimensional reprs. (Sklyanin: FA 16:4, 17:4).

$n=2$ $\quad S_0,S_1,S_2,S_3$

$$[S_i,S_j]=\{S_0,S_k\},\qquad [S_i,S_0]=J_i\{S_j,S_k\}.$$

GELFAND-P001167

27/II-84 Gelfand Sem.

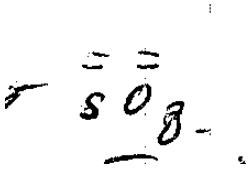
A. Zelevinsky (joint work with I. M. Gelfand)

Models of representations of classical groups and their hidden symmetries.

1. All reprs. of SO_3 are realized in functions on S^2 (in $L^2(S^2)$).

2. L. C. Biedenharn, D. Flath:

In the model of reprs. of SU_3 there is a mysterious action of $\overline{\overline{s}}\,\overline{o}_8$ — (the algebra only!) and the corresponding repr. of so_8 is irreducible.



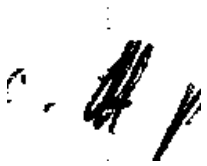
Original source details: GELFAND-P001167

Uncertain reading. There are two short strokes over each of s and o. The later so_8 is written without them.

3. First step in Penrose's twistor construction

$$\mathbb{R}^4 \subset \mathbb{C}^4 \subset G_2^4$$

Content [overwritten mark] of the work: all 3 constructions extend to all classical groups $\Rightarrow$



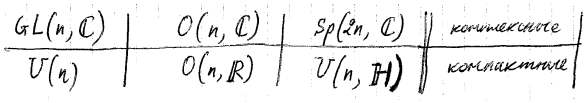
Original source details: GELFAND-P001167

Uncertain reading. Between "Content" and "of the work" is a heavily overwritten mark, possibly 4.

$\Rightarrow$ there arise by themselves supergroups, superalgebras and so on!!

1. Complex algebraic groups will take the place of compact groups

Model: direct sum of all finite-dimensional irreduc. reprs., each once



Original source details: GELFAND-P001167

Diagram and typed description.

$GL(n,\mathbb{C})$	$O(n,\mathbb{C})$	$Sp(2n,\mathbb{C})$	complex
$U(n)$	$O(n,\mathbb{R})$	$U(n,\mathbb{H})$	compact

G — one of the groups $GL(n,\mathbb{C})$, $O(n,\mathbb{C})$, $Sp(2n,\mathbb{C})$.

1 realization: induced representation (in functions or in sections of a bundle on the homogeneous space G/M).

GELFAND-P001168

[source number: 60]

$$\mathrm{Ind}_M^G \tau$$

Model (Bernstein + 2 Gelfands):

(1)

G	$GL(n)$	$O(2n+1)$	$O(2n)$	$Sp(2n)$
M	$O(n)$	$O(n+1)\times O(n)$	$O(n)\times O(n)$	$GL(n)$
τ	$\lambda(n)$	$\lambda(n+1)\otimes 1$	$\lambda(n)\otimes 1$	$\lambda(n)$

Original source details: GELFAND-P001168

Diagram and typed description. (1)

G	$GL(n)$	$O(2n+1)$	$O(2n)$	$Sp(2n)$
M	$O(n)$	$O(n+1)\times O(n)$	$O(n)\times O(n)$	$GL(n)$
τ	$\lambda(n)$	$\lambda(n+1)\otimes 1$	$\lambda(n)\otimes 1$	$\lambda(n)$

(all complex)

$\lambda(n)$ — exterior algebra of the standard representation

(actually because of supergroups)

Dual models (superduality)

(2)

G	$GL(2n)$	$O(2n)$	$Sp(4n+2)$	$Sp(4n)$
M	$Sp(2n)$	$GL(n)$	$Sp(2n+2)\times Sp(2n)$	$Sp(2n)\times Sp(2n)$
τ	$\sigma(2n)$	$\sigma(n)$	$\sigma(2n+2)\otimes 1$	$\sigma(2n)\otimes 1$

Original source details: GELFAND-P001168

Diagram and typed description.

(2)

G	$GL(2n)$	$O(2n)$	$Sp(4n+2)$	$Sp(4n)$
M	$Sp(2n)$	$GL(n)$	$Sp(2n+2)\times Sp(2n)$	$Sp(2n)\times Sp(2n)$
τ	$\sigma(2n)$	$\sigma(n)$	$\sigma(2n+2)\otimes 1$	$\sigma(2n)\otimes 1$

Here $\sigma(n)$ — the symmetric algebra of the standard representation.

(3) Transition:

$$G(n) \longrightarrow GL(2n)$$
$$O(n) \longrightarrow Sp(2n)$$
$$Sp(2n) \longrightarrow O(4n)$$
$$\lambda(n) \longrightarrow \sigma(2n)$$

$G(n) \longrightarrow GL(2n)$

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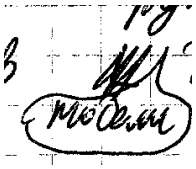
Uncertain reading. The left side of the first line reads $G(n)$. The L may have been omitted.

(4) Theorem 1. Table (2) consists of models.

Rem. It does not contain all the classical groups (unlike (1)): it lacks $GL(2n+1)$ and $O(2n+1)$, this comes after the comments.

$G \subset L$ — overgroup. L — is also a group.

Model in the sense of an overgroup: in [struck-out mark] [inserted below and circled: the model] a repr. of the Lie alg. ℓ of the group L acts.



Original source details: GELFAND-P001168

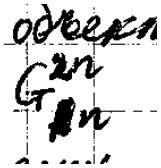
Uncertain reading. The writing before "acts" is struck out. The mark under the strokes may be M. Below it "the model" is inserted.

The models are constructed by one geom. scheme.

$\Gamma(L)$ — the basic Grassmannian of the group L.

I. M. G. Remarkable object G_2^4 — a book should be written about it (basic building block)

In general G_n^{2n} — is the main thing.



Original source details: GELFAND-P001168

Uncertain reading. Before n in the subscript is a heavily overwritten mark. It may be a cancelled 2.

What polyhedra are there?

What takes the place of their volumes?

Where does the dilogarithm come from?

Char. classes and why they do not localize, what takes the place of localization?

See the medieval book on the icosahedron (and Klein's book on it).

GELFAND-P001169

A. Z. List of Grassmannians (they are hom. spaces of the corresp. group):

$$\left[\begin{array}{rl} \Gamma(GL(2n)) = \Gamma_n^{2n} & (n\text{-dimensional planes in } 2n\text{-dimensional space}). \\ \Gamma(GL(2n+1)) = \Gamma_{n+1}^{2n+1} & \\ \Gamma(O(2n)) = & \text{Grassmannian of } n\text{-dimensional isotropic subspaces} \\ & \text{in } \mathbb{C}^{2n} \\ & \text{(isotr. — all vectors have length } 0) \\ & \text{It is disconnected.} \\ \Gamma(O(2n+1)) = & \text{Grassmannian of } n\text{-dimensional isotropic subspaces} \\ & \text{in } \mathbb{C}^{2n+1} \end{array}\right. \tag{5}$$

In $\mathbb{C}^2$ $\quad z_1^2 + z_2^2 = 0$, $z_1 = \pm i z_2 \Rightarrow \Gamma(O(2))$ consists of 2 points.

(6) Exercise. The map

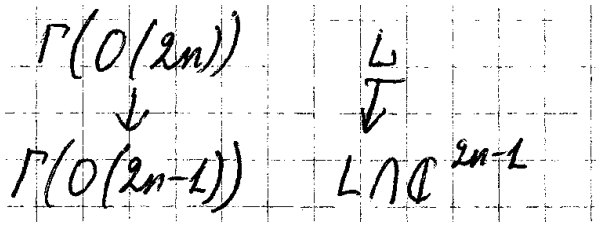


Diagram and typed description.

$$\begin{array}{ccc} \Gamma(O(2n)) & & L \\ \downarrow & & \downarrow \\ \Gamma(O(2n-1)) & & L \cap \mathbb{C}^{2n-1} \end{array}$$

Original source details: GELFAND-P001169

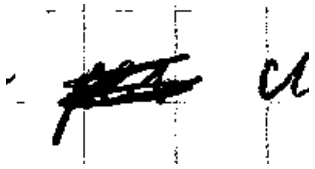
is an isomorphism on each component.

(5) cont. $\Gamma(Sp(2n))$ — Grassmannian of n-dimensional Lagrangian subspaces in $\mathbb{C}^{2n}$.

Next: $G \subset L$.

$\Gamma(L) \supset \Omega$. [Below Ω: open subset (so that there are holom. functions).]

It is in functions on Ω [struck-out writing] that the representation is constructed.

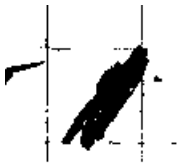


Uncertain reading. After the first Ω the beginning of a word is struck out, possibly Russian “p...”.

Original source details: GELFAND-P001169

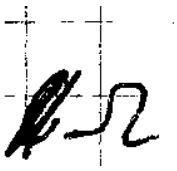
Ω — the unique open orbit.

L does not act, but ℓ acts [struck-out writing] on functions on [struck-out writing] Ω (infinites. shifts!).



Uncertain reading. After “acts” there is an illegible struck-out mark.

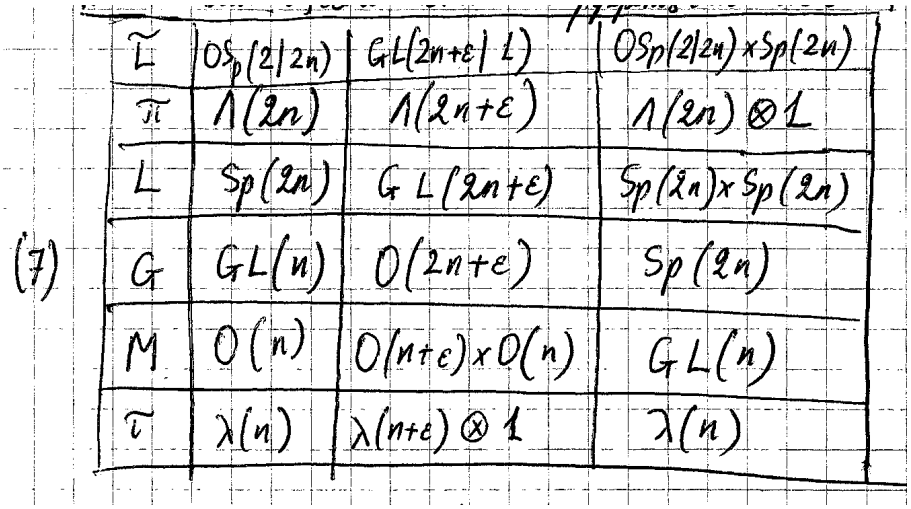
Original source details: GELFAND-P001169



Original source details: GELFAND-P001169

Uncertain reading. Immediately before Ω a mark is struck out, possibly ℓ.

I. M. G. Apparently there is an even differential in place of d (with $d^2 = 0$).



Original source details: GELFAND-P001169

Diagram and typed description. (7)

$\widetilde{L}$	$OSp(2\|2n)$	$GL(2n+\varepsilon\|1)$	$OSp(2\|2n) \times Sp(2n)$
$\widetilde{\pi}$	$\Lambda(2n)$	$\Lambda(2n+\varepsilon)$	$\Lambda(2n) \otimes 1$
L	$Sp(2n)$	$GL(2n+\varepsilon)$	$Sp(2n) \times Sp(2n)$
G	$GL(n)$	$O(2n+\varepsilon)$	$Sp(2n)$
M	$O(n)$	$O(n+\varepsilon) \times O(n)$	$GL(n)$
τ	$\lambda(n)$	$\lambda(n+\varepsilon) \otimes 1$	$\lambda(n)$

$\varepsilon = 0, 1$

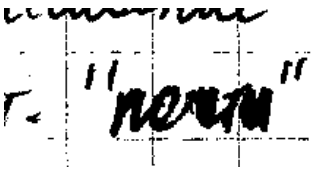
[To the right of the table:] $\Omega = G/M$.

τ — repr. in the fibre.

$$\mathrm{Ind}_M^G \tau$$

(8) Thm. Under the action of G on $\Gamma(L)$ there is a unique open orbit and a vect. bundle π on which there is an action of ℓ. (hidden symmetry)

The repr. is "almost" irreducible.

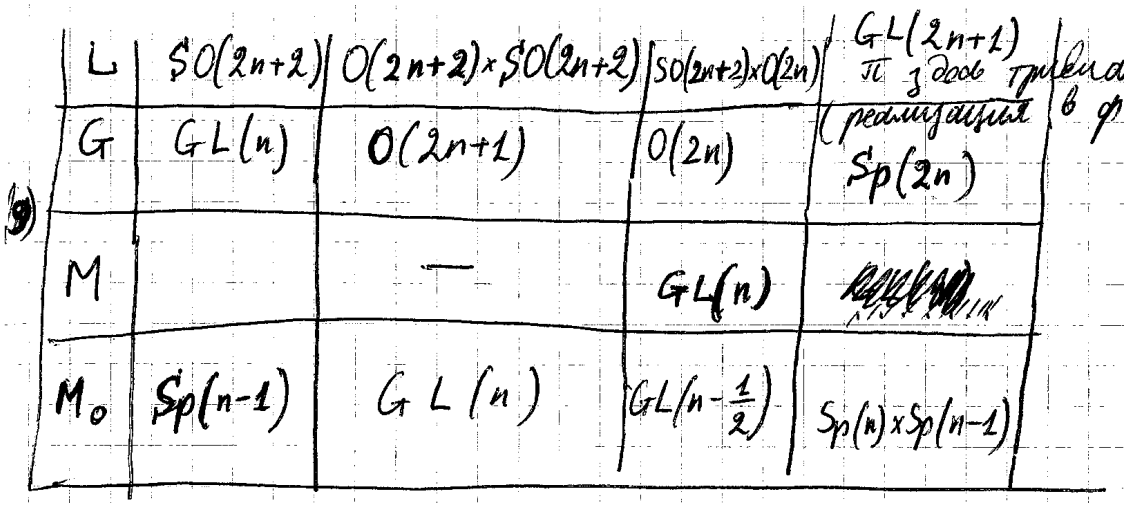


Original source details: GELFAND-P001169

Uncertain reading. The word in quotation marks is heavily retraced. It reads "almost".

GELFAND-P001170

[source number: 61]



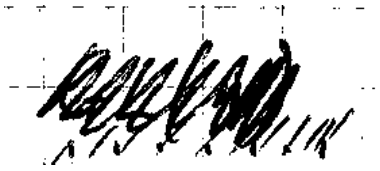
Original source details: GELFAND-P001170

Diagram and typed description.

(9)

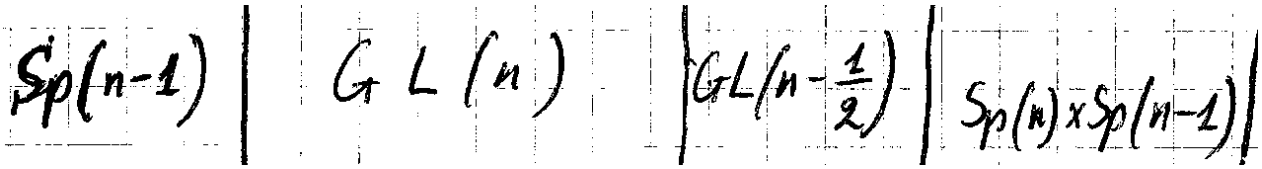
L	$SO(2n+2)$	$O(2n+2) \times SO(2n+2)$	$SO(2n+2) \times O(2n)$	$GL(2n+1)$
G	$GL(n)$	$O(2n+1)$	$O(2n)$	$Sp(2n)$
M		—	$GL(n)$	[struck out]
M_0	$Sp(n-1)$	$GL(n)$	$GL(n-\frac{1}{2})$	$Sp(n) \times Sp(n-1)$

[By the last column:] π is trivial here (realization in functions on the Grassmannian).



Uncertain reading. The last cell of row M is heavily struck out. It may read $GL(n)$.

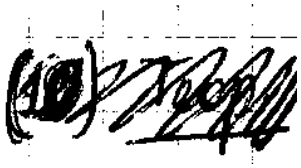
Original source details: GELFAND-P001170



Original source details: GELFAND-P001170

Uncertain reading. In row M_0 the manuscript shows $Sp(n-1)$, $GL(n-\frac{1}{2})$ and $Sp(n) \times Sp(n-1)$.

[struck out] ~~(10) Thm.~~

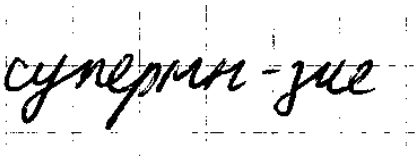


Uncertain reading. The struck-out heading appears to read “(10) Thm.”

Original source details: GELFAND-P001170

$$\tau = \mathrm{Ind}_{M_0}^{M} 1 \quad \Rightarrow \quad \text{table (7) reads } \mathrm{Ind}_{M_0}^{G} 1.$$

$\Rightarrow$ actually on L there arises a sheaf $\widetilde{L} \Rightarrow$ supermanif. and a super Grassmannian.



Uncertain reading. The abbreviation appears to mean “supermanifold”. The letters after the hyphen are uncertain.

Original source details: GELFAND-P001170

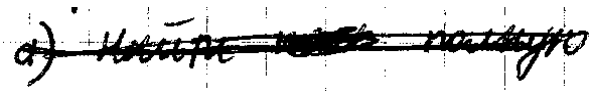
[struck out] There are many unusual symmetries. For example, the top row in (9) resembles the bottom row in (7).



Uncertain reading. Before “There are” there is a short illegible struck-out entry.

Original source details: GELFAND-P001170

I. M. G. Open question. On G/M a large symmetry group acts.



Undeciphered passage. The crossed-out line begins “a)”. It may read “find ... such a”. The middle is heavily crossed out.

Original source details: GELFAND-P001170

What other symmetries are there?

Nikiforov. (joint with Uvarov and Suslov)

$$\sigma(x)y'' + \tau(x)y' + \lambda y = 0, \tag{1}$$

σ — polyn. of degree 2, τ — polyn. of degree ≤ 1.

For

$$\lambda = \lambda_n = -n\tau' - \frac{n(n-1)}{2}\sigma'' \tag{2}$$

— orthog. polynomials.

Many spec. functions are contained here:

$$P_n^{(\alpha,\beta)}(x), \quad L_n^{\alpha}(x), \quad H_n(x)$$

(orthog. polynomials), depending on the degrees.

Gen. formula:

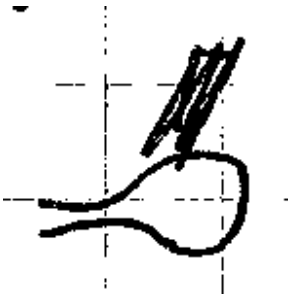
$$y_n(x) = \frac{A_n}{\rho(x)}\frac{d^n}{dx^n}[\sigma^n(x)\rho(x)]. \tag{3}$$

ρ — from the eq.: $(\sigma\rho)' = \tau\rho$.

[At the right:] The whole theory of spec. functions is in here — hypergeom., confluent hypergeom., and so on. We must take n nonintegral.

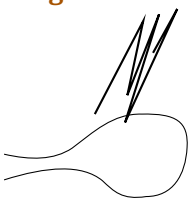
$$y(x) = \frac{A}{\rho(x)}\frac{1}{\Gamma(\alpha)}\int \frac{\sigma^{\alpha}(s)\rho(s)}{(s-x)^{\alpha+1}}. \tag{4}$$

α from a relation of type (2).



Original source details: GELFAND-P001170

Diagram and typed description.



There are heavy strokes at the upper branch. Their meaning and the direction of traversal are unclear.

(5) [crossed out with two intersecting strokes]

$$\frac{\sigma^{\alpha+1}(s)\rho(s)}{(s-x)^{\alpha+2}}.$$

Property of (1): the eq. for y' has the same form!! (with other params.)

Generalization: pass to a difference eq. keeping property (5).

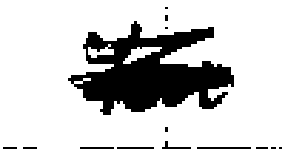
[In the left margin:]



Original source details: GELFAND-P001170

Uncertain reading. Struck out. The first letter may be “B”.

I. M. G. What is a multivariate hypergeom. function? I. M. G., Gindikin and others tried. [struck out] Cond.: that matrix entries be written in terms of it (via monodromy).



Uncertain reading. A short word before "Cond." is struck out.

Original source details: GELFAND-P001170

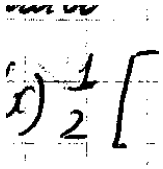


Uncertain reading. The abbreviation reads "Cond.:".

Original source details: GELFAND-P001170

N.

$$\tau(x)\frac{1}{2}\left[\frac{y(x+h)-y(x)}{h}+\frac{y(x)-y(x-h)}{h}\right]+\lambda y(x)$$
$$+\sigma(x)\frac{1}{2}\left[\frac{y(x+h)-y(x)}{h}-\frac{y(x)-y(x-h)}{h}\right]=0. \tag{6}$$



Uncertain reading. The factor before the second square bracket is written $\frac{1}{2}$.

Original source details: GELFAND-P001170

GELFAND-P001171

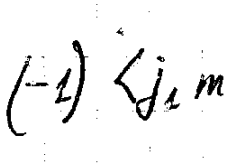
$\Rightarrow$ we obtain polynomials of a discr. variable.

In place of Jacobi polyns. we get Hahn polyns. $h_n(x)$ or $h_n^{(\alpha,\beta)}(x,N)$ $\left(\frac{2}{N}=h\right)$.

They are orthog. with a weight on a discr. set of points.

$\Rightarrow$ Clebsch–Gordan coeffs.

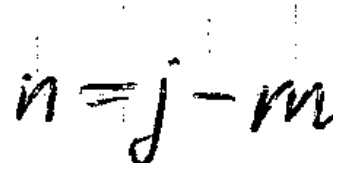
$$(-1)\quad \langle j_1 m_1\ j_2 m_2 \mid jm\rangle = \frac{\sqrt{\rho(x)}}{d_n} h_n^{(\alpha,\beta)}(x,N).$$



Uncertain reading. At the left is (-1). A small detached mark sits above the gap by the angle bracket. No exponent can be read.

Original source details: GELFAND-P001171

$$n = j - m, \quad x = j_2 - m_2, \quad N = j_1 + j_2 - m + 1,$$
$$\alpha = m - m', \quad \beta = m + m', \quad m' = j_1 - j_2.$$

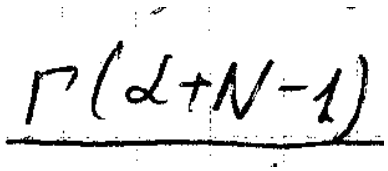


Uncertain reading. There is a dot above the left stroke of the initial n. Its purpose is unclear.

Original source details: GELFAND-P001171

Weight

$$\rho(x) = \frac{\Gamma(\alpha + N - 1)\Gamma(\beta + x + 1)}{\Gamma(x + 1)\Gamma(N - x)}.$$



Uncertain reading. The first factor in the numerator is written $\Gamma(\alpha + N - 1)$.

Original source details: GELFAND-P001171

In the book by Gelf.–Minl.–Shapiro — an expr. in terms of Jacobi polynomials. But the formula there is a polynomial depending on n (?)

(incorrect $\rho(x)$)



Uncertain reading. The word before $\rho(x)$ reads “incorrect”, possibly “incorrectly”.

Original source details: GELFAND-P001171

One can pass to a variable grid and write the diff. operator so that polynomials give polynomials of the corresp. degrees.

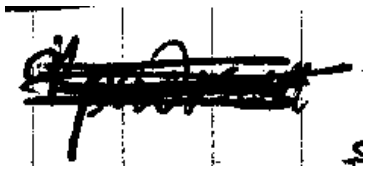
A quadr. grid gives Racah coeffs. in place of Clebsch–Gordan.

GELFAND-P001172

[source number: 66]

5/III–84 Gelfand Sem.

[struck out] ~~Fradkin~~ Berezansky Yu. M.



Uncertain reading. An entry before Berezansky’s name is struck out. It appears to read “Fradkin”.

Original source details: GELFAND-P001172

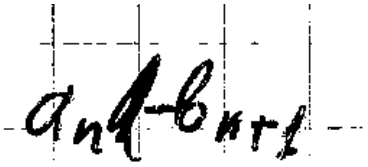
Applications of the ordinary inverse spectral problem in place of the inverse problem of scattering theory.

$$\begin{cases} \dot{a}_n = \frac{1}{2} a_n (-b_{n+1} \\ \dot{b}_n = a_n^2 - a_{n-1}^2 \end{cases} \tag{1}$$

(Toda latt.)

$n = 0, 1, \ldots$

$a_{-1} = 0.$



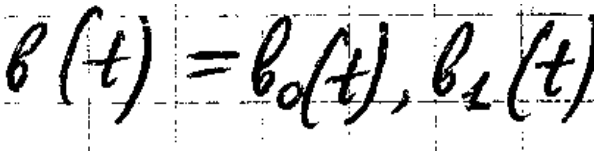
Uncertain reading. Before b_{n+1} are a retraced bracket-like stroke and a dash. The reading $(-$ is provisional.

Original source details: GELFAND-P001172

[At the right:]

$$a(t) = (a_0(t), a_1(t), \ldots)$$
$$b(t) = b_0(t), b_1(t), \ldots)$$

$a(0), b(0)$ are known.



Uncertain reading. There is no visible opening parenthesis before $b_0(t)$. The closing parenthesis is present.

Original source details: GELFAND-P001172

In the class of seqs.: $a(t) \in \ell^\infty$, $b(t) \in \ell^\infty$, $\forall t$ and $a_j(t) > 0$.

Prop. The soln. exists, is unique and is obtained by the following procedure:

Consider the Jacobi matr.

$$\begin{pmatrix} b_0(0) & a_0(0) & & 0 \\ a_0(0) & b_1(0) & a_1(0) & \\ \text{[struck out]} & a_1(0) & & \\ 0 & & & \end{pmatrix} = L(0). \tag{2}$$



Uncertain reading. The first entry in row three is struck out. The ending (0) is visible. Possible readings are $a_1(0)$ and $a_0(0)$.

Original source details: GELFAND-P001172

$$\mu(t) = \left[\int_{-\infty}^{\infty} e^{\lambda t} \, d\rho(\lambda, 0) \right]^{-1}; \tag{3}$$

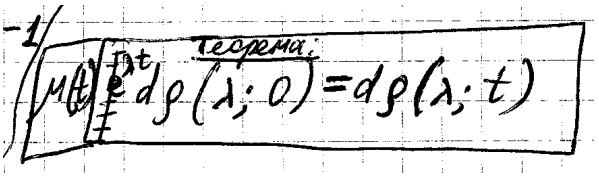


Diagram and typed description.

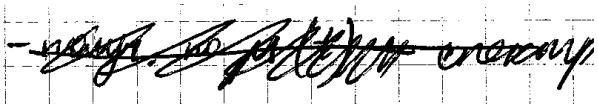
$$\mu(t) e^{\lambda t} \, d\rho(\lambda; 0) = d\rho(\lambda; t)$$

Theorem: under the evolution of $L(t)$ according to (1).

Original source details: GELFAND-P001172

$\rho(\lambda, 0)$ — spectr. function of the Jacobi matr. $L(0)$.

$\rho(\lambda, t)$ — [struck-out phrase] spectr. function of $L(t)$, its entries $a_n(t), b_n(t)$.



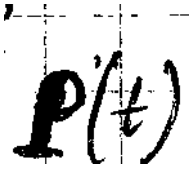
Original source details: GELFAND-P001172

Undeciphered passage. Several words before "spectr. function" are struck out. Writing resembling $\rho(t)$ is visible beneath the strokes.

I. M. G. (explanation)

$$\int_0^\infty t^n \, d\rho(t) = a_n \qquad \text{(suppose given)}$$

$$\int_0^\infty P'(t) \, d\rho(t) \qquad \text{is known too}$$



Original source details: GELFAND-P001172

Uncertain reading. A short stroke appears between the retraced P and the parenthesis. It may be a prime.

$\Rightarrow (P, Q)_{L^2_\rho}$ is known.

Condition: nonnegativity of this inner prod..

We obtain a Hilbert space and multiplication in it by $t \Rightarrow$ orthogonalizing, we construct an orthog. basis $P_n(t)$.

$$tP_n(t) = a_{n-1}P_{n-1}(t) + b_nP_n(t) + a_nP_{n+1}(t).$$

(there is nothing below P_{n-1} because the operator is self-adjoint).

$\{a_n\} \rightsquigarrow$ Jacobi matr..

Jacobi matr. $\rightsquigarrow$ moment problem.

There is a spectr. decomposition

$$\underline{(E(\lambda)1, 1) = \rho(\lambda)}$$

— this function solves the moment problem.

Uniqueness and nonuniqueness are deep matters.

Essential self-adj. does not always hold!

ρ nonunique $\Leftrightarrow$ the op. L has deficiency indices $\Leftrightarrow$

$\Leftrightarrow a_n$ does not grow too fast.

Connection with quasianalyt. classes and so on.

~~Spectr. decomp. of a self-adj. operator~~

Yu. M. B. a, b are bounded $\Rightarrow$ self-adjointness.

$$a_n(t) = \int_{-\infty}^{\infty} \lambda P_n(\lambda; t) P_{n+1}(\lambda, t)\, d_\lambda \rho(\lambda, t)$$

$$b_n(t) = \int_{-\infty}^{\infty} \lambda P_n^2(\lambda, t)\, d_\lambda \rho(\lambda, t)$$

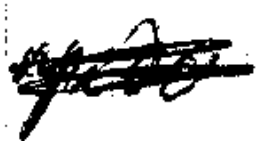
Difference: no asymptotics at infinity!!

Conditions: e^{tA} makes sense for the original Jacobi matr..

Fradkin E. S. How can one build a unified theory of all interactions?

Current point of view:

$\exists$ a broad group and all elem. [struck out] particles are embedded in one representation of it. (at high energies)



Original source details: GELFAND-P001173

Uncertain reading. A word between "elem." and "particles" is heavily struck out.

Basic particle. The others are its projections.

Breaking of the group [inserted above: (symmetry)] must be spontaneous.

It breaks down at low energies.

Gravitons will come in too.

Space/time remains

there will be a Riem. metric too.

The group will be a supergroup

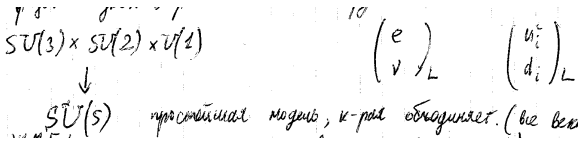
(superext. of the Poincaré group or of the conformal group).

1. Unification of the electromagn. and weak interactions (Weinberg, Salam, Glashow)

see the review: Phys. Reports, 1981, Lancaster.

The group $SU(2) \times U(1)$ unifies the weak and electromagn. interactions.

All low-energy physics is described by the group



Original source details: GELFAND-P001173

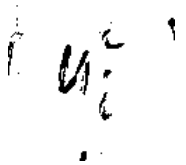
Diagram and typed description.

$SU(3) \times SU(2) \times U(1)$ $\begin{pmatrix} e \\ \nu \end{pmatrix}_L$ $\begin{pmatrix} u_i \\ d_i \end{pmatrix}_L$ — quarks

$\downarrow$

$SU(5)$

[To the right of $SU(5)$:] the simplest model that unifies. (all vector particles)



Uncertain reading. There is a separate hook above the index i of u_i. Its purpose is unclear.

Original source details: GELFAND-P001173

I. M. G. (see also Coleman's review in "gauge fields")

Supergroup — ext. of the Poincaré group Golfand, Likhtman

review Ferrari, Fayet Phys. Report, 1978.



Uncertain reading. The first surname in the manuscript reads "Ferrari".

Original source details: GELFAND-P001173

$$[M, M] = M \qquad \text{(Lorentz)}$$
$$[P, M] = P \qquad \text{(Poincaré)}$$
$$[Q^\alpha, \bar{Q}^\beta] = i(\gamma^\mu \bar{c})^\alpha_\beta P_\mu.$$

GELFAND-P001174

[source number: 67]

Gravitation is a gauge th. of the Poincaré grp..

Supergravity: review: Nieuwenhauzen

Phys. Report, vol. 68, $N = 4$ (1981)

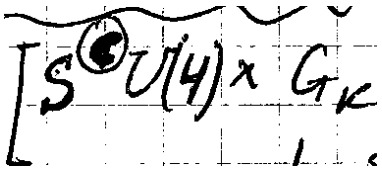
There are no other reviews.

Large unification (without grav.) — 24-dim. group in which all particles are unified.

(Grand unification (with gravitation))

Compare large unif. with the group:

$$\left[S^{(?)}U(4) \times \underbrace{G_k}_{\text{gauge}} \right.$$



Uncertain reading. A dark mark between S and $U(4)$ is circled. The reading is unresolved.

Original source details: GELFAND-P001174

rem. [struck out] the group that solves grand unif.



Uncertain reading. An illegible mark is crossed out after "rem."

Original source details: GELFAND-P001174

$G_k = E_8$ [inserted above: Fundam. repr.] (coincides with the adj.)

[At the right:] $E_8 \supset SU(5)$ as a subgroup.

Lagrangian: Letters to JETP vol. 32, 575 (Fradkin, Konshtein)

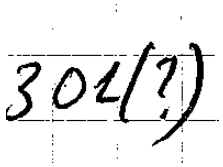
($SU(5)$ — "Quarks-80" (Fradkin))

E_8-model — is finite — has no divergences at all!!! —

— at all loops

Vainshtein, Shifman,

Fradkin, Tseytlin — Phys. Letters, 135 (1984), p. 301(?)



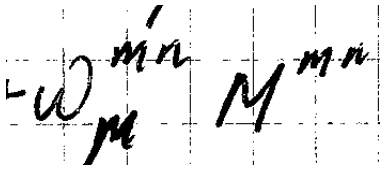
Original source details: GELFAND-P001174

Uncertain reading. The last digit resembles 1, and the number is followed by (?).

Gauging is an art (there is no technique yet)

From the group G we construct a connection V_μ (or vect. field)

$$V_\mu = e^a_\mu P_a + \omega^{mn}_\mu M^{mn} + \bar{\psi}^\alpha_\mu \Theta_\alpha.$$



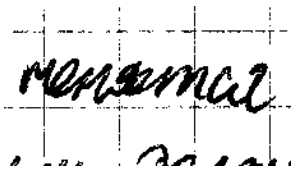
Original source details: GELFAND-P001174

Uncertain reading. In the first formula mn is written above M. In the next formula the same indices are below it.

$$\Lambda = \epsilon^a P_a + \lambda^{mn} M_{mn} + \bar{\epsilon}^\alpha \Theta_\alpha \qquad \text{— infinitesimal}$$

$$\left.\begin{matrix} \omega_{mn} & M_{mn} \text{ — gener.,} & \omega_{mn} \text{ — coeffs.} \\ e^a_\mu & P_\mu \end{matrix}\right]$$

To gauge: odd group and so on.



Original source details: GELFAND-P001174

Uncertain reading. The word before "group" is provisionally read as "odd". Another possible reading is "a certain".

BUT: translations must be replaced by something local.

Grav.: theory of particles with spins $\begin{pmatrix} 2 \\ 3/2 \end{pmatrix}$

gravitation is spin 2.

I. M. G. believes that space is neither 3- nor 4-dimensional, but ∞-dimensional.

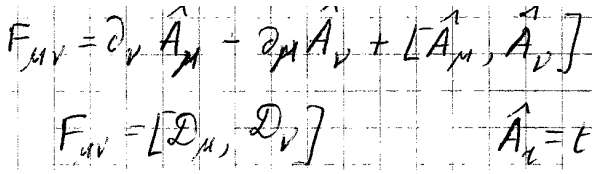
GELFAND-P001175

12/III–84 Gelfand Sem.

Fradkin E. S.

$$F_{\mu\nu} = \partial_\nu \hat{A}_\mu - \partial_\mu \hat{A}_\nu + [\hat{A}_\mu, \hat{A}_\nu]$$

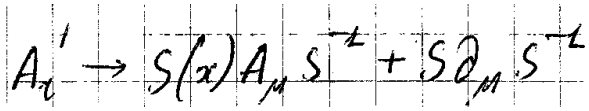
$$F_{\mu\nu} = [D_\mu, D_\nu] \qquad \hat{A}_\tau = t^a A^a_\tau$$



Uncertain reading. The index order of both formulas for $F_{\mu\nu}$ follows the manuscript.

Original source details: GELFAND-P001175

$$A'_\tau \longrightarrow S(x) A_\mu S^{-1} + S \partial_\mu S^{-1}$$



Uncertain reading. The index on the left resembles τ. On the right μ is written twice.

Original source details: GELFAND-P001175

How to introduce mass?

Adding $m^2 A^2_\mu$ does not work (no gauge invariance)

[At the right, with an arrow to the parenthesis:] and it is also nonrenormalizable.

It must be by [struck out] spontaneous symmetry breaking. //



Uncertain reading. An illegible mark is struck out in the phrase meaning “by”.

Original source details: GELFAND-P001175

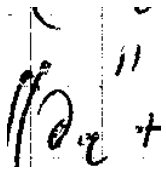
I. M. G.: we must state precisely what renormalizability is.

F. Higgs introduced a scalar (add.) field.

$$(D_\tau \varphi)^2 + \dots$$

$$\left((\partial''_\tau + t^\alpha \hat{A}_\mu)\varphi\right)^2$$

(covar. deriv.)



Uncertain reading. Two short strokes are written to the upper right of ∂_τ. Their purpose is unclear.

Original source details: GELFAND-P001175

If $\langle\varphi\rangle \neq 0$ (average over the vacuum), then

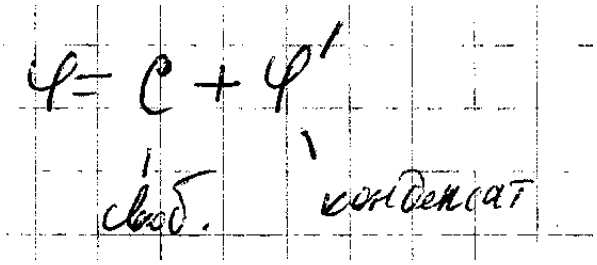


Diagram and typed description.

$\varphi = C + \varphi'$

free — condensate

Original source details: GELFAND-P001175

We obtain $(t^\alpha A_\mu \varphi(0))^2$

One can write:

$$(D_\tau \varphi)^2 + m_1^2 \varphi^2 + \lambda \varphi^4$$

We vary:

$$D_\tau^2 \varphi(0) + m^2 \varphi(0) + 4\lambda \varphi(0)^3 = 0$$

Simple soln. 0

There can also be a 2nd soln.:

$$m^2 \varphi(0) + 4\lambda \varphi(0)^3 = 0,$$

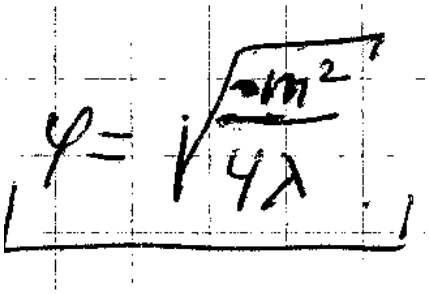


Diagram and typed description. $\varphi = \sqrt{\dfrac{-m^2}{4\lambda}}$

Original source details: GELFAND-P001175

$\Rightarrow$ a mass term for all the particles will arise.

GELFAND-P001176

[source number: 71]

Lagr.

$$\mathcal{L} = F_{\mu\nu}^2 + (D_\mu \varphi)^2 + m_1^2 \varphi^2 + \lambda \varphi^4$$

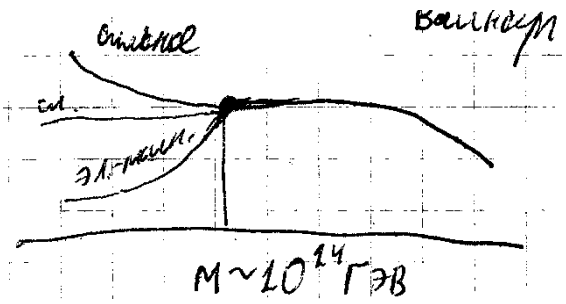
We want: the remaining [abbr.]$^?$ massless grp. to be

Uncertain reading. remaining [abbr.]



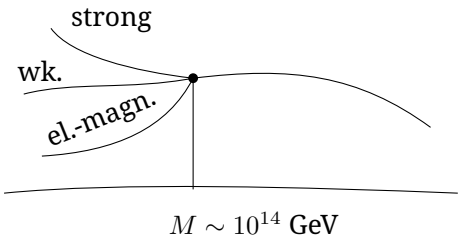
Original source details: GELFAND-P001176

$$SU(3) \times \underbrace{SU(2) \times U(1)}_{\text{Weinberg–Salam}}$$



Original source details: GELFAND-P001176

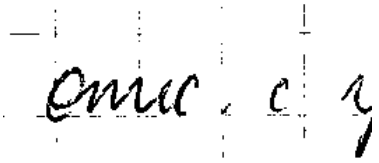
Diagram and typed description.



Sketch of the convergence of the interaction couplings

— · —

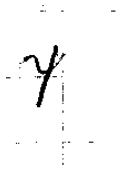
Abbreviated heading: list / description[?] with gravity:



Original source details: GELFAND-P001176

Uncertain reading. Abbreviated heading: list / description

				$N = 8$
$g_{\mu\nu}$	— 1 particle			1
ψ	N particles	(from $SO(N)$)	(spin $3/2$)	8
A_μ	$SO(N)$	(vector part.)		28
χ	·		(spin $1/2$)	56
φ	— scal. particles			70



Original source details: GELFAND-P001176

Uncertain reading. The symbol on the spin-$1/2$ row is tentatively read as χ.

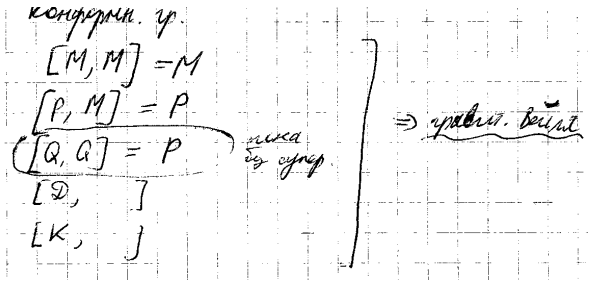
max spin determ. $N/4$ (taking chirality into account) $\Rightarrow N = 8$.

— · —

In 11-dimensional sp. there arises a multiplet with $N = 1$ and there are no conformal anomalies.

— · —

conform. grp.



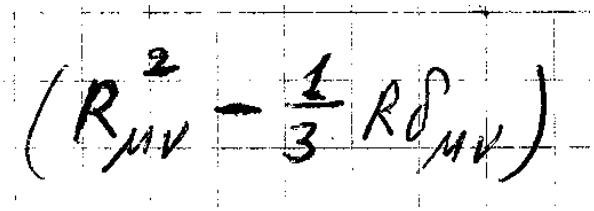
Original source details: GELFAND-P001176

Diagram and typed description.

$[M, M] = M$
$[P, M] = P$
$[Q, Q] = P$ for now without super
$[D, \ ]$
$[K, \ ]$
$\Rightarrow$ Weyl gravit.

Conformal group brackets and the note beside the circled line

$$\left(R^2_{\mu\nu} - \frac{1}{3} R\delta_{\mu\nu}\right)$$



Uncertain reading. The second term reads as $-\frac{1}{3} R\delta_{\mu\nu}$.

Original source details: GELFAND-P001176

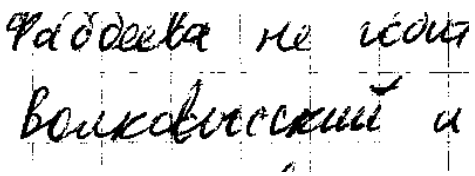
The simplest problem: compute the integral

$$\int e^{i \int d^4x\, R}\, dg_{\mu\nu}$$

highly degenerate. quantization?

Faddeev's method does not work.

Fradkin, Vilkovisskii [spelling uncertain]$^?$ and others —



Uncertain reading. Vilkovisskii [spelling uncertain]

Original source details: GELFAND-P001176

[Inserted above the line: to] gravity and others.

An answer arises, not expr. through det —

— the Lie bracket is not closed (closed only on the extremals).

— · —

GELFAND-P001177

19/III–84 yr. Gelfand Sem.

Leites.

Lie superalgebras.

$gl(n|m)$

Supersp. $V = V_{\bar{0}} \oplus V_{\bar{1}}$, grad. $\{\bar{0}, \bar{1}\} = \mathbb{Z}_2$, dim $V = (n|m)$, dim $V_{\bar{0}} = n$, dim $V_{\bar{1}} = m$.

Hom(V, W) — supersp.;

(W — another supersp. $W = W_{\bar{0}} \oplus W_{\bar{1}}$)

Hom$(V, W)_{\bar{0}}$ — pres. parity;

$$\backslash\backslash$$
$$\left\{\begin{pmatrix} A & 0 \\ 0 & D \end{pmatrix}\right\}$$

Hom$(V, W)_{\bar{1}}$ — changing parity.

$$//$$
$$\begin{pmatrix} 0 & B \\ C & 0 \end{pmatrix}$$

Lie superalg.:

$$[x, y] = -(-1)^{\bar{x}\bar{y}}[y, x]$$
$$[x, [y, z]] = [[x, y], z] + (-1)^{\bar{x}\bar{y}}[y, [x, z]]$$

[In the bracket on the left before x, crossed out: illegible mark?]

Uncertain reading. illegible mark



Original source details: GELFAND-P001177

(Quillen's rule:)

if letters of parity p and q, are permuted, the sign arises $(-1)^{pq}$.

Superalg.: $A = A_{\bar{0}} \oplus A_{\bar{1}}$

$$\begin{matrix} m\colon A \otimes A \to A \\ p(m) = \bar{0} \end{matrix} \quad // \quad \text{mult. must be even}$$

Assoc. superalg. $\rightsquigarrow$ Lie superalg.

$$[a, b] = ab - (-1)^{\bar{a}\bar{b}}ba$$

[Crossed out: illegible writing?]

Uncertain reading. illegible writing



Original source details: GELFAND-P001177

$gl(n|m)$ — superalg. obt. from $\mathrm{Hom}(\mathbb{R}^{n/m}, \mathbb{R}^{n/m})$.

$$sl(n|m) = \{a \mid a \in gl(n|m),\ \mathrm{str}\, a = 0\},$$

$$\mathrm{str} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \mathrm{tr}\, A - \mathrm{tr}\, D$$

(Euler char. — supertrace)

Classif. in the ordinary case:

[Inserted below the first word: of algebras]

$$gl,\ sl,\ o,\ sp, \quad w,\ S,\ H,\ K.$$

$$osp(n|2m) = \{X \in gl(n|2m) \mid X^{st} B_{n,2m} + B_{n,2m} X = 0\},$$

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}^{st} = \begin{pmatrix} A^t & -C^t \\ B^t & D^t \end{pmatrix}$$

(one can also use $\begin{pmatrix} A^t & C^t \\ -B^t & D^t \end{pmatrix}$, this is equivalent)

GELFAND-P001178

$$B_{n,2m} = \left(\begin{array}{c|cc} 1_n & 0 & \\ \hline 0 & 0 & 1_m \\ & -1_m & 0 \end{array}\right) \qquad \text{(even form pres.)}$$

matrix — even.

(matrix of an even symm. or odd skew-symm. form)

I.M.G. isomorphic notation:

$$osp(2n|2m) = \left(\begin{array}{cc|cc} 0 & 1_n & & \\ 1_n & 0 & 0 & \\ \hline & & 0 & 1_m \\ 0 & & -1_m & 0 \end{array}\right)$$

$osp(2n+1|2m)$

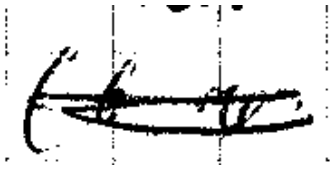
— · —

L.

$$\Pi(n) = \{X \in gl(n|n) \mid X^{st} J_{2n} + (-1)^{\bar{X}} J_{2n} X = 0\}$$

$$= \left\{ \begin{pmatrix} A & B \\ C & -A^t \end{pmatrix} \,\middle|\, \begin{array}{l} B = B^t \\ C = -C^t \\ A \in gl(n) \end{array} \right\}.$$

[Crossed out below the formula: illegible?]



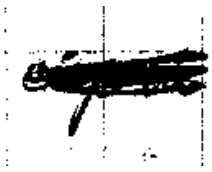
Uncertain reading. illegible

Original source details: GELFAND-P001178

(Π — in honor of Palamodov)

$$J_{2n} = \begin{pmatrix} 0 & 1_n \\ -1_n & 0 \end{pmatrix}$$

(odd [Crossed out: illegible word?] form pres.)



Uncertain reading. illegible word

Original source details: GELFAND-P001178

matrix odd.

$$q(n) = \{X \in gl(n|n) \mid [X, J_{2n}] = 0\} = \left\{ \begin{pmatrix} A & B \\ B & A \end{pmatrix} \right\}$$

$$\downarrow \quad q \cdot \mathrm{tr}$$
$$\mathrm{tr}\, B = \mathrm{str}(X \cdot J_{2n}) \qquad (q \text{ from queer, strange})$$

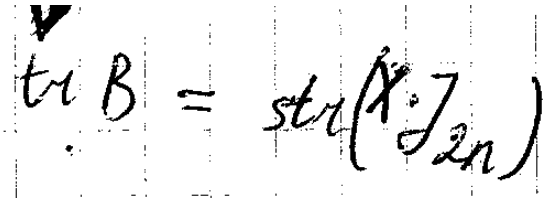


Diagram and typed description. The manuscript equation for the q-trace

Original source details: GELFAND-P001178

I.M.G. asks: what does the theor. on Jordan form look like?

[Crossed out: sq?]



Uncertain reading. sq?

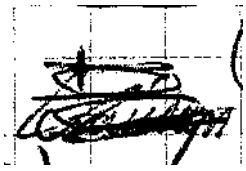
Original source details: GELFAND-P001178

$sq(n)$ also exists.

Projectivizations exist.

$$AB_3 = \underbrace{sl(2) \oplus O(7)}_{(AB_3)_{\bar{0}}} \oplus (AB_3)_{\bar{1}}$$

[Two short lines crossed out under the first term: illegible fragments?]



Uncertain reading. illegible fragments

Original source details: GELFAND-P001178

For $sl(2)$ there is a stand. [Inserted above: 2-dimensional] rep. V

$O(7)$ has the rep. spin_7 (stand.)

$$V \otimes \mathrm{spin}_7 \text{ — rep. } sl(2) \oplus O(7)$$

$$\mathcal{D}(\alpha) = osp(4|2;\alpha)$$

$$\mathcal{D}(\alpha)_{\bar{0}} = O(4) \oplus sp(2) = sl(2)_1 \oplus sl(2)_2 \oplus sl(2)_3$$

$$\mathcal{D}(\alpha)_{\bar{1}} = V_1 \otimes V_2 \otimes V_3$$

V_i — stand. rep. $sl(2)_i$

$$[v_1 \otimes v_2 \otimes v_3,\ u_1 \otimes u_2 \otimes u_3] = \sum_{\substack{(i,j,k)= \\ =\sigma(1,2,3)}} \lambda_k \underbrace{(v_i, u_i)_i}_{\text{skew-symm. prod.}} (v_j, u_j)_j \underbrace{v_k \cdot u_k}_{\text{symm. prod.}} .$$

Jacobi ident. satisf. when: $\lambda_1 + \lambda_2 + \lambda_3 = 0$.

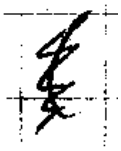
GELFAND-P001179

[source number: 79]

Penkov I.

$$G = SL(T^{n/m}) \qquad n \neq m$$

[Below T, a downward stroke labelled: compl.] [Mark crossed out at right: illegible?]



Uncertain reading. illegible

Original source details: GELFAND-P001179

$B =$ max. solv. closed subsupergroup, all of whose irred. finite-dimensional rep. have dimension $1|0$ or $0|1$.

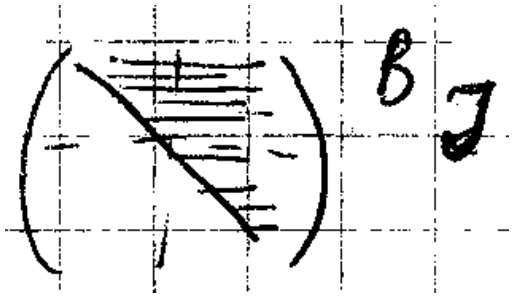
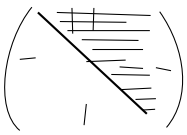


Diagram and typed description.



B_J

Matrix sketch B_J. The meaning of some short strokes is unclear.

Original source details: GELFAND-P001179

$$J \in \mathcal{J} = \{\text{permutations of } n+ \text{ and } m-\}$$

$$G/B_J = \underset{\text{(flags)}}{Fl}\,(J, T^{n/m})$$

$$\dim G/B_J = \frac{n(n-1)}{2} + \frac{m(m-1)}{2} \mid mn$$

$$\begin{array}{c} \operatorname{Pic} G/B \\ \| \\ \operatorname{Mor}(H, G_m) \end{array}$$ — Picard grp.

[Under G_m, a downward stroke: mult. group] [Crossed out: illegible?] $\mathbb{C}^*$



Uncertain reading. illegible

Original source details: GELFAND-P001179

$$\mathrm{Mor}(H, G_m) \xrightarrow{\sim} \mathrm{Pic}^G G/B_J$$
$$\stackrel{df}{=} \left\{ \begin{array}{c} \text{grp. of sheaves of rank } 1|0 \\ \text{with } G\text{-linearization} \end{array} \right\}.$$

[Under H, a downward stroke: Cartan subgrp. of the even part]

Theorem

$$\chi \in \mathrm{Mor}(H, G_m), \quad (\chi, \varepsilon_i - \varepsilon_j) \neq 0,$$
$$(\chi, \delta_i - \delta_j) \neq 0, \quad (\chi, \varepsilon_i - \delta_j) \neq 0.$$

Then $\forall J\ H^i(\mathcal{O}_J(\chi - \rho_J)) = 0$ for $i \neq l(w)$.

$$H^{l(w)}(\mathcal{O}_{J_1}(\chi - \rho_{J_1})) = H^{l(w)}(\mathcal{O}_{J_2}(\chi - \rho_{J_2}))$$
$$= V_{J_1}(\mu - \rho_{J_1}) = V_{J_2}(\mu - \rho_{J_2})$$

$\Rightarrow$ highest-weight rep. are obtained in cohomology.

$(\chi, \varepsilon_i - \delta_j) \neq 0$, but $(\chi, \alpha_0) = 0 \Rightarrow \forall J$ the sheaf $\mathcal{O}_J(\chi - \rho_J)$ is acyclic.

GELFAND-P001180

26/III–84. Gelfand Sem.

Penkov.

$$G = SL(n|m), \qquad G/B_J = Fl(J, T)$$

T — stand. rep.,

J — flag index

$$\mathrm{Pic}^G_J = \mathrm{Pic}^G(G/B_J)$$

Claim 1 $\mathrm{Pic}^G_J =$ characters of H

Thm. Let χ — a character, χ-regular; [Crossed out: illegible marks?]



Uncertain reading. illegible marks

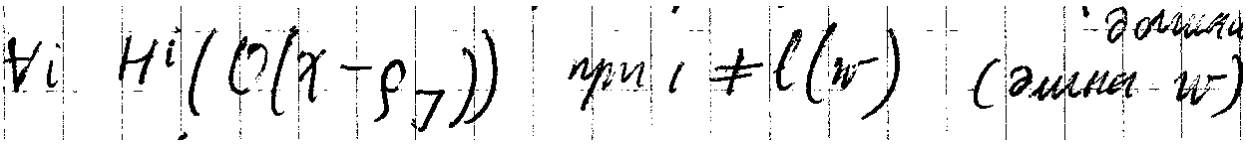
Original source details: GELFAND-P001180

$\exists \mu = w(\chi)$

[Under μ, a stroke labelled: dominant]

Then

$$\forall i \quad H^i(\mathcal{O}(\chi - \rho_J)) \quad \text{for } i \neq l(w) \qquad (\text{length of } w)$$



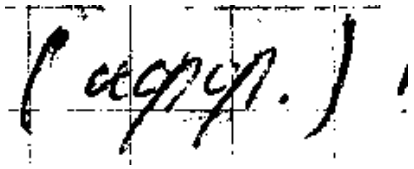
Original source details: GELFAND-P001180

Uncertain reading. There is a gap between $H^i(\mathcal{O}(\chi - \rho_J))$ and "for".

$$H^{l(w)}(\mathcal{O}(\chi-\rho_J))=V_J(\mu-\rho_J).$$

And there is a D-module version of this theorem.

Affine flags: also with area elements (aff.?[?]) on $\forall$ plane of the flag



Uncertain reading. aff.?

Original source details: GELFAND-P001180

T — lin. supersp.,

$$T=T^{n/m}=T^{n/0}_{\text{even}}\oplus T^{0/m}_{\text{odd}}$$

[Small mark crossed out above T: illegible[?]]



Uncertain reading. illegible

Original source details: GELFAND-P001180

Type.

$$T^{1/0}\subset T^{2/0}\subset\dots$$
$$\subset T^{1/1}\subset\dots$$



Uncertain reading. The last superscript looks like $1/1$.

Original source details: GELFAND-P001180

GELFAND-P001181

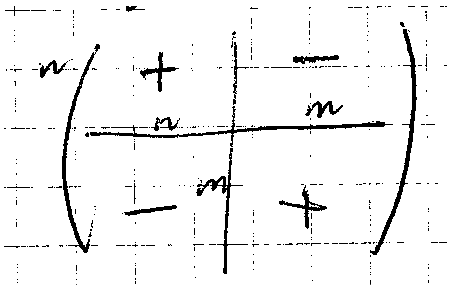
[source number: 87]

Underlying var.: prod. of two flag var.
(even part) (indep. of type)

$$Fl(J,T)_{\text{even}} = Fl(1,2,\dots,T^{n/0}) \times Fl(1,2,\dots,T^{0/m})$$

$$\dim Fl(J,T^{n/m}) = \frac{n(n-1)+m(m-1)}{2} \mid mn$$

(From the equality: $G/B_J = Fl(J,T)$).



Original source details: GELFAND-P001181

Diagram and typed description.

$$n\left(\begin{array}{c|c} + & - \\ n & m \\ \hline - & + \\ & m \end{array}\right)$$

GELFAND-P001182

2/IV–84 yr. Gelfand Sem.
Fuks.



Original source details: GELFAND-P001182

Uncertain reading. The name reads as “Fuks”, with overlapping middle strokes.

$\widehat{sl(2)}$ — algebra, gen. by 6 generators f_j, e_j, h_j $(j = 1, 2)$
K.-M.

$$\begin{array}{c} \| \\ N_- \oplus H \oplus N_+ \end{array}$$

$$[e_i, f_j] = \delta_{ij} h_i$$
$$[e_1[e_1[e_1, e_2]]] = 0$$
$$[e_2[e_2[e_2, e_1]]] = 0$$

and something else.

$$[h_1, e_2] = -2e_2, \quad [h_2, e_1] = -2e_1,$$
$$[h_1, e_1] = 2e_1, \quad [h_2, e_2] = 2e_2$$

and the same fmls. with e replaced by f
and with the sign chang.,

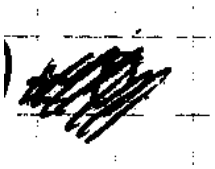
$$[h_1, h_2] = 0$$

I.M.G. This alg. almost coinc. with the algebra

$$\begin{pmatrix} a(t) & b(t) \\ c(t) & d(t) \end{pmatrix}, \qquad a + d = 0, \qquad t \in S^1.$$

— · —

F. Claim. Verma modules over it depend on 2 compl. parameters. $M(\lambda_1, \lambda_2)$ [Crossed out: illegible fragment?]



Uncertain reading. illegible fragment

Original source details: GELFAND-P001182

GELFAND-P001183

N_- — words, comp. of f_j.

N_+ — words, comp. of e_j.

$\dim H < \infty$.

Verma module — rep. in which $\exists$ a gen. vector for which

$$\begin{cases} N_- v = 0 \\ h_j v = \lambda_j v \end{cases}$$

$e_1 v, \quad [e_1, e_2]v, \ldots$

May turn out to be reducible for certain λ_1, λ_2.

Thm. It is reducible $\Longleftrightarrow \exists w \in M(\lambda_1, \lambda_2)$; $f_i w = 0$.

It is also an eigenvector (sing. vector).

[In the left margin: then $\exists w \nparallel v$.]

(Kac — The spelling looks like "Katsdan", probably referring to Kazhdan.[?];



Uncertain reading. The spelling looks like "Katsdan", probably referring to Kazhdan.

Original source details: GELFAND-P001183

∞-dim. version of the theor. of Bernsh. — Gelf. — Gelf.)

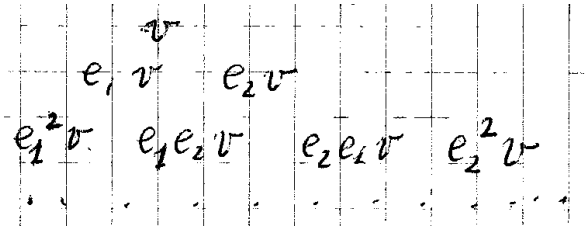


Diagram and typed description.

$$\begin{array}{ccccccccccc} & & & & v & & & & & & \\ & & & e_1 v & & & e_2 v & & & & \\ & & e_1^2 v & & & e_1 e_2 v & & & e_2 e_1 v & & & & e_2^2 v \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \end{array}$$

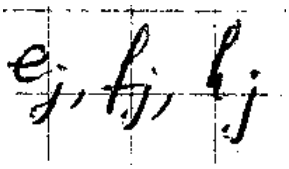
Original source details: GELFAND-P001183

Thm. $M(\lambda_1, \lambda_2)$ is reducible $\Longleftrightarrow \lambda_1, \lambda_2$ satisfy one of the eqs.

$$\begin{aligned} &(1) \quad (k+1)(\lambda_1 + 1) + k(\lambda_2 + 1) = n, \quad k \geq 0,\ n \geq 0. \\ &(2) \quad \lambda_1 \longleftrightarrow \lambda_2 \\ &(3) \quad \lambda_1 + \lambda_2 + 2 = 0 \end{aligned}$$

Problem. (I.M.G.) Find the Casimir op. of this algebra.

(polynomials (noncommut.) in the letters e_j, f_j, l_j



Uncertain reading. The last sign resembles l_j, possibly h_j.

Original source details: GELFAND-P001183

Idea of the constr.:

$$w = e_1^{n+\ell t} \dots e_2^{n+t} e_1^n e_2^{n-t} e_1^{n-2t} \dots e_1^{n-\ell t} v$$
$$t \in \mathbb{C} \ (!!)$$

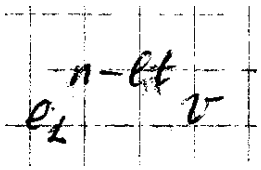
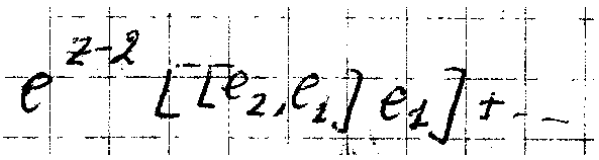


Diagram and typed description. The last factor has a subscript that reads 1.

Original source details: GELFAND-P001183

But: it turns out that this element lies in the enveloping algebra.

$$[e_2, e_1^z] = z e_1^{z-1} [e_2, e_1] + \frac{z(z-1)}{2} e^{z-2} [[e_2, e_1] e_1] + - -$$



Uncertain reading. There is no visible subscript on the base of e^{z-2}. The outer bracket is written without a comma, and two strokes follow the plus sign.

Original source details: GELFAND-P001183

Beilinson.

Hodge structures.

$n \in \mathbb{Z}$.

pure Hodge struct. of weight n — $H_{\mathbb{R}}$, [Under $H_{\mathbb{R}}$: real vect. sp.]

$$H_{\mathbb{C}} := H_{\mathbb{R}} \otimes \mathbb{C} = \bigoplus_{p+q=n} H^{p,q}, \qquad \bar{H}^{pq} = H^{qp}.$$

[Under $H_{\mathbb{C}}$, a stroke leading to “its c”]

Hodge filtration:

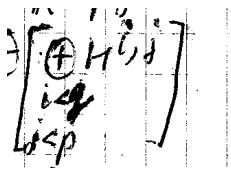
$$F^i H = \bigoplus_{p \geq i} H^{p,q}.$$

The structure is recovered from it:

$$H^{p,q} = F^p H \cap \overline{F^q H}.$$

Mixed Hodge struct. $H_{\mathbb{R}}$ and decomp. $H_{\mathbb{C}} = \bigoplus H^{p,q} \dots$, satisf. the conditions

$$(1) \qquad \bar{H}^{p,q} \subset H^{q,p} \oplus \left[\bigoplus_{\substack{i<q \\ j<p}} H^{i,j} \right].$$



Uncertain reading. The q in the bound $i < q$ is heavily retraced.

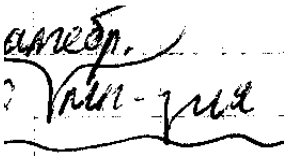
Original source details: GELFAND-P001183

Hereafter Hodge struct. means a mixed Hodge struct..

GELFAND-P001184

[source number: 91]

On the cohom. of any [Inserted above: algebr.]



Uncertain reading. var. [source abbr. uncertain]

Original source details: GELFAND-P001184

over $\mathbb{C}$ (not nec. smooth or compact) there is a mixed Hodge struct. (construction)

Split Hodge struct.: if $\bar{H}^{pq} = H^{qp}$.

This is a direct sum of pure ones (easy exercise). (the weight is not fixed at n).

Again — filtrations (Hodge filtr.).

$$F^i := \bigoplus_{p \geq i} H^{p,q} \qquad (\text{decr.})$$

Mixed Hodge struct. form an abelian category.

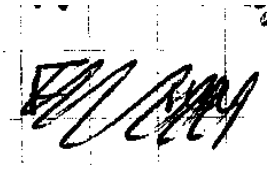
$$W_n := \bigoplus_{p+q \leq n} H^{p,q} \qquad \begin{matrix}(\text{weight filtr.}) \\ \text{incr.}\end{matrix}$$

Ex.* 1) This pair of filtrations uniquely determines the Hodge struct..

2) For a pair of filtr. to determ. a mixed Hodge struct., it is nec. and suff. to [Inserted above: fulf.] the foll. conditions:

A. W is real (=complexif. of a real filtration).

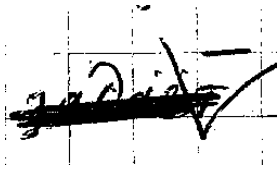
B. [Crossed out: $F \ldots$; the rest is illegible beneath the cancellation?]



Uncertain reading. $F \ldots$; the rest is illegible beneath the cancellation

Original source details: GELFAND-P001184

$(\mathrm{Gr}_n^W H, \mathrm{Gr}_n^W F)$ [Crossed out: defines? A short dash is visible alongside.?]



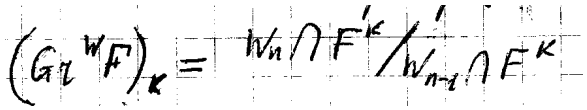
Uncertain reading. defines? A short dash is visible alongside.

Original source details: GELFAND-P001184

— pure Hodge struct. of weight n.

[Bracket under $\mathrm{Gr}_n^W F$:] this is a filtration on $\mathrm{Gr}_n^W H = W_n/W_{n-1}$.

$$(\mathrm{Gr}^W F)_k = \frac{W_n \cap F^k}{W_{n-1} \cap F^k}.$$



Original source details: GELFAND-P001184

Diagram and typed description. The left side is written $(\mathrm{Gr}^W F)_k$.

— · —

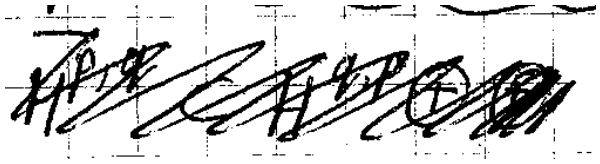
[Crossed out: Problem.; inserted above: Ex.] nontrivial.

(see Deligne, Hodge Theory II. Coll. of trans. “Matem.”, 1973)

— · —

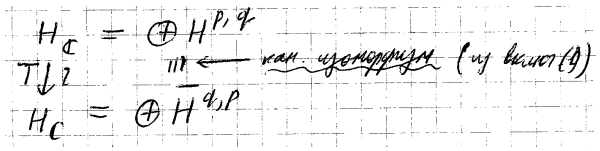
Arithmetic monodromy.

[Crossed out: $\bar{H}^{pq} \subset H^{qp} \oplus$ End of the crossed-out formula: $\bigoplus ?^{?}$]



Original source details: GELFAND-P001184

Uncertain reading. End of the crossed-out formula: $\bigoplus ?$



Original source details: GELFAND-P001184

Diagram and typed description.

$$\begin{array}{ccc} H_{\mathbb{C}} = \bigoplus H^{p,q} & & \\ T \downarrow 2 & \quad \| \leftarrow & \text{can. isomorphism (from incl. (1))} \\ H_{\mathbb{C}} = \bigoplus \bar{H}^{q,p} & & \end{array}$$

The meaning of the short strokes between the rows is uncertain.

$\bar{H}^{pq} \simeq H^{qp}$ (there is a can. isomorphism.), [Below: from (1)] T — the direct sum of these isomorphisms.

T is called the arithmet. monodromy operator.

$$T(H^{p,q}) \subset H^{pq} \oplus \bigoplus_{\substack{i<p \\ j<q}} H^{i,j}.$$

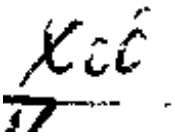
T is strictly unipotent (and even the identity on W_n/W_{n-2}) [Below: from the def.]

Claim.

The Hodge struct. is split $\iff T = \mathrm{Id}$.

GELFAND-P001185

Structures



Original source details: GELFAND-P001185

Uncertain reading. Hod. [source abbr. uncertain]

on [Crossed out: illegible fragment?]



Original source details: GELFAND-P001185

Uncertain reading. illegible fragment

(more gen.) over A (A-structure).

$A = \mathbb{Z}, \mathbb{Q}$.

$$(H_A,\ W_\bullet H_A \otimes \mathbb{C},\ H_{\mathbb{R}} = H_A \otimes \mathbb{R})$$

[Under $W_\bullet$ stroke leading to: incr. filtr.]

Defin. M — Hodge struct.. Abs. cohomology with coeff. in M.

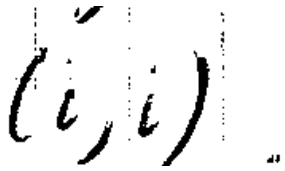
$$H^i_{\mathcal{H}}(M) = \mathrm{Ext}^i(\mathbb{R}, M) \qquad \text{(in the categ. of Hodge structures)}$$

Here $\mathbb{R}$ — the triv. Hodge struct. $\mathbb{R} = H^{0,0}$.

Tate structure

$$i \in \mathbb{Z} \ \Rightarrow \mathbb{R}(i) = (\sqrt{-1})^i \mathbb{R} \subset \mathbb{C}$$

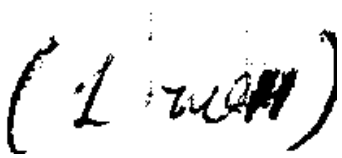
bigrading (i, i)



Original source details: GELFAND-P001185

Diagram and typed description. The manuscript writes (i, i).

(1 term [ending retraced]?)



Original source details: GELFAND-P001185

Uncertain reading. 1 term [ending retraced]

A-Hodge struct.:

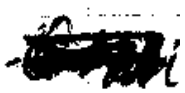
$$(2\pi\sqrt{-1})^i \cdot A = A(i).$$

Def.

$$H^i_{\mathcal{H}_A}(M) := \mathrm{Ext}^i_{\mathcal{H}_A}(A, M).$$

Lemma [Crossed out: illegible word or symbol?]

Uncertain reading. illegible word or symbol



Original source details: GELFAND-P001185

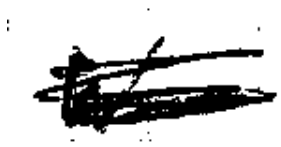
$H^i_{\mathcal{H}} = 0$ for all i, except 0 and 1.

Let $A = \mathbb{Q}, \mathbb{R}$; then

$$H^1_{\mathcal{H}}(M) = {}^{W_{\mathbb{C}\,0}}\!/_{W_{A\,0}+F^0\cap W_{\mathbb{C}\,0}}$$

[Under $W_{A\,0}$ stroke leading to: lattice in $W_{\mathbb{C}\,0}$.] [Crossed out at right: illegible mark?]

Uncertain reading. illegible mark

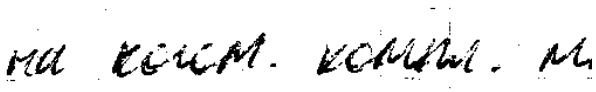


Original source details: GELFAND-P001185

Ex. Prove this fmla. (it is simple).

Aim: to put on the cohom. of

Uncertain reading. comp. compl. [tentative reading of the abbreviations]



Original source details: GELFAND-P001185

var. the same struct. as [Inserted above: there is] on étale cohom..